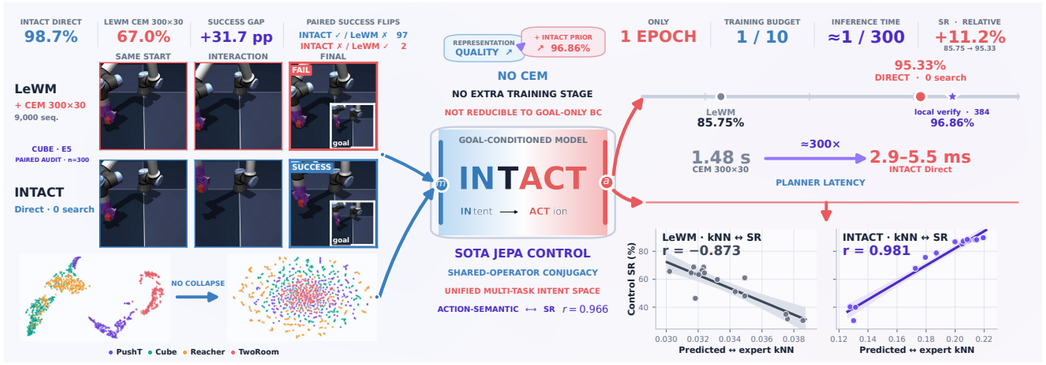
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Isomorphic Intent-to-Action Learning for Search-Free World Models

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INTACT turns action-aligned latent intent into search-free control. Left: a fixed E5 Cube audit compares INTACT Direct with LeWM CEM 300×30 from 300 matched starts. INTACT achieves 98.7% versus 67.0% success (+31.7 points), with 97 favorable and only two adverse success flips. The lower panels show that INTACT prevents representational collapse and preserves a unified multi-task intent space. Center: one shared goal-conditioned operator maps latent motion intent directly to an action chunk, requiring neither CEM nor an additional policy-training stage; its joint physical and goal calls make the learned interface action-semantic rather than goal-only behavior cloning. Right: one-epoch INTACT Direct reaches 95.33% four-task macro SR, while local verification reaches 96.86%. Inference latency falls from 1.48 s for CEM 300×30 to 2.9–5.5 ms. Across 15 checkpoints, predicted–expert kNN alignment strongly tracks INTACT Direct SR ($r = 0.981$), whereas the corresponding LeWM–CEM relation is negative ($r = -0.873$).

Abstract

Forward latent world models predict how actions change a scene, but recover actions for a desired change only through expensive test-time search. We introduce INTACT (**IN**tent-**T**o-**ACT**ion), an end-to-end JEPA that turns action-labeled, reward-free trajectories into a deployable intent-to-action interface. Each transition supplies physical intent $z_{t+1} - z_t$, while a future goal supplies deployment intent $\mathrm{sg}(z_g) - z_t$. The architecture is **isomorphic** in two explicit between–and senses: **isomorphic between the local and goal motion-intent backbone-input graphs** through an identical four-slot grammar and shared parameters, and **isomorphic between the supported local and goal motion-intent families** through the action-law semantics induced by the same predictor, rather than pointwise latent equality. INTACT also realizes two complementary forms of **intact** transfer: **intact between RGB evidence and latent intent coordinates**, because end-to-end action gradients retain action-effective information while filtering nuisance unrelated to motion intent; and **intact between latent intent families and their corresponding action-law families**, because the shared predictor learns the supported family mapping. Asymmetric endpoint gradients ground physical successors and fix future goals as anchors, joining representation learning and control without pointwise latent matching or globally linear dynamics. The resulting intent coordinates support a robust distributional action law: its conditional mean serves directly as a **search-free** policy, while sampling remains available for diversity or optional verification rather than being required for goal-directed control. On the four official LeWM tasks, one-epoch, zero-search models reach 85.78%, 100.00%, 97.67%, and 97.89% success. Optional local CEM centered on the Direct plan reaches 96.86% macro success using 384 instead of 9,000 candidate sequences, reducing sampling by 23.44× while improving pure CEM by 16.00 points. One shared four-task encoder reaches 89.39% E5 Direct macro and improves every task over jointly trained LeWM, while predicted–expert action-family kNN tracks Direct success at $r = 0.954$. Direct inference takes 2.9–5.5 ms.

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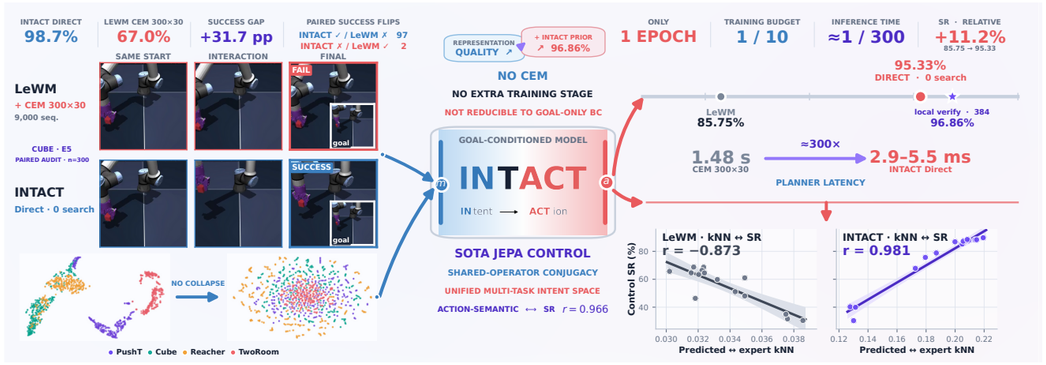
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用于无需搜索世界模型的同构意图到动作学习

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INTACT 将动作对齐的潜在意图转为无搜索控制。左：固定 E5 Cube 审计，从 300 个匹配起点比较 INTACT Direct 与 LeWM CEM 300×30 。INTACT 成功率 98.7%，LeWM 为 67.0%（差 +31.7 个百分点），97 次成功结果向好翻转，仅两次向坏翻转。下方图表明模型防止表征坍塌，并保持统一多任务意图空间。中：同一个目标条件算子，将潜在运动意图直接映射为动作块，无需 CEM 或额外策略训练；物理与目标两类联合调用，使接口具有动作语义，而不只是仅目标行为克隆。右：单轮 INTACT Direct 的四任务宏平均成功率为 95.33%，局部验证为 96.86%；推理延迟从 CEM 300×30 的 1.48 秒降至 2.9–5.5 毫秒。15 个检查点中，预测–专家 k 近邻对齐与 Direct 成功率高度一致（$r = 0.981$），对应 LeWM–CEM 关系则为负（$r = -0.873$）。

摘要

前向潜在世界模型预测动作如何改变场景，但要恢复实现期望变化的动作，仍需昂贵的测试时搜索。我们提出 INTACT（INtent-To-ACTion），一种端到端 JEPA，将带动作标注、无奖励的轨迹转为可部署的意图到动作接口。每次转移提供物理意图 $z_{t+1} - z_t$，未来目标提供部署意图 $\mathrm{sg}(z_g) - z_t$。架构在两种明确的“之间”关系上同构：局部与目标运动意图骨干输入图，经相同四槽语法及共享参数同构；受支持的局部与目标意图族，经同一预测器诱导的动作规律语义同构，而非逐点潜变量相等。INTACT 还实现两种互补的完整传递：RGB 证据与潜在意图坐标之间完整，因为端到端动作梯度保留动作有效信息，过滤与运动意图无关的干扰；潜在意图族与

相应的动作规律族，因为共享预测器学习受支持的族映射。非对称端点梯度锚定物理后继，将未来目标固定为锚点，在不逐点匹配潜变量、也不要求全局线性动力学的情况下，结合表征学习与控制。所得意图坐标支持稳健的分布式动作规律：条件均值直接成为无搜索策略，采样仍可用于多样性或可选验证，却不再是目标导向控制的必要条件。四项官方 LeWM 任务中，单轮无搜索模型成功率为 85.78%、100.00%、97.67%、97.89%。以 Direct 计划为中心的可选局部 CEM，使用 384 而非 9000 条候选序列，宏平均成功率达到 96.86%，采样减少 23.44× ，并比纯 CEM 高 16.00 个百分点。共享四任务编码器在 E5 的 Direct 宏平均为 89.39%，所有任务均优于联合训练 LeWM；预测–专家动作族 k 近邻与 Direct 成功率相关为 $r = 0.954$。Direct 推理为 2.9–5.5 毫秒。

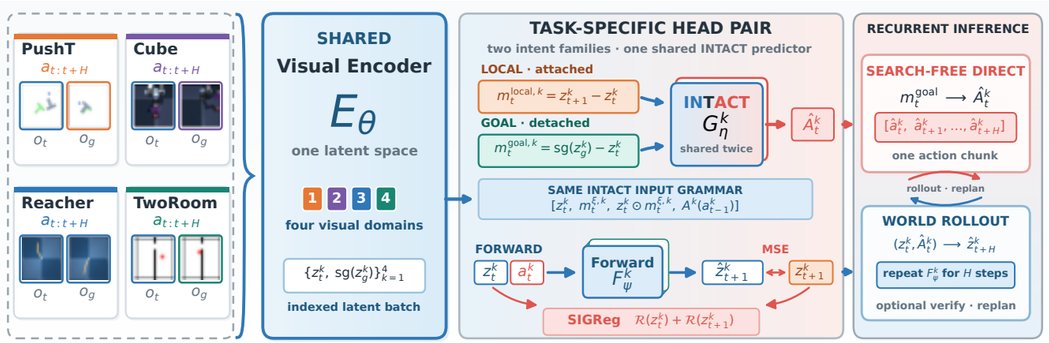


Figure 1: **Shared-encoder INTACT training and recurrent control.** Four visual domains use one encoder and task-specific predictor pairs. Within each task, graph-isomorphic local and goal calls enter the same INTACT Predictor through one matched input grammar, using attached local intent $z_{t+1}^k - z_t^k$ or detached goal intent $\mathrm{sg}(z_g^k) - z_t^k$. The Forward Predictor retains latent-dynamics and SIGReg supervision. At inference, the goal-conditioned call emits one action chunk for search-free Direct control; the world rollout is retained only for optional verification and replanning.

1 Introduction

Latent world models learn a forward conditional: given a visual state and an action, predict what happens next. Goal-conditioned control then numerically inverts this map by proposing action sequences, rolling them through the model, and retaining those that approach a goal [1–4]. This division creates a representation–control asymmetry. Actions shape the latent during training, yet CEM or MPPI begins deployment from generic random or Gaussian actions [5, 6]. Proposals become task-directed only after repeated model rollout, scoring, and refitting. The world model learns what an action will do, but not which action should realize a requested latent change. Action and latent are coupled in the forward predictor but semantically uncalibrated in the inverse direction needed for control.

The missing supervision is already present in offline trajectories. Our core observation is that **even when demonstrations are unordered and training uses no reward, every action-labelled transition still reveals a state-conditioned motion intent and the action that realizes it**. Across many trajectories, these labels reveal families of state-conditional motion intents and their corresponding action families. Our insight is to represent a desired motion by a latent intent m_t, train $\pi_\eta(a_t \mid z_t, m_t)$ jointly with the world model, and let observed actions organize which latent intents require compatible control. The same intent construction is available at deployment, so the learned representation becomes an action proposal interface rather than only a scoring space.

Forward prediction already gives a latent predictive meaning; calling that latent wholly meaningless would be incorrect. What it does not specify is an intent-to-action semantics: which compressed distinctions should be immediately useful for converting a requested motion into control. In this sense, a forward-only bottleneck behaves like an uncalibrated funnel: it gathers the information needed to forecast observations while leaving its action geometry incidental. Intent-conditioned likelihood turns that funnel into a calibrated filter. Action-equivalent requests are organized for rapid readout, while the forward objective retains contacts, topology, and visual residuals needed for future prediction. Representation shaping is therefore not a cosmetic regularizer; it determines whether the learned state can be queried directly by a downstream controller.

The name INTACT encodes three complementary notions of intactness. First, *representation intactness* asks whether RGB-to-latent compression keeps the controllable distinctions needed by the intent-to-action conditional. This is not synonymous with higher rank, more training epochs, or retaining every pixel detail. Second, *conversion intactness* asks whether relations among state-conditioned intent families survive their map into the corresponding action families. Third, ***isomorphic** intactness* has two levels. Structurally, the local and goal motion-intent calls use **isomorphic** backbone-input graphs: the same typed slots, parameter-labelled predictor, and action-law codomain. Semantically, their supported motion-intent families are **isomorphic** through the conditional action relation induced by that shared predictor: each conditional quotient class corresponds bijectively to its realizable action-law image, without requiring local and goal latents to be pointwise equal. Within either call, the first-order intent m_t and its state-dependent interaction $z_t \odot m_t$ retain one matched feature grammar **intact**. The latter is a fixed state–intent bilinear feature—a low-cost second-order interaction in the inputs—not a learned second-order dynamics model. A frozen downstream head can read only what its encoder already exposes; it cannot reconstruct controllable distinctions discarded during pretraining. End-to-end action likelihood therefore shapes both what enters the latent filter and how it is converted into control. In the first sense, action-relevant intent must survive RGB compression **intact**; in the second, the relation between an intent family and its corresponding action family must survive conversion **intact**.

This idea is not an additive recipe of two auxiliary losses. Taken alone, the local branch is physical inverse dynam-

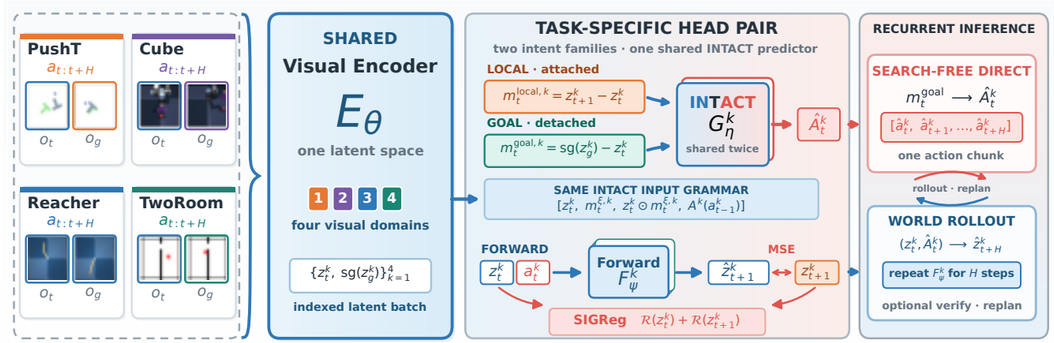


图 1：共享编码器 INTACT 训练与递归控制。四视觉领域共用编码器，各有专用预测器对。任务内，图同构的局部与目标调用，通过匹配输入语法进入同一预测器，使用保留梯度局部意图 $z_{t+1}^k - z_t^k$，或分离梯度目标意图 $\mathrm{sg}(z_g^k) - z_t^k$。前向预测器保留潜在动力学和 SIGReg 监督。推理时目标条件调用输出单动作块，实现无搜索 Direct；世界展开仅保留用于可选验证和重新规划。

1 引言

潜在世界模型学习前向条件关系：给定视觉状态和动作，预测后续。目标条件控制再通过提出动作序列、在模型中展开、保留接近目标的序列，数值求逆 [1–4]。这造成表征–控制不对称：训练时动作塑造潜变量，部署时 CEM 或 MPPI 却从通用随机／高斯动作开始 [5, 6]；反复展开、评分、重新拟合后，提案才具有任务方向。世界模型学习动作会做什么，却没学习哪种动作能实现请求的潜在变化。动作与潜变量在前向预测器中耦合，但在控制所需逆方向未进行语义校准。

缺失监督已存在于离线轨迹。核心观察是：即使示范无序、训练无奖励，每个动作标注转移仍揭示状态条件运动意图及实现它的动作。跨多轨迹，这些标签揭示意图族与对应动作族。我们用潜在意图 m_t 表示期望运动，与世界模型联合训练 $\pi_\eta(a_t \mid z_t, m_t)$，让观测动作组织哪些意图需要兼容控制。同样意图构造在部署可用，所学表征因而成为动作提案接口，而不仅是评分空间。

前向预测已经赋予潜变量预测意义；把这种潜变量说成完全没有意义是不正确的。它尚未规定的是意图到动作的语义：哪些被压缩的差异应该能立即用于把所请求的运动转化为控制。从这个意义上说，只有前向目标的信息瓶颈像一个未经校准的漏斗：它汇集预测观测所需的信息，却让动作几何结构偶然形成。以意图为条件的似然则把这个漏斗变成经过校准的过滤器。动作等价的请求被组织成便于快速读出的形式，同时前向目标保留未来预测所需的接触、拓扑和视觉残差信息。因此，表征塑造不是表面修饰的正则项；它决定了下游控制器能否直接查询学习得到的状态。

INTACT 名称包含三种互补的完整性。第一，表征完整性：RGB 到潜变量的压缩，是否保留意图到动作条件所需的可控制区别；这不等于更高秩、更多训练轮次或保留所有像素。第二，转换完整性：状态条件意图族之间的关系，映射为相应动作族后是否保持。第三，同构完整性有两层：结构上，局部与目标运动意图调用具有同构的骨干输入图，即相同类型槽、参数标记预测器及动作规律陪域；语义上，在共享预测器诱导的条件动作关系下，受支持的局部与目标意图族同构，每个条件商类与其可实现动作规律像一一对应，无须逐点潜变量相等。每次调用中，一阶意图 m_t 及其状态依赖交互 $z_t \odot m_t$ 保持同一匹配特征语法。后者是固定状态–意图双线性特征，是低成本的输入二阶交互，而非学习出的二阶动力学。冻结下游头只能读取编码器已有信息，无法恢复预训练丢弃的可控区别。因此端到端动作似然同时塑造进入潜在滤波器的信息及其控制转换。第一层含义是动作相关意图完整通过 RGB 压缩，第二层是意图族与对应动作族之间的关系完整通过转换。

这一思想不是两个辅助损失的简单相加。单独看局部分支，它是物理逆动力

ics; taken alone, the deployment branch is GCSL-like goal-conditioned imitation. Neither isolated objective expresses the coupling that INTACT needs. The underlying asymmetry is

$$\underbrace{z_{t+1}}_{\substack{\text{physically grounded}\\ \text{unavailable before acting}}} \quad \text{vs.} \quad \underbrace{z_g}_{\substack{\text{available before acting}\\ \text{not a one-step successor}}}. \tag{1}$$

INTACT resolves it by asking one conditional action operator to interpret both families, with $m \in \{m^{\text{local}}, m^{\text{goal}}\}$,

$$G_\eta : (z, m, a^-) \longmapsto p_\eta(a \mid z, m, a^-). \tag{2}$$

The local successor remains attached and supplies physical reachability and representation shaping; the future goal is a stop-gradient deployment anchor. The two conditions are never forced together by latent L_2, while the Forward Predictor preserves world information richer than the immediate action. This shared, graph-isomorphic dual call is the coupling mechanism, not mere coexistence of two losses.

We therefore treat the deployment condition as a *motion request*, not an imagined trajectory. Our waypoint attribution uses the horizon-normalized intent $m^{\text{waypoint}}_{t,h} = (\text{sg}(z_g) - z_t)/h$; the final model instead exposes the raw goal intent $m^{\text{goal}}_t = \text{sg}(z_g) - z_t$. The former imposes an equal-progress scale, while the latter retains the complete terminal residual and lets the nonlinear INTACT Predictor choose state-dependent progress. In both cases, the demonstrated action at the same state supervises the intent, and the same intent semantics are used at deployment. After their conditions are constructed, the two calls form **isomorphic typed predictor graphs**; their upstream gradient routes remain asymmetric. They do not pretend that a terminal goal is the next physical frame. This supervision is obtained *without reward labels or additional semantic annotations; multi-task training uses the known domain index only to select task-specific heads*.

The resulting geometry is conditional. At fixed current state z, two endpoint conditions are equivalent when they induce the same expert action law:

$$y \sim_z y' \iff p^\star_E(a \mid z, y) = p^\star_E(a \mid z, y'). \tag{3}$$

Distinct goals may share the same first action before a doorway and separate after a junction; a real successor and a goal-derived intent may be equivalent without being close in Euclidean latent space. This quotient applies only to the action condition. The complete JEPA latent must still preserve contacts, obstacles, and future visual information required by forward prediction. We call the resulting equivalence class a **conditional action quotient**.

We instantiate this view in INTACT (**IN**tent-**T**o-**ACT**ion learning). INTACT comprises a *Forward Predictor* and one shared *INTACT Predictor*. The latter is called with $m^{\text{local}}_t = z_{t+1} - z_t$ to preserve action-recoverable physical change and with $m^{\text{goal}}_t = \text{sg}(z_g) - z_t$ to map a pre-action goal intent to the demonstrated action block. These calls share feature grammar and parameters, but not necessarily their numerical distributions or encoder gradients. At inference, INTACT alternates the INTACT Predictor mean with the unchanged Forward Predictor and replans from real observations. Its DIRECT mode evaluates no candidates and calls no terminal latent cost; CEM remains available as an optional verifier.

To our knowledge, INTACT is the first end-to-end latent-control architecture to combine all three ingredients: one shared proper action likelihood for observed physical-transition and deployable goal-intent condition families, attached physical-successor endpoints but stop-gradient future-goal anchors, and no pointwise loss that collapses the two latent condition families. The novelty is the complete statistical construction rather than inverse dynamics alone. It aligns the two families through the behavior they induce under the same conditional action operator, not by making their latent coordinates equal. The goal call itself is a goal-conditioned imitation objective. Our claim is not that INTACT lies outside behavior cloning; it is that goal-only, post-hoc frozen-representation, and independent-actor explanations are insufficient for the measured behavior of the complete system. Accordingly, goal-intent-only serves as the GCSL-like ablation in our factorial; the contribution of full INTACT is the end-to-end coupling of this deployment likelihood with JEPA dynamics and a shared local physical anchor. Equivalently, goal-intent-only identifies one familiar branch objective; Full INTACT begins when the attached physical and stop-gradient deployment conditions must be explained by the same parameterized action-law operator.

This distinction also rules out a tempting shortcut. Making a physical successor close to a hand-designed latent waypoint can make a trajectory look straighter while discarding velocity, contact, or action-identifiable information. INTACT does not regress a goal intent onto a successor. It uses a probabilistic action likelihood to preserve the conditional family relation induced by the demonstrated action.

Nor is this merely pointwise interpolation behavior cloning. The goal intent is constructed inside the jointly learned world representation, and its likelihood updates the current-state encoding while the future goal is stop-gradient. This goal-intent path has no physical-successor gradient; successor-based shaping is supplied by the separate physical inverse loss in the full objective. The Forward Predictor remains responsible for rolling the next request. Most importantly, actor-disabled CEM improves after local/goal-intent training, and conversion-fidelity metrics predict closed-loop SR across objective designs. The evidence therefore separates representation shaping, intent-to-action conversion, and optional search rather than assigning all gains to a terminal policy head.

We deliberately separate the theory and evidence into single-task and multi-task regimes. In one task, physical inverse and goal-intent supervision are complementary: the inverse should improve the shared representation even when its actor is removed, while goal intent should primarily improve the deployed conditional. With a shared encoder across heterogeneous tasks, however, individually valid local- and goal-intent gradients can interact. We therefore test the minimal 2×2 factorial—neither action loss, physical inverse only, goal intent only, or both—with one shared encoder and task-specific small heads. This isolates whether local physical anchoring and deployable goal conditioning remain complementary under heterogeneous dy-

力学；单独看部署分支，它是类似 GCSL 的目标条件模仿。两个孤立目标都不能表达 INTACT 所需的耦合。其根本不对称性是

$$\underbrace{z_{t+1}}_{\substack{\text{physically grounded}\\ \text{unavailable before acting}}} \quad \text{vs.} \quad \underbrace{z_g}_{\substack{\text{available before acting}\\ \text{not a one-step successor}}}. \tag{1}$$

INTACT 通过让一个条件动作算子解释两类条件来解决它，其中 $m \in \{m^{\text{local}}, m^{\text{goal}}\}$，

$$G_\eta : (z, m, a^-) \longmapsto p_\eta(a \mid z, m, a^-). \tag{2}$$

局部后继保留梯度，提供物理可达性并塑造表征；未来目标则是停止梯度的部署锚点。两种条件从未被潜变量 L_2 强制拉近，而 Forward Predictor 保留比即时动作更丰富的世界信息。共享且计算图同构的双重调用是耦合机制，绝非两个损失的简单共存。

因此，部署条件被视为运动请求，而非想象轨迹。航点归因使用按时域归一化的意图 $m^{\text{waypoint}}_{t,h} = (\text{sg}(z_g) - z_t)/h$；最终模型则暴露原始目标意图 $m^{\text{goal}}_t = \text{sg}(z_g) - z_t$。前者施加等进度尺度，后者保留完整终点残差，让非线性 INTACT 预测器选择状态依赖的进度。两者都由同一状态的示范动作监督意图，并在部署复用相同意图语义。条件构造后，两次调用形成同构的有类型预测器图；上游梯度路径仍非对称。它们不假装终点目标就是下一物理帧。监督不需要奖励标签或额外语义标注；多任务训练仅使用已知领域索引选择任务专用头。

所得几何结构是条件性的。在固定当前状态 z 下，若两个端点条件诱导相同的专家动作规律，它们就等价：

$$y \sim_z y' \iff p^\star_E(a \mid z, y) = p^\star_E(a \mid z, y'). \tag{3}$$

不同目标可能在门口之前具有相同的首个动作，到岔路后才分开；真实后继与从目标得到的意图也可能等价，却不必在潜空间中具有较小欧氏距离。该商结构仅适用于动作条件。完整 JEPA 潜变量仍须保留前向预测所需的接触、障碍物及未来视觉信息。我们将所得等价类称为条件动作商。

INTACT（意图到动作学习）实现此观点，由前向预测器和一个共享 INTACT 预测器构成。后者以 $m^{\text{local}}_t = z_{t+1} - z_t$ 调用，保持动作可恢复的物理变化；以 $m^{\text{goal}}_t = \text{sg}(z_g) - z_t$ 调用，将动作前目标意图映射到示范动作块。两者共享特征语法和参数，却不必共享数值分布或编码器梯度。推理交替使用预测器均值与未改变的前向预测器，并根据真实观测重新规划。DIRECT 模式不评估候选，也不调用终端潜空间代价；CEM 仍可作为可选验证器。

据我们所知，INTACT 是首个结合以下三项要素的端到端潜在控制架构：对观测物理转移和可部署目标意图两类条件使用同一个适当动作似然；物理后继端点保留梯度，未来目标锚点停止梯度；不使用使两类潜在条件坍缩到一起的逐点损失。创新在于完整的统计构造，而不只是逆动力学。它通过同一条件动作算子下两类条件诱导的行为来对齐两者，而非令潜在坐标相等。目标分支本身是目标条件模仿目标。我们并非声称 INTACT 不属于行为克隆，而是认为，仅目标、事后冻结表征或独立 actor 的解释，不足以解释完整系统的实测行为。因此，仅目标意图是析因实验中类似 GCSL 的消融；完整 INTACT 的贡献，是将该部署似然与 JEPA 动力学及共享局部物理锚点端到端耦合。换言之，仅目标意图对应一种熟悉的分支目标；当保留梯度的物理条件和停止梯度的部署条件必须由同一个参数化动作规律算子解释时，才构成完整 INTACT。

这种区分也排除了一种诱人的捷径。让物理后继接近手工设计的潜空间航点，可能使轨迹看上去更直，却丢失速度、接触或可辨识动作的信息。INTACT 不将目标意图回归到后继状态，而是用概率动作似然保留示范动作所诱导的条件族关系。

这也不只是逐点插值行为克隆。目标意图在联合学习的世界表征内构造，其似然更新当前状态编码，未来目标停止梯度。目标意图路径没有物理后继梯度；后继塑造由完整目标中的独立物理逆损失提供。前向预测器仍负责展开下一请求。关键是局部／目标意图训练后，关闭 actor 的 CEM 也改善，转换保真度指标还能跨目标设计预测闭环成功率。因此证据分离表征塑造、意图到动作转换与可选搜索，而非把全部收益归给终端策略头。

刻意将理论与证据分为单任务、多任务两类。单任务中物理逆与目标意图监督互补：即使去除 actor，逆监督也应改善共享表征，目标意图主要改善部署条件。但异质任务共享编码器时，各自有效的局部和目标梯度可能交互。因此测试最小 2×2 析因：无动作损失、仅物理逆、仅目标意图、两者都有，使用共享编码器和任务专用小头。由此隔离异质动力学下，局部物理锚定与可部署目标条件是否仍互补，

namics.

Here "unified" has an operational meaning: all four domains update one visual encoder in one training run, and the same deployment-supported action loss that produces control also shapes that encoder. This differs from training one world model per domain or freezing a pretrained LeWM and fitting a downstream controller or sampling prior. The published LeWM, C-JEPA, Fast-LeWM, GC-IDM, PRISM, and Qantara control results use task-specific checkpoints or cover only a subset of the suite. Among the interfaces surveyed in Table 1, none previously reports this four-domain joint setting; our novelty claim is deliberately restricted to that comparison set.

The official four-task LeWM benchmark supports both parts of the account. With one full-data epoch and no candidate search, the final distance-preserving single-task model reaches 85.78%, 100.00%, 97.67%, and 97.89% on PushT, Cube, Reacher, and TwoRoom, with three training seeds per task. Earlier matched PushT attribution shows that actor-disabled CEM 30×10 improves from 42.22% for LeWM to 57.67% with inverse supervision, 61.44% with a matched goal-intent objective, and $69.44\pm1.64\%$ for the Waypoint INTACT model with SIGReg 0.03. Because the actor is disabled, this is functional evidence beyond the INTACT Predictor proposal; frozen probes provide direct readout evidence. Direct uses no candidate search and only 2.9–5.5 ms in our audited planner-side measurements; matched PushT is 4.8 ms. Relative to the measured actor-initialized CEM 300×30 average of 1.48 s, this is about a $300\times$ planner-latency reduction. It removes a major model-based search bottleneck for high-frequency control, although it is not an end-to-end VLA latency benchmark. Over all 12 final checkpoints, the complete 576-job planner audit finds that Guarded A, a coherent Direct plan followed by reference-preserving 128×3 local raw-action CEM at $\sigma_0 = 0.25$, reaches 96.86% macro and 92.22% worst-task SR. Relative to matched pure CEM 300×30, it gains 16.00 macro points while reducing sampled candidate sequences from 9,000 to 384 (23.44×). Ordinary $\sigma_0 = 0.25$ is only 0.08 macro point lower, making it a particularly simple strong default.

For multi-task learning, we use the first complete controlled Math-SDPA matrix: six objective cells, three training seeds, and three evaluation seeds per checkpoint. At E1, Full INTACT reaches $42.69\pm7.19\%$ Direct macro SR, versus $35.75\pm7.58\%$ for matched waypoint-intent-only and $33.47\pm3.98\%$ for inverse-only. Relative to goal-intent-only, adding physical inversion raises PushT and TwoRoom by 12.89 and 15.33 points and reduces TwoRoom training-seed standard deviation from 31.13 to 10.58 points. At E5, one shared Goal-displacement INTACT encoder reaches $89.39\pm0.77\%$ macro Direct SR. Relative to the matched shared-encoder LeWM, it improves all four tasks by 5.66/32.23/12.56/42.44 points; it also exceeds the published task-specific LeWM macro and its Cube/Reacher scores. Against the matched three-seed goal-intent-only cell, the physical call adds 8.78 ± 1.63 macro points. With every action head disabled, pure CEM 300×30 rises from $66.17\pm2.67\%$ for LeWM to $70.08\pm1.13\%$ for Goal-displacement INTACT; adding Guarded A to the learned Direct plan further raises $89.39\pm0.77\%$ to $90.47\pm0.84\%$. These matched controls separate representation/Forward usability, direct intent conversion, and optional verification. Separately, a 45-checkpoint E1–E5 simultaneous-dual-likelihood audit supplies evidence beyond objective rankings: predicted–expert action-family kNN and CKA have Pearson $r = 0.954$ and 0.897 with official SR, respectively, compared with 0.815 for pointwise action R^2. After controlling epoch and cohort, kNN remains $r = 0.902$. This gap indicates that preserving local and global action-family geometry is more informative than recovering every expert action by a single point estimate.

Our contributions are:

1. We define a state-conditional action quotient whose classes are canonically isomorphic to the realizable action-law image, represent endpoint requests by latent displacement coordinates, and introduce an end-to-end *isomorphic implicit-training architecture*: one shared proper likelihood aligns observed-transition and deployment-intent condition families, with asymmetric gradient routing and without forcing their latent endpoints to coincide. A matched sharing ablation shows that two independent actors remain worse even at double capacity, directly supporting the shared-operator choice.

2. We identify the complementary matched intra-call grammar required by the INTACT head: the first-order intent m_t must be paired with its matched state–intent bilinear interaction $z_t \odot m_t$. The new A–G control shows that centering the product alone gives only $+0.89$ points, whereas matching the centered main condition and interaction yields a $+3.89$-point residual–product interaction with the same direction in all three training seeds. This is a state-conditioned inverse-control factor, not a second-order dynamics model.

3. We develop a non-diffusion, autoregressive intent–forward controller that emits a state-conditioned action distribution rather than a single static answer. It supports zero-search Direct control and plan-centered local CEM: the best fixed 128×3 rule reaches 96.86% macro and 92.22% worst-task SR while using only 4.27% of the action-step proposals used by CEM 300×30.

4. We introduce native-coordinate action-family diagnostics—predicted–expert CKA and kNN overlap—and show across 45 eligible E1–E5 checkpoints that their correlations with official SR ($r = 0.897/0.954$) exceed that of pointwise action R^2 ($r = 0.815$); kNN remains $r = 0.902$ after controlling epoch and cohort, exposing family-level intent-to-action preservation as the relevant mechanism. A 21,600-episode gauge intervention further shows that paired correspondence recovers 68.04% SR versus 9.46% after pair shuffling, while a Full backbone cannot give a LeWM actor an untrained deployment map (5.08%).

5. We show that joint intent supervision improves not only the deployed actor but also the shared representation: after removing the INTACT Predictor at evaluation, matched actor-disabled pure CEM still improves over LeWM, with frozen probes independently confirming more recoverable state and intervention information.

力学。这里"统一"有操作性定义：四领域在同次训练中更新一个视觉编码器，产生控制的部署支持动作损失也塑造编码器。这不同于各领域单独训练世界模型，或冻结预训练 LeWM 再拟合下游控制器／采样先验。已发表 LeWM、C-JEPA、Fast-LeWM、GC-IDM、PRISM、Qantara 控制结果采用任务专用检查点或仅覆盖部分任务。表 1 接口中没有此前报告的四领域联合设置；创新声明刻意限制于该比较集合。

官方四任务 LeWM 基准支持上述两方面解释。完整数据训练一轮，无候选搜索，最终保持距离的单任务模型在 PushT、Cube、Reacher、TwoRoom 达到 85.78%、100.00%、97.67%、97.89%，每任务三个训练种子。较早匹配的 PushT 归因显示，关闭 actor 的 CEM 30×10 从 LeWM 42.22%，提高至逆监督 57.67%、匹配目标意图 61.44%、SIGReg 0.03 的航点 INTACT 69.44 ± 1.64 %。因 actor 关闭，这是超出 INTACT 预测器提案作用的功能证据；冻结探测器提供直接读出证据。Direct 无候选搜索，审计的规划器侧延迟仅 2.9–5.5 毫秒，匹配 PushT 为 4.8 毫秒；相比 actor 初始化 CEM 300×30 平均 1.48 秒，延迟约减少 $300\times$。它消除高频控制的一大模型搜索瓶颈，但并非端到端 VLA 延迟基准。全部 12 个最终检查点的 576 项规划审计表明，Guarded A 先生成一致 Direct 计划，再保留参考，以 128×3 局部原始动作 CEM、$\sigma_0 = 0.25$ 验证，宏平均 96.86%、最差任务 92.22%。相比匹配纯 CEM 300×30，宏平均高 16.00 个百分点，候选序列从 9000 减至 384（23.44×）。普通 $\sigma_0 = 0.25$ 仅低 0.08 个宏平均百分点，是尤其简单且强的默认方案。

多任务学习使用首个完整受控 Math-SDPA 矩阵：六个目标组合、三个训练种子，每检查点三个评估种子。E1 时，完整 INTACT Direct 宏平均成功率为 $42.69\pm7.19\%$，匹配的仅航点意图为 $35.75\pm7.58\%$，仅逆模型为 $33.47\pm3.98\%$。相较仅目标意图，加入物理逆推使 PushT 和 TwoRoom 分别提高 12.89、15.33 个百分点，TwoRoom 训练种子标准差从 31.13 降至 10.58。E5 时，共享目标位移 INTACT 编码器 Direct 宏平均为 $89.39\pm0.77\%$。相较匹配共享编码器 LeWM，四任务分别提高 5.66／32.23／12.56／42.44 个百分点；也超过已发表任务专用 LeWM 的宏平均及 Cube／Reacher 分数。相较匹配三种子的仅目标意图组合，物理调用增加 8.78 ± 1.63 个宏平均百分点。关闭全部动作头，纯 CEM 300×30 从 LeWM 的 $66.17\pm2.67\%$ 升至目标位移 INTACT 的 $70.08\pm1.13\%$；为学习到的 Direct 计划加入 Guarded A，又使 $89.39\pm0.77\%$ 升至 $90.47\pm0.84\%$。这些匹配对照分离了分离表征／前向模型可用性、直接意图转换及可选验证。另有 45 检查点 E1–E5 同时双似然审计，提供目标排名之外的证据：预测–专家动作族 k 近邻和 CKA 与官方成功率 Pearson 分别为 $r = 0.954$ 、0.897，逐点动作 R^2 为 0.815。控制轮次和组别后 k 近邻仍 $r = 0.902$。差距说明，保持局部和全局动作族几何，比单点估计恢复每个专家动作更具信息量。

我们的贡献如下：

1. 定义状态条件动作商，其类与可实现动作规律像规范同构；以潜在位移表示端点请求，提出端到端同构隐式训练架构：同一共享适当似然，通过非对称梯度路由对齐观测转移与部署意图条件族，不强迫潜在端点重合。匹配共享消融说明，即使独立 actor 容量翻倍仍更差，直接支持共享算子。

2. 确定头所需的互补调用内部匹配语法：一阶意图 m_t 必须配其匹配状态–意图双线性交互 $z_t \odot m_t$。新 A–G 对照说明，仅中心化乘积只提高 $+0.89$ ，而匹配中心化主条件与交互产生 $+3.89$ 个残差–乘积交互百分点，三个训练种子方向一致。这是状态条件逆控制因子，并非二阶动力学模型。

3. 我们提出一个非扩散式、自回归的意图–前向控制器，输出以状态为条件的动作分布，而非单一静态答案。它支持无需搜索的 Direct 控制和以计划为中心的局部 CEM：最佳固定 128×3 规则达到 96.86% 的宏平均成功率和 92.22% 的最差任务成功率，动作步候选数量仅为 CEM 300×30 的 4.27%。

4. 提出原生坐标动作族诊断：预测–专家 CKA 与 k 近邻重叠。45 个合格 E1–E5 检查点中，与官方成功率的相关（$r = 0.897/0.954$）超过逐点动作 R^2 （$r = 0.815$）；控制轮次与组别后 k 近邻仍为 $r = 0.902$，表明族层面的意图到动作保持是相关机制。21600 回合坐标规范干预还显示，配对对应恢复 68.04% 成功率，打乱后仅 9.46%；完整骨干也不能给 LeWM actor 创造未经训练的部署映射（5.08%）。

5. 我们表明，联合意图监督不仅改善部署时的 actor，也改善共享表征：即使评估时移除 INTACT Predictor，在匹配条件下禁用 actor 的纯 CEM 仍优于 LeWM；冻结探针也独立证实状态与干预信息更容易恢复。

6. We give latent space an operational semantics by extending a Gaussian-regularized self-supervised predictor into a multi-task mapping between motion-intent families and action families. This principle is not specific to LeWM's encoder–encoder JEPA: it applies to general learned representations whenever action-labelled transitions are available.

2 Related Work

Latent world models and JEPAs. PlaNet and Dreamer learn recurrent latent dynamics through generative objectives [1, 7]; TD-MPC2 learns compact dynamics jointly with control-relevant predictions [2]. JEPA-style world models instead predict in representation space. DINO-WM plans with a frozen DINOv2 encoder [3], PLDM trains end-to-end with a VICReg-style objective [8], and LeWM combines latent prediction with Sketched Isotropic Gaussian Regularization (SIGReg) for stable end-to-end learning [4, 9]. We retain LeWM's predictive backbone but ask how an action-conditioned deployment interface can shape, rather than collapse, that same representation. Two recent extensions improve the forward-model side without changing this search interface. C-JEPA uses object-level latent masking and reports efficient CEM on PushT [10]; Fast-LeWM predicts all action-prefix horizons in parallel and reduces the matched TwoRoom CEM solve from 54.4 to 28.3 seconds while retaining the same candidate budget [11]. Both train a separate world model per reported control domain. INTACT instead asks whether a single shared encoder can expose a directly deployable action conditional across the four LeWM domains. The distinction is important: a forward-only encoder is not meaningless, but its compression criterion is prediction sufficiency rather than immediate action sufficiency. INTACT uses labeled transitions to calibrate that bottleneck without replacing its predictive role.

Inverse dynamics and representation learning. Inverse models have long been used to isolate controllable visual structure [12, 13] and support multitask imitation [14]. An early robotic precursor jointly learned forward and inverse models from unlabeled poking interaction, using inverse prediction to shape visual features and forward prediction to regularize their dynamics [15]. SMWM shows that one-step inverse supervision can regularize an end-to-end JEPA toward sensorimotor features [16]; broader probing work likewise links temporal pretraining to action recoverability [17]. These results concern action-relevant representation, but a local inverse evaluated only on observed successors does not identify its behavior on a goal-derived intent. INTACT makes this support mismatch explicit and keeps the learned conditional as a deployment interface.

Hindsight goal-conditioned imitation. GCSL relabels a trajectory with a future state that it actually reaches and maximizes the corresponding goal-conditioned action likelihood, iterating between collection under the current policy and supervised updates [18]. Complementary formulations cast goal reaching as recursive future-state classification or learn a contrastive representation whose inner product is a goal-conditioned value function [19, 20]. INTACT's deployment branch has this GCSL-like statistical form, and goal-intent-only is the corresponding hindsight-imitation ablation in our factorial. It is not the original GCSL algorithm: it retains the JEPA forward objective and SIGReg and uses fixed offline expert trajectories rather than iteratively collecting its own. INTACT is not goal-only GCSL. Its deployment branch is a GCSL-like hindsight action likelihood inside a jointly trained JEPA, while the complete method additionally uses the same conditional operator on attached one-chunk physical successors to anchor local reachability. This distinction is structural rather than terminological. Goal-intent-only captures one branch call and is therefore the GCSL-like baseline reported in our tables; physical-inverse-only captures the other branch call. Only Full INTACT evaluates both on each transition through one weight-shared, graph-isomorphic predictor while routing endpoint gradients asymmetrically. Thus the shared operator is not cosmetic parameter tying: it is the mechanism that gives physically grounded and deployable condition families a common action-law semantics without matching their latent coordinates.

Options, empowerment, and unsupervised skill discovery. The options framework treats closed-loop policies as temporally extended action units [21], while empowerment measures the information capacity from an agent's action channel to reachable outcomes [22]. Modern unsupervised RL often combines these ideas with a latent skill variable: DIAYN makes skills distinguishable from visited states [23]; DADS favors skills with predictable dynamics [24]; CIC contrastively aligns skills with state transitions [25]; and METRA learns a temporally grounded metric space in which a policy moves along latent directions [26]. These methods share the useful interface $\pi(a \mid s, \xi)$, but solve a different statistical problem. Their code ξ is generally sampled or discovered through interaction and optimized by intrinsic reward, mutual information, coverage, or temporal distance. Our m is not a free behavior index: it is a state-conditional motion intent grounded in an offline demonstration, and the recorded action is its supervised target. Rather than discovering behaviors that make latent codes identifiable, we identify which latent requests are equivalent under the demonstrated action law.

Metric learning and action-induced equivalence. Classical metric learning uses pair or triplet labels to make Euclidean embeddings invariant within a semantic class and separated across classes [27, 28]. In control, bisimulation metrics learn task-relevant invariances without pixel reconstruction, while goal-conditioned bisimulation extends this idea to functional equivariance between analogous goals [29, 30]. Action labels play an analogous organizational role here: across otherwise unordered trajectories, they reveal families of motion intents requiring similar control. The distinction is that INTACT does not impose a pairwise distance or collapse complete JEPA latents with the same action. Equivalence is defined at fixed state by equality of the conditional action law and learned through a proper action likelihood. Forward prediction and SIGReg

6. 我们将高斯正则化的自监督预测器扩展为运动意图族与动作族之间的多任务映射，从而赋予潜空间操作语义。该原理并不限于 LeWM 的编码器–编码器 JEPA：只要存在带动作标签的转移，它就适用于一般的学习表征。

2 相关工作

潜在世界模型与 JEPA。PlaNet、Dreamer 用生成目标学习递归潜在动力学 [1, 7]；TD-MPC2 将紧凑动力学与控制相关预测联合学习 [2]。JEPA 世界模型则在表征空间预测：DINO-WM 使用冻结 DINOv2 编码器规划 [3], ，PLDM 以 VICReg 风格目标端到端训练 [8], ，LeWM 将潜在预测与随机压缩各向同性高斯正则化 SIGReg 结合，实现稳定端到端学习 [4, 9]。保留 LeWM 预测骨干，同时考察动作条件部署接口如何塑造同一表征而不使其坍塌。近期两项扩展改善前向模型，但未改变搜索接口：C-JEPA 用物体级潜在遮挡，在 PushT 报告高效 CEM [10]；Fast-LeWM 并行预测全部动作前缀时域，在相同候选预算下，将 TwoRoom CEM 从 54.4 秒降至 28.3 秒 [11]。两者为各控制领域分别训练世界模型。INTACT 则研究同一共享编码器，能否在四个 LeWM 领域暴露可直接部署的条件动作接口。前向编码器并非没有意义，但其压缩准则是预测充分性，而非即时动作充分性。INTACT 以标注转移校准瓶颈，不替代其预测角色。

逆动力学与表征学习。逆模型长期用于分离可控视觉结构 [12, 13] 、支持多任务模仿 [14]。早期机器人研究从无标注戳碰交互中联合学习前向与逆模型，以逆预测塑造视觉特征，以前向预测正则化动力学 [15]。SMWM 表明单步逆监督能使端到端 JEPA 向感觉运动特征正则化 [16]；更广泛探测工作也将时间预训练与动作可恢复性联系起来 [17]。这些结果讨论动作相关表征，但仅在观测后继上评估的局部逆模型，不能识别其在目标生成意图上的行为。INTACT 明确这一支持不匹配，保留所学条件接口用于部署。

事后目标条件模仿。GCSL 用轨迹实际到达的未来状态重新标记轨迹，并最大化相应的目标条件动作似然，在当前策略收集数据与监督更新之间迭代 [18]。互补方法将目标到达表述为递归的未来状态分类，或学习对比式其内积为目标条件价值函数的表征 [19, 20]。INTACT 部署分支具有这一类似 GCSL 的统计形式，仅目标意图是析因实验中对应的事后模仿消融。它并非原始 GCSL 算法：保留 JEPA 前向目标和 SIGReg，采用固定离线专家轨迹，而非迭代采集自身轨迹。INTACT 也不等于仅目标 GCSL。部署分支是在联合训练 JEPA 内的类似 GCSL 事后动作似然，而完整方法还将同一条件算子用于保留梯度的单块物理后继，以锚定局部可达性。区别是结构性的，而非术语性的。仅目标意图对应一类分支调用，是表中类似 GCSL 的基线；仅物理逆模型对应另一类。只有完整 INTACT 对每次转移，经过同一个权重共享、计算图同构的预测器评估两者，同时非对称路由端点梯度。因此共享算子并非装饰性的参数绑定，而是在不匹配潜在坐标的情况下，赋予物理基础条件与可部署条件共同动作规律语义的机制。

选项、赋能与无监督技能发现。options 框架把闭环策略视为时间延展的动作单元 [21], ；赋能衡量智能体动作通道到可达结果的信息容量 [22]。现代无监督强化学习常将其与潜在技能变量结合：DIAY N 使技能可从访问状态区分 [23]；DADS 偏好动态可预测的技能 [24]；CIC 对比对齐技能与状态转移 [25]；METRA 学习时间锚定的度量空间，让策略沿潜在方向移动 [26]。这些方法共享有效接口 $\pi(a \mid s, \xi)$，但统计问题不同。其代码 ξ 通常经交互采样或发现，用内在奖励、互信息、覆盖率或时间距离优化。我们的 m 并非自由行为索引，而是以离线示范为基础的状态条件运动意图，记录动作就是监督目标。我们不是发现使潜在代码可识别的行为，而是识别在示范动作规律下等价的潜在请求。

度量学习与动作诱导等价。经典度量学习用成对或三元组标签，使欧氏嵌入在语义类内不变、类间分离 [27, 28]。控制中，双模拟度量不重建像素而学习任务相关不变性；目标条件双模拟将其扩展为类似目标间的功能等变性 [29, 30]。动作标签在此有类似组织作用：在无序轨迹中揭示需要相似控制的运动意图族。但 INTACT 不施加成对距离，也不把相同动作的完整 JEPA 潜变量压缩到一起。等价在固定状态下，以条件动作规律相等定义，通过适当动作似然学习。前向预测和 SIGReg

preserve variations within an action fiber that remain important for future dynamics. In this sense our quotient is an output-distribution geometry, not a metric-learning loss on observations.

Amortized and prior-guided world-model control. GLAMOR learns goal-conditioned inverse and prior models to guide random shooting [31]. Biased-MPPI provides a general mechanism for fusing ancillary sampling distributions into MPPI [32]. More recently, PRISM predicts a current/goal-conditioned Gaussian trajectory prior from a frozen JEPA encoder and fuses it with MPPI by precision-weighted Product-of-Gaussians updates [33]. It addresses a train–deployment goal shift through the planner's deployment-time cost, but still evaluates sampled trajectories; its reported limitations include task-specific near-expert demonstrations and a unimodal local prior for potentially multimodal long-horizon behavior.

GC-IDM first studies a pairwise inverse over consecutive frozen LeWM latents with plans produced by latent linear interpolation [34]. Its oracle action $R^2 = 0.993$ does not rescue invalid straight latent paths; on TwoRoom, its appendix reports 34% for interpolation, 48% after refinement and candidate selection, and 87% for CEM. Conversely, its final goal-conditioned controller reaches 100% TwoRoom SR with training R^2 around 0.20 by predicting useful directions and replanning. GC-IDM thus already shows that scalar coordinate recovery is not a sufficient theory of control. It avoids imagined latent paths in its final controller and conditions on encoded current/goal observations plus remaining horizon. INTACT takes another route: the waypoint intent is a request coordinate rather than a physical successor, and is placed on training support by pairing it with the demonstrated current action. INTACT also updates the JEPA encoder end to end and evaluates both search-free execution and the actor-disabled world model. Our claim is not the first inverse controller or learned sampling prior; it is that deployment-supported intent supervision can organize the representation and directly amortize action search.

Statistically, the frozen-encoder stage used by GC-IDM and PRISM is closer to behavior cloning or prior fitting on top of a fixed visual representation: it can exploit information that the encoder already exposes, but its action loss cannot decide which distinctions that encoder should preserve. INTACT is unified in a stricter sense. The visual encoder, forward dynamics, and deployment-supported action conditional are optimized together, and the same encoder is shared across all four domains. Among LeWM and the LeWM-derived systems compared here, we are not aware of a prior report that jointly trains one four-domain encoder while allowing the deployed action conditional to shape that encoder. This is a scoped claim about the published interfaces in Table 1, not a claim that frozen-feature controllers are ineffective.

Action quotients and visual fibers. QuoVLA formalizes a quotient of VLM prompt representations induced by the Bayes-optimal action-trajectory law and approximates it with quantization and a dual-branch VLA design [35]. We share the general principle that representations distinguished by perception may be equivalent for action, and do not claim to originate action quotients. Our quotient acts on local/goal motion *intents at fixed JEPA state*, not on the complete VLM prompt representation; it predicts a transition-aligned action block and coexists with an explicit forward world model. FiberTune identifies the complementary risk that action supervision leaves visually meaningful directions within an action fiber unconstrained and therefore vulnerable to collapse [36]. This supports our architectural separation: the actor may quotient action-equivalent conditions, while forward prediction and SIGReg preserve the richer world latent.

Unified video–action models. UWM couples video and action diffusion to expose forward, inverse, policy, and generation queries [37]; DreamGen converts imagined futures to actions through a separate inverse or latent-action model [38]. Qantara trains a JEPA for several dispatch-dependent control paradigms using bridge flow [39]. These systems motivate a unified conditional view. Our scope is deliberately smaller and mechanistic: a compact reward-free JEPA, a proper endpoint action likelihood, and controlled evidence linking the statistical unit of supervision to latent geometry and direct control.

3 INTACT: Theory and Method

INTACT augments a forward JEPA with one conditional action operator used twice: once on a realized physical transition and once on a condition available before acting. The two calls share parameters and feature grammar but deliberately have different endpoint gradients. We first give the single-task construction and then extend it to a shared multi-task encoder.

3.1 Problem Setting and the Forward-Action Gap

Figures 1 and 2 show the same system from its training and deployment views. In Fig. 1, superscript k indexes a visual domain and its task-specific heads; below we rename that index τ and reserve k for temporal position inside a window. For a single task we omit τ. This removes a notational collision between the four colored task routes in Fig. 1 and the recurrent planning steps in Fig. 2.

For domain τ, let the offline data be pixel–action trajectories $\mathcal{D}^\tau = \{(\boldsymbol{o}^\tau_{i,s}, \boldsymbol{u}^\tau_{i,s})\}$, where i indexes trajectories and s indexes environment steps. We index model-chunk boundaries by t and form the fixed action block $\boldsymbol{a}^\tau_{i,t} = [\boldsymbol{u}^\tau_{i,Bt}; \dots; \boldsymbol{u}^\tau_{i,Bt+B-1}] \in \mathbb{R}^{Bd_a}$ with $B = 5$. A training window spans $H = 5$ such chunks from o_t to o_{t+H}, and the shared encoder-projector gives $z_j = E_\theta(o_j) \in \mathbb{R}^d$ and $z_g = z_{t+H}$. At teacher step $k \in \{0, \dots, H-1\}$, the orange local call in Fig. 1 uses $z_{t+k+1} - z_{t+k}$, while the green goal call uses $\mathrm{sg}(z_g) - z_{t+k}$. They enter the same four typed input slots of G_η and both predict the fixed one-chunk target a_{t+k}; the $\widehat{A}^\tau_t$ box in the figure denotes this conditional action output. The goal endpoint stays fixed while its remaining offset is $r = H - k \in \{5, 4, 3, 2, 1\}$ chunks. Thus "multi-horizon" means multiple remaining goal offsets,

保留动作纤维内部对未来动力学仍然重要的变化。因此，我们的商是输出分布的几何结构，而不是观测上的度量学习损失。

摊销与先验引导的世界模型控制。GLAMOR 学习目标条件逆模型与先验，指导随机射击 [31]；Biased-MPPI 提供融合辅助采样分布到 MPPI 的通用机制 [32]。近期 PRISM 从冻结 JEPA 编码器预测当前／目标条件高斯轨迹先验，以精度加权高斯乘积更新融合 MPPI [33]。通过部署代价解决训练–部署目标偏移，但仍评估采样轨迹；其报告局限包括任务专用近专家示范，以及用单峰局部先验描述可能多峰的长时域行为。

GC-IDM 首先研究连续冻结 LeWM 潜变量间的成对逆模型，其计划来自潜在空间线性插值 [34]。oracle 动作 $R^2 = 0.993$ 无法补救无效直线路径：TwoRoom 附录报告插值 34%，细化与候选选择后 48%，CEM 为 87%。反之，其最终目标条件控制器在训练 R^2 约 0.20 时，通过预测有效方向并重新规划达到 TwoRoom 100%。这已说明标量坐标恢复不足以作为控制理论。最终控制器避免想象的潜在路径，采用当前／目标观测编码及剩余时域为条件。INTACT 走另一条路线：航点意图是请求坐标而非物理后继，通过与示范当前动作配对，将其放到训练支持范围内。INTACT 还端到端更新 JEPA 编码器，评估无搜索执行及关闭 actor 的世界模型。我们不声称首创逆控制器或学习采样先验，而是指出受部署支持的意图监督可以组织表征并直接摊销动作搜索。

统计上，GC-IDM 和 PRISM 的冻结编码器阶段，更接近在固定视觉表征上做行为克隆或先验拟合：能利用编码器已有信息，但动作损失不能决定编码器应保留哪些区别。INTACT 在更严格意义上统一：视觉编码器、前向动力学及部署支持的动作条件共同优化，同一编码器在四领域共享。在此比较的 LeWM 及衍生系统中，据我们所知，此前没有联合训练四领域编码器、并让部署动作条件塑造编码器的报告。这是针对表 1 已发表接口的有限范围声明，并非认为冻结特征控制器无效。

动作商与视觉纤维。QuoVLA 将由贝叶斯最优动作轨迹规律诱导的 VLM 提示表征商形式化，并以量化和双分支 VLA 设计近似它 [35]。我们共享以下一般原则：感知上能够区分的表征可能

在动作上等价，并不声称首创动作商。我们的商在固定 JEPA 状态下作用于局部／目标运动意图，而非完整 VLM 提示表征；预测与转移对齐的动作块，与显式前向世界模型共存。FiberTune 指出互补风险：动作监督未约束动作纤维内有视觉意义的方向，易使其坍塌 [36]。这支持架构分工：actor 可将动作等价条件取商，前向预测和 SIGReg 则保留更丰富世界潜变量。

统一视频–动作模型。UWM 耦合视频和动作扩散，支持前向、逆、策略、生成查询 [37]；DreamGen 经独立逆模型或潜在动作模型将想象未来转为动作 [38]；Qantara 用桥流训练 JEPA，支持不同调度的控制范式 [39]。这些系统启发统一条件观点。我们的范围刻意更小且重机制：紧凑无奖励 JEPA、适当端点动作似然，以及将监督统计单元与潜在几何和直接控制联系的受控证据。

3 INTACT：理论与方法

INTACT 在前向 JEPA 上加入一个条件动作算子，并调用两次：一次作用于已实现的物理转移，另一次作用于行动前可获得的条件。两次调用共享参数与特征语法，却有意采用不同的端点梯度。我们先给出单任务构造，再扩展到共享的多任务编码器。

3.1 问题设定与前向–动作差距

图 1 和图 2 分别从训练与部署角度展示同一个系统。图 1 的上标 k 索引视觉领域及其专用头部；下文将该索引改为 τ，并用 k 表示窗口内的时间位置。单任务时省略 τ。这样可避免图 1 的四条彩色任务路径与图 2 的递归规划步骤发生符号冲突。

对于领域 τ，离线数据为像素–动作轨迹 $\mathcal{D}^\tau = \{(\boldsymbol{o}^\tau_i, s, \boldsymbol{u}^\tau_i, s)\}$，其中 i 为轨迹索引，s 为环境步索引。以 t 索引模型动作块边界，构造固定动作块 $\boldsymbol{a}^\tau_{i,t} = [\boldsymbol{u}^\tau_{i,Bt}; \dots; \boldsymbol{u}^\tau_{i,Bt+B-1}] \in \mathbb{R}^{Bd_a}$，其中 $B = 5$。训练窗口从 o_t 至 o_{t+H}，跨越 $H = 5$ 个动作块；共享编码器–投影器输出 $z_j = E_\theta(o_j) \in \mathbb{R}^d$ 和 $z_g = z_{t+H}$。在教师步骤 $k \in \{0, \dots, H-1\}$，图 1 橙色局部分支使用 $z_{t+k+1} - z_{t+k}$，绿色目标分支使用 $\mathrm{sg}(z_g) - z_{t+k}$。两者进入 G_η 相同的四个有类型输入槽，并预测相同的固定 $\widehat{\ }$ 单块目标 a_{t+k}；图中 A^τ_t 方框表示条件动作输出。目标端点固定，其剩余偏移为 $r = H - k \in \{5, 4, 3, 2, 1\}$ 个块。因此，"多时域"指多个剩余目标偏移，

Table 1: Positioning among closely related methods. "Action→ E" means that action supervision updates the visual/world encoder, and "Shared-4T E" means one encoder is jointly trained over all four official LeWM domains. The final column states each method's actual training-to-control path; a cross denotes an unreported interface, not an impossible extension.

Method	Action→ E	Shared-4T E	Condition	Action obj.	Direct	Search	Training-to-control path
DINO-WM [3]	✗	✗	action	none	✗	✓	Frozen image encoder; CEM ranks latent rollouts
C-JEPA [10]	✗	✗	action/proprio.	none	✗	✓	Frozen object features; CEM searches masked rollouts
LeWM [4]	✗	✗	action	none	✗	✓	Joint JEPA prediction; CEM inverts the forward model
Fast-LeWM [11]	✗	✗	action prefix	none	✗	✓	LeWM training; parallel action prefixes are searched
SMWM [16]	✓	✗	successor	local action	✗	✓	Local inverse shapes the encoder; deployment still uses CEM
GC-IDM [34]	✗	✗	goal + horizon	next action	✓	✗	Frozen LeWM; a separate goal/horizon actor acts directly
PRISM [33]	✗	✗	state/goal	Gaussian prior	✗	✓	Frozen LeWM; a separately trained prior initializes MPPI
Qantara [39]	✓	✗	dispatch-dependent	action flow	✓	✓	Joint JEPA; bridge-flow heads serve dispatch modes
QuoVLA [35]	✓	✗	vi-sual/language	action chunk	✓	✗	VLM action quotient; quantized intent drives a VLA actor
INTACT	✓	✓	local/goal intent	action block	✓	opt.	One shared JEPA/actor law; Direct by default, search verifies

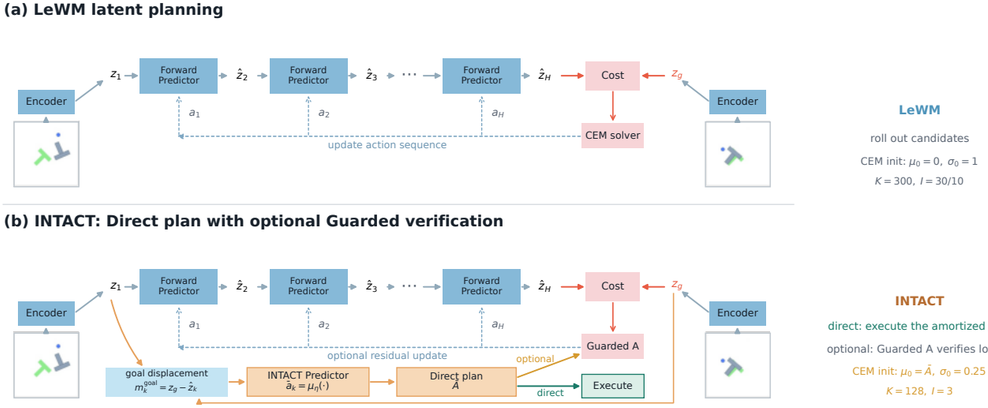


Figure 2: **INTACT converts mandatory search into optional verification.** LeWM samples raw actions from an uninformed Gaussian and uses forward rollouts to become goal-directed. INTACT alternates the raw goal displacement, the shared conditional action mean, and the unchanged Forward Predictor to produce a Direct plan. It can execute immediately or locally verify that plan with $K = 128$, $I = 3$, and $\sigma_0 = 0.25$.

not multiple output chunk lengths, and the final raw goal policy does not receive r explicitly.

At deployment, the lower route of Fig. 2 reuses exactly these symbols: encode the current and goal images as z_t and z_g, set $\widetilde{z}_0 = z_t$, then recurrently compute $m_k = z_g - \widetilde{z}_k$, $\bar{a}_k = \mu_\eta(\widetilde{z}_k, m_k, \bar{a}_{k-1})$, and $\widetilde{z}_{k+1} = F_\phi(\widetilde{z}_k, \bar{a}_k)$. The one-chunk means form the plan $\bar{A} = (\bar{a}_0, \ldots, \bar{a}_{H-1})$ corresponding to the training-time action output in Fig. 1. Direct executes its first block and replans from the next observation; Guarded A only verifies this plan locally. The upper route in Fig. 2 instead initializes raw actions independently of these learned intent coordinates.

Across both routes, the unchanged LeWM Forward Predictor consumes the last $N = 3$ latents and action blocks,

$$\widehat{z}_{t+1} = F_\phi(z_{t-N+1:t}, A_\omega(a_{t-N+1:t})), \tag{4}$$

and is trained with prediction plus SIGReg,

$$\mathcal{L}_{\text{world}} = \mathbb{E}[\|\widehat{z}_{t+1} - z_{t+1}\|_2^2/d] + \lambda_{\text{sig}}\mathcal{R}_{\text{SIG}}. \tag{5}$$

This learns how a supplied action changes the latent, but not how to obtain an action from a deployable latent request. LeWM consequently samples actions and uses the forward model only to rank them.

3.2 Action-Aligned Intent Coordinates

The center of Fig. 1 gives the graphical counterpart of this section: its attached local path carries a realized one-chunk successor, while its stop-gradient goal path carries a condition available before acting. In Fig. 2, only the latter can enter the lower deployment route before the next action is chosen. The endpoint y, displacement $m_z(y)$, and conditional law below formalize those two pictured paths and explain why neither one alone closes the forward–action gap.

For current latent z and endpoint condition y, let $p_E^\star(a \mid z, y)$ be the expert action law.

表 1：与相关方法的定位比较。"Action→ E"表示动作监督会更新视觉/世界编码器；"Shared-4T E"表示一个编码器在全部四个官方 LeWM 领域上联合训练。最后一列列出各方法实际采用的训练到控制路径；叉号表示未报告的接口，而非不可能的扩展。

Method	Action→ E	Shared-4T E	Condition	Action obj.	Direct	Search	Training-to-control path
DINO-WM [3]	✗	✗	action	none	✗	✓	Frozen image encoder; CEM ranks latent rollouts
C-JEPA [10]	✗	✗	action/proprio.	none	✗	✓	Frozen object features; CEM searches masked rollouts
LeWM [4]	✗	✗	action	none	✗	✓	Joint JEPA prediction; CEM inverts the forward model
Fast-LeWM [11]	✗	✗	action prefix	none	✗	✓	LeWM training; parallel action prefixes are searched
SMWM [16]	✓	✗	successor	local action	✗	✓	Local inverse shapes the encoder; deployment still uses CEM
GC-IDM [34]	✗	✗	goal + horizon	next action	✓	✗	Frozen LeWM; a separate goal/horizon actor acts directly
PRISM [33]	✗	✗	state/goal	Gaussian prior	✗	✓	Frozen LeWM; a separately trained prior initializes MPPI
Qantara [39]	✓	✗	dispatch-dependent	action flow	✓	✓	Joint JEPA; bridge-flow heads serve dispatch modes
QuoVLA [35]	✓	✗	vi-sual/language	action chunk	✓	✗	VLM action quotient; quantized intent drives a VLA actor
INTACT	✓	✓	local/goal intent	action block	✓	opt.	One shared JEPA/actor law; Direct by default, search verifies

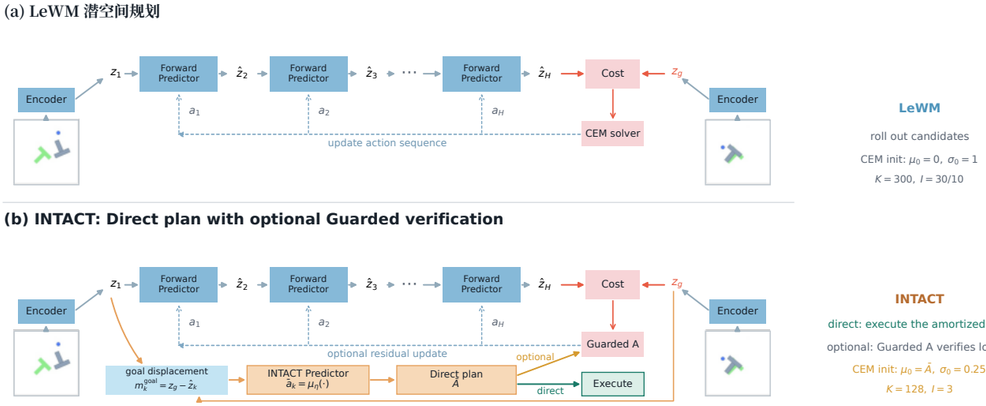


图 2：INTACT 将必需的搜索变为可选的验证。LeWM 从不含目标信息的高斯分布采样原始动作，再通过前向滚动预测获得目标导向性。INTACT 交替使用原始目标位移、共享的条件动作均值和未改动的 Forward Predictor，生成 Direct 计划。它可以立即执行，也可以通过 $K = 128$、$I = 3$ 和 $\sigma_0 = 0.25$ 在局部验证该计划。

并非多种输出动作块长度，且最终原始目标策略不显式接收 r。

部署时，图 2 下方路线复用完全相同的符号：将当前与目标图像编码为 z_t 和 z_g，设置 $\widetilde{z}_0 = z_t$，递归计算 $m_k = z_g - \widetilde{z}_k$，$\bar{a}_k = \mu_\eta(\widetilde{z}_k, m_k, \bar{a}_{k-1})$ 及 $\widetilde{z}_{k+1} = F_\phi(\widetilde{z}_k, \bar{a}_k)$。单块均值形成计划 $\bar{A} = (\bar{a}_0, \ldots, \bar{a}_{H-1})$，对应图 1 的训练动作输出。Direct 执行首块，从下一观测重新规划；Guarded A 仅局部验证该计划。图 2 上方路线则独立于所学意图坐标初始化原始动作。

两条路径中，未改动的 LeWM Forward Predictor 均接收最近 $N = 3$ 个潜变量与动作块，

$$\widehat{z}_{t+1} = F_\phi(z_{t-N+1:t}, A_\omega(a_{t-N+1:t})), \tag{4}$$

并以预测和 SIGReg 进行训练，

$$\mathcal{L}_{\text{world}} = \mathbb{E}[\|\widehat{z}_{t+1} - z_{t+1}\|_2^2/d] + \lambda_{\text{sig}}\mathcal{R}_{\text{SIG}}. \tag{5}$$

这学习了给定动作如何改变潜变量，却没有学习如何从可部署的潜空间请求得到动作。因此 LeWM 需要采样动作，并仅用前向模型为其评分。

3.2 动作对齐的意图坐标

图 1 中间是本节图示：保留梯度局部路径携带已实现单块后继，停止梯度目标路径携带动作前可获得条件。图 2 中，选择下一动作前只有后者能进入下方部署路线。下文端点 y、位移 $m_z(y)$、条件规律，形式化这两条路径，并解释为何单独任一路径都不能闭合前向–动作差距。

对于当前潜变量 z 与端点条件 y，设 $p_E^\star(a \mid z, y)$ 为专家动作规律。

Definition 1 (Conditional action equivalence)**.**

$$y \sim_z y' \iff p_E^\star(a \mid z, y) = p_E^\star(a \mid z, y'). \tag{6}$$

The quotient $\mathcal{Y}/\!\sim_z$ is in one-to-one correspondence with the realizable action-law image at z.

Proposition 1 (Quotient-to-law identification)**.** $\Phi_z([y]_z) = p_E^\star(\cdot \mid z, y)$ *is well defined and bijective from* $\mathcal{Y}/\!\sim_z$ *onto* $\{p_E^\star(\cdot \mid z, y) : y \in \mathcal{Y}\}$.

The displacement

$$m_z(y) = y - z \tag{7}$$

is a convenient motion-intent coordinate, not a claim that latent dynamics are globally linear or that equivalent endpoints must coincide.

A physical inverse likelihood,

$$\mathcal{L}_{z^+} = \mathbb{E}[-\log p_\eta(a_t \mid z_t, z_{t+1}, a_{t-1})], \tag{8}$$

asks the encoder to preserve action-recoverable transitions, but it is circular at deployment because z_{t+1} is observed only after acting. Conversely, a goal-conditioned likelihood is deployable but, by itself, is GCSL-like hindsight imitation without an attached one-chunk physical anchor. This is the central asymmetry:

$$\underbrace{z_{t+1}}_{\substack{\text{physical but unavailable}\\ \text{before acting}}} \quad \text{versus} \quad \underbrace{z_g}_{\substack{\text{available but not a}\\ \text{one-step successor}}}.$$

INTACT uses one nonlinear conditional operator for both. This is the single INTACT Predictor drawn twice in Fig. 1; its goal invocation is the action-generating operator on the lower route of Fig. 2:

$$G_\eta : (z, m, a_{t-1}) \longmapsto p_\eta(a \mid z, m, a_{t-1}), \tag{9}$$

while retaining the separate Forward Predictor $F_\phi : (z, a) \mapsto z^+$. The final goal-displacement route constructs

$$\boxed{m_t^{\text{local}} = z_{t+1} - z_t, \qquad m_t^{\text{goal}} = \text{sg}(z_g) - z_t.} \tag{10}$$

The local endpoint remains attached; stop-gradient applies only to the future goal occurrence in the deployment loss. The same z_g is therefore plastic when it appears as a real successor and fixed when it serves as an intent anchor. Current-state latents, G_η, and the independent forward/physical branches continue to receive gradients.

For route χ, the physical and deployable condition families are

$$\mathcal{C}^\chi_{\text{phy}} = \{(z_t, c_t^{\text{phy},\chi}, a_{t-1})\}, \tag{11}$$
$$\mathcal{C}^\chi_{\text{dep}} = \{(z_t, c_t^{\text{dep},\chi}, a_{t-1})\}, \tag{12}$$

and one shared action operator serves their union,

$$G_\eta : \mathcal{C}^\chi_{\text{phy}} \cup \mathcal{C}^\chi_{\text{dep}} \to \mathcal{P}(\mathcal{A}). \tag{13}$$

Algorithm 1 INTACT training

Require: window $(o_{t:t+H}, a_{t-1:t+H-1})$; shared E_θ, F_ϕ, G_η
$z_{t:t+H} \leftarrow E_\theta(o_{t:t+H})$; $z_g^{\text{anchor}} \leftarrow \text{sg}(z_{t+H})$
for $k = 0, \dots, H-1$ **do**
  $j \leftarrow t+k$; $(m_j^{\text{local}}, m_j^{\text{goal}}) \leftarrow (z_{j+1} - z_j, z_g^{\text{anchor}} - z_j)$
  accumulate physical and goal NLLs through the same G_η
update (θ, ϕ, η) with $\mathcal{L}_{\text{world}} + \lambda_{\text{inv}}\mathcal{L}_{z^+} + \lambda_{\text{goal}}\mathcal{L}_{\text{goal}}$

Its paired proper likelihood is

$$\mathcal{L}^\chi_{\text{I2A}} = \lambda_{\text{inv}}\mathbb{E}[-\log p_\eta(a_t \mid z_t, c_t^{\text{phy},\chi}, a_{t-1})] + \lambda_{\text{goal}}\mathbb{E}[-\log p_\eta(a_t \mid z_t, c_t^{\text{dep},\chi}, a_{t-1})]. \tag{14}$$

The first NLL is the attached physical call in Fig. 1, and the second is its stop-gradient deployment-goal call. Thus Eqs. (12) to (14) are the set, operator, and loss views of the same two-arrow construction in the figure. No term minimizes $\|m_t^{\text{goal}} - m_t^{\text{local}}\|^2$. The two calls may have different input scales and output Gaussians. Here **isomorphic** again has two levels: their backbone-input graphs are **isomorphic** by typed slots and shared parameters, while their supported motion-intent families are **isomorphic** through the conditional action-law relation represented by G_η. Neither level asserts numerical latent equality.

Proposition 2 (Off-support non-identifiability)**.** *If inverse risk is evaluated only on realized successors, any two actors that agree on that support have identical risk even when they disagree on a deployable goal condition outside it.*

Proposition 3 (Supported quotient recovery)**.** *For a well-specified conditional actor, population proper-NLL minimization recovers* $p_E^\star(\cdot \mid z, y)$ *on every supported condition. Hence action-equivalent representatives receive the same predicted action law.*

Proofs, the horizon-normalized waypoint derivation, the Bernoulli endpoint variant, and the direct-linearity negative control are collected in Sections A and C. The waypoint request is an ablation; the headline model uses the raw displacement in Eq. (10) because it preserves direction and remaining distance without imposing equal progress.

The single-task objective is

$$\boxed{\mathcal{L}_{\text{ST}} = \mathcal{L}_{\text{world}} + \lambda_{\text{inv}}\mathcal{L}_{z^+} + \lambda_{\text{goal}}\mathcal{L}_{\text{goal}}.} \tag{15}$$

This end-to-end coupling makes action-law aliasing visible to the encoder: a frozen head cannot recover a controllable distinction already collapsed by E_θ.

3.3 INTACT Predictor and Control

Figure 1 depicts the identical input wiring that makes the two INTACT Predictor invocations structurally isomorphic; the four incoming quantities are written explicitly as

$$h_t^\Delta(m_t) = [z_t; m_t; z_t \odot m_t; A_\omega(a_{t-1})]. \tag{16}$$

定义 1（条件动作等价）。

$$y \sim_z y' \iff p_E^\star(a \mid z, y) = p_E^\star(a \mid z, y'). \tag{6}$$

商 $\mathcal{Y}/\!\sim_z$ 与 z 处可实现的动作规律像一一对应。

命题 *1*（商到规律的辨识）。 $\Phi_z([y]_z) = p_E^\star(\cdot \mid z, y)$ 是从 $\mathcal{Y}/\!\sim_z$ 到 $\{p_E^\star(\cdot \mid z, y) : y \in \mathcal{Y}\}$ 的良定义双射。

位移

$$m_z(y) = y - z \tag{7}$$

是便利的运动意图坐标，并不声称潜动力学全局线性，也不要求等价端点重合。

物理逆向似然，

$$\mathcal{L}_{z^+} = \mathbb{E}[-\log p_\eta(a_t \mid z_t, z_{t+1}, a_{t-1})], \tag{8}$$

要求编码器保留能够恢复动作的转移，但在部署时会形成循环依赖，因为 z_{t+1} 只有行动后才能被观测到。相反，目标条件似然可以部署，但若单独使用，就只是类似 GCSL 的事后模仿，缺少一个保留梯度的单动作块物理锚点。这是核心的不对称性：

$$\underbrace{z_{t+1}}_{\substack{\text{physical but unavailable}\\ \text{before acting}}} \quad \text{versus} \quad \underbrace{z_g}_{\substack{\text{available but not a}\\ \text{one-step successor}}}.$$

INTACT 对两者使用同一个非线性条件算子。它就是图 1 中画了两次的单个 INTACT Predictor；其目标调用是图 2 下方路径中生成动作的算子：

$$G_\eta : (z, m, a_{t-1}) \longmapsto p_\eta(a \mid z, m, a_{t-1}), \tag{9}$$

同时保留独立的 Forward Predictor $F_\phi : (z, a) \mapsto z^+$。最终目标位移路径构造

$$\boxed{m_t^{\text{local}} = z_{t+1} - z_t, \qquad m_t^{\text{goal}} = \text{sg}(z_g) - z_t.} \tag{10}$$

局部端点保留梯度；停止梯度仅作用于部署损失中出现的未来目标。因此，同一个 z_g 作为真实后继时具有可塑性，作为意图锚点时则固定。当前状态潜变量、G_η 以及独立的前向/物理分支继续接收梯度。

对于路径 χ，物理与可部署条件族为

$$\mathcal{C}^\chi_{\text{phy}} = \{(z_t, c_t^{\text{phy},\chi}, a_{t-1})\}, \tag{11}$$
$$\mathcal{C}^\chi_{\text{dep}} = \{(z_t, c_t^{\text{dep},\chi}, a_{t-1})\}, \tag{12}$$

且一个共享动作算子作用于两者的并集，

$$G_\eta : \mathcal{C}^\chi_{\text{phy}} \cup \mathcal{C}^\chi_{\text{dep}} \to \mathcal{P}(\mathcal{A}). \tag{13}$$

算法 1：INTACT 训练

输入：窗口 $(o_{t:t+H}, a_{t-1:t+H-1})$；共享的 E_θ、F_ϕ、G_η
$z_{t:t+H} \leftarrow E_\theta(o_{t:t+H})$；$z_g^{\text{锚点}} \leftarrow \text{sg}(z_{t+H})$
对于 $k = 0, \dots, H-1$ 执行
  $j \leftarrow t+k$；$(m_j^{\text{local}}, m_j^{\text{goal}}) \leftarrow (z_{j+1} - z_j, z_g^{\text{anchor}} - z_j)$
  通过同一个算子累积物理与目标负对数似然 G_η
使用 $\mathcal{L}_{\text{world}} + \lambda_{\text{inv}}\mathcal{L}_{z^+} + \lambda_{\text{goal}}\mathcal{L}_{\text{goal}}$ 更新 (θ, ϕ, η)

其配对的适当似然为

$$\mathcal{L}^\chi_{\text{I2A}} = \lambda_{\text{inv}}\mathbb{E}[-\log p_\eta(a_t \mid z_t, c_t^{\text{phy},\chi}, a_{t-1})] + \lambda_{\text{goal}}\mathbb{E}[-\log p_\eta(a_t \mid z_t, c_t^{\text{dep},\chi}, a_{t-1})]. \tag{14}$$

第一项 NLL 是图 1 保留梯度的物理调用，第二项是停止梯度部署目标调用。因此式（12）至（14）分别是同一双箭头构造的集合、算子与损失视角。没有项最小化 $\|m_t^{\text{goal}} - m_t^{\text{local}}\|^2$。两调用可有不同输入尺度和输出高斯。同构仍有两层：骨干输入图经类型槽与共享参数同构，受支持运动意图族经 G_η 表示的条件动作规律关系同构。两层都不要求潜变量数值相等。

命题 *2*（支持集外不可辨识性）。若逆向风险仅在真实后继上评估，那么两个 *actor* 只要在该支持集上一致，风险就相同，即使它们在支持集外的可部署目标条件上不同。

命题 *3*（支持集内的商恢复）。若条件 *actor* 的模型设定正确，最小化总体适当负对数似然可在每个受支持条件上恢复 $p_E^\star(\cdot \mid z, y)$。因此，动作等价的代表元获得相同的预测动作规律。

证明、时域归一化航点推导、伯努利端点变体及直接线性化负对照见附录 A、C。航点请求是一项消融；主要模型使用式（10）的原始位移，因为它保留方向与剩余距离，无需施加等进度假设。

单任务目标为

$$\boxed{\mathcal{L}_{\text{ST}} = \mathcal{L}_{\text{world}} + \lambda_{\text{inv}}\mathcal{L}_{z^+} + \lambda_{\text{goal}}\mathcal{L}_{\text{goal}}.} \tag{15}$$

这种端到端耦合让编码器能够发现动作规律混叠：被冻结的头部无法恢复已经被 E_θ 压缩掉的可控差异。

3.3 INTACT Predictor 与控制

图 1 展示了使 INTACT Predictor 两次调用在结构上同构的相同输入连接；四个输入量显式写为

$$h_t^\Delta(m_t) = [z_t; m_t; z_t \odot m_t; A_\omega(a_{t-1})]. \tag{16}$$

Algorithm 2 INTACT Direct control with optional local verification

Require: latent history, goal, and inference mode v

$\widetilde{z}_0 \leftarrow z_t$
for $k = 0, \dots, H-1$ **do**
  $m_k \leftarrow z_g - \widetilde{z}_k$; $\bar{a}_k \leftarrow \mu_\eta(\widetilde{z}_k, m_k, \bar{a}_{k-1})$
  $\widetilde{z}_{k+1} \leftarrow F_\phi(\widetilde{z}_k, \bar{a}_k)$
if $v =$ Direct **then**
  return $\bar{A}$ with zero candidates and cost calls
if $v =$ one-shot **then**
  score $\bar{A}$ and clipped local perturbations; return the observed minimum
else
  initialize $(\mu, A_{\text{best}}) \leftarrow (\bar{A}, \bar{A})$ and $\sigma \leftarrow 0.25$
  for $i = 1, \dots, 3$ **do**
    score 128 samples plus $\bar{A}, A_{\text{best}}$; refit on 16 elites
  re-score μ; return the best observed reference or candidate

The product is a cheap state–intent interaction:

$$w^\top (z_t \odot m_t) = z_t^\top \operatorname{diag}(w) m_t, \tag{17}$$

allowing the same displacement to be interpreted differently across contact, pose, and obstacle states. A three-layer MLP outputs a diagonal Gaussian with log standard deviation clipped to $[-5, 2]$. Detailed endpoint grammar and the matched A–G product ablation are in Section D.

At deployment, the lower INTACT route in Fig. 2 is the recurrent composition of Eq. (16) with the unchanged Forward Predictor: the Gaussian mean generates an H-block plan and re-encodes after executing the next real action block. Direct uses no sampled candidate and no terminal cost. Optional Guarded A centers a small 128×3 raw-action search at this plan, retains the Direct reference and the globally best candidate, and uses $\sigma_0 = 0.25$. Pure CEM disables the INTACT Predictor and recovers the upper search-first route in the figure; it is retained only as a compatibility and actor-disabled control. Figure 2 and Algorithm 2 therefore give the visual and algorithmic views of the same Direct controller and optional local verification. The complete verifier matrix appears in Section J.

3.4 Multi-Task Extension

For heterogeneous tasks $\tau \in \{1, \dots, T\}$, INTACT shares the encoder and projector but keeps task-specific Forward, action-embedding, and INTACT heads. The known domain index selects a head; no reward or additional semantic label is used. The objective is

$$\mathcal{L}_{\text{MT}} = \sum_{\tau=1}^{T} \rho_\tau \left[\mathcal{L}^\tau_{\text{world}} + \lambda_{\text{inv}} \mathcal{L}^\tau_{z+} + \lambda_{\text{goal}} \mathcal{L}^\tau_{\text{goal}} \right]. \tag{18}$$

Every task therefore shapes the same visual coordinates without forcing heterogeneous actions into one padded space. The controlled factorial compares forward-only LeWM, physical inverse-only, intent-only, and Full INTACT under matched deterministic Math-SDPA. We test both waypoint and raw-goal intent coordinates. Direct SR measures the deployed conditional; actor-disabled CEM tests the encoder–Forward stack under a fixed optimizer; frozen probes and predicted–expert CKA/kNN diagnose representation and family conversion. None of these diagnostics replaces closed-loop SR.

4 Experimental Design

Tasks and evaluation. We use the four official LeWM domains: PushT, OGBench Cube, DMC Reacher, and TwoRoom [4, 40]. Models receive RGB observations and five-step action blocks; simulator state is used only by probes and success predicates. Headline SR follows the official evaluator for comparability. Unless stated otherwise, every checkpoint is evaluated with seeds $\{0, 1, 42\}$ and 100 episodes per seed. We separately report the public CLEAR-LeWM v0.5.1 Moderate audit [41], which removes initially solved pairs and repairs known Reacher and TwoRoom evaluator defects. Official and CLEAR scores are never pooled.

4.1 Single-Task Protocol

Each task-specific model is trained from scratch on its complete dataset for one epoch with batch size 256, AdamW learning rate 5×10^{-4}, forward weight 1.0, inverse weight 0.1, goal weight 0.05, and three training seeds. The final goal-displacement model uses SIGReg 0.02 and the shared grammar in Eq. (16). A matched waypoint chain isolates forward-only LeWM, physical inverse supervision, deployment-intent supervision, and the low-SIGReg setting. Actor sharing is tested against parameter-matched, double-capacity, two-output, and condition-token alternatives.

Direct asserts zero candidate sequences and zero terminal-cost calls. Pure CEM disables the INTACT Predictor and evaluates the encoder–Forward stack under a fixed raw-action optimizer; it is a functional control, not an encoder-only statistic. Guarded A evaluates 128 sequences for three iterations around the Direct plan. The complete 576-job planner matrix, coordinate grammar, SIGReg, straightening, optimizer, and search-radius selection studies are deferred to Sections D and J because they select a final configuration rather than establish the principal result.

Frozen probes use episode-disjoint train/test episodes. We measure physical state and transition-action R^2, effective rank, per-dimension standard deviation, collapsed-dimension fraction, and mean cosine similarity. Together with actor-disabled CEM, these distinguish representation shaping from the benefit of executing the learned action conditional.

4.2 Multi-Task Protocol

One distributed job traverses all four datasets. The ViT-Tiny/14 encoder, projector, and 192-dimensional latent are shared; each task retains a small Forward Predictor, action embedding, and INTACT Predictor. Training uses batch size 256, learning rate

算法 2：INTACT Direct 控制及可选局部验证

输入：潜变量历史、目标与推理模式 v

$\widetilde{z}_0 \leftarrow z_t$
对于 $k = 0, \dots, H-1$ **执行**
  $m_k \leftarrow z_g - \widetilde{z}_k$；$\bar{a}_k \leftarrow \mu_\eta(\widetilde{z}_k, m_k, \bar{a}_{k-1})$
  $\widetilde{z}_{k+1} \leftarrow F_\phi(\widetilde{z}_k, \bar{a}_k)$
若 $v =$ 为 Direct，则
  返回 $\bar{A}$，候选与代价调用均为零
若 $v =$ **为单次模式，则**
  分 $\bar{A}$ 及截断的局部扰动；返回已观测到的最小值
否则
  初始化 $(\mu, A_{\text{best}}) \leftarrow (\bar{A}, \bar{A})$ 与 $\sigma \leftarrow 0.25$
  对于 $i = 1, \dots, 3$ **do**
    评分 128 个样本以及 A, A_{best}；用 16 个精英样本重新拟合
  重新评分 μ；返回已观测到的最佳参考或候选

乘积是低成本的状态–意图交互项：

$$w^\top (z_t \odot m_t) = z_t^\top \operatorname{diag}(w) m_t, \tag{17}$$

使同一位移在不同接触、姿态和障碍物状态下得到不同解释。三层 MLP 输出对角高斯分布，对数标准差截断至 $[-5, 2]$。端点特征语法及匹配的 A–G 乘积消融详见附录 D。

部署时，图 2 下方 INTACT 路线是式（16）与未改变的前向预测器递归组合：高斯均值生成 H 块计划，执行下一真实动作块后重新编码。Direct 不采样候选，也不用终端代价。可选 Guarded A 以该计划为中心进行小规模 128×3 原始动作搜索，保留 Direct 参考和全局最佳候选，使用 $\sigma_0 = 0.25$。纯 CEM 关闭 INTACT 预测器，恢复图中上方先搜索路线，仅作兼容性和 actor 关闭对照。图 2 与算法 2 因而是同一 Direct 控制器及可选局部验证的视觉和算法表述。完整验证器矩阵见附录 J。

3.4 多任务扩展

对于异构任务 $\tau \in \{1, \dots, T\}$，INTACT 共享编码器和投影器，但保留任务专用的 Forward、动作嵌入及 INTACT 头部。已知领域索引选择相应头部，不使用奖励或额外语义标签。目标为

$$\mathcal{L}_{\text{MT}} = \sum_{\tau=1}^{T} \rho_\tau \left[\mathcal{L}^\tau_{\text{world}} + \lambda_{\text{inv}} \mathcal{L}^\tau_{z+} + \lambda_{\text{goal}} \mathcal{L}^\tau_{\text{goal}} \right]. \tag{18}$$

因此，每个任务都塑造相同的视觉坐标，而不把异构动作强制填充进同一个空间。受控析因实验比较仅前向 LeWM、物理

仅逆向、仅意图和完整 INTACT 均在匹配的确定性 Math-SDPA 下比较。我们同时测试航点与原始目标意图坐标。Direct 成功率衡量部署时的条件模型；禁用 actor 的 CEM 在固定优化器下检验编码器–Forward 组合；冻结探针及预测–专家 CKA/kNN 诊断表征与族转换。这些诊断均不能替代闭环成功率。

4 实验设计

任务与评估。使用四个官方 LeWM 领域：PushT、OGBench Cube、DMC Reacher、TwoRoom[4, 40]。模型接收 RGB 观测和五步动作块；仿真状态仅用于探测器与成功判定。主要成功率沿用官方评估器保证可比。除另有说明，每检查点使用种子 $\{0, 1, 42\}$，每种子 100 回合。另报告公开 CLEAR-LeWM v0.5.1 Moderate 审计 [41]，移除初始已解对，修复已知 Reacher、TwoRoom 缺陷。官方与 CLEAR 分数绝不合并。

4.1 单任务协议

每个任务专用模型在完整数据上从头训练一轮，批量 256，AdamW 学习率 5×10^{-4}，前向权重 1.0、逆权重 0.1、目标权重 0.05，三个训练种子。最终目标位移模型采用 SIGReg 0.02 和式（16）共享语法。匹配航点链分离仅前向 LeWM、物理逆监督、部署意图监督及低 SIGReg 设置。actor 共享与参数匹配、双容量、双输出、条件词元等替代设计比较。

Direct 保证候选序列数与终端代价调用数均为零。纯 CEM 关闭 INTACT 预测器，以固定原始动作优化器评估编码器–前向模型，是功能对照而非仅编码器统计。Guarded A 在 Direct 附近每次评估 128 条、三次迭代。576 项完整规划矩阵、坐标语法、SIGReg、拉直、优化器和搜索半径选择见附录 D、J，因为它们选择最终配置，而非建立主要结果。

冻结探针使用训练/测试回合互不重叠的数据。我们衡量物理状态及转移动作的 R^2、有效秩、各维标准差、坍塌维度比例和平均余弦相似度。结合禁用 actor 的 CEM，这些指标可区分表征塑造与执行学习所得动作条件模型的收益。

4.2 多任务协议

一个分布式作业遍历全部四个数据集。ViT-Tiny/14 编码器、投影器与 192 维潜变量共享；各任务保留小型 Forward Predictor、动作嵌入及 INTACT Predictor。训练批量大小为 256，学习率为

Table 2: **Published landscape and matched task-specific INTACT inference matrix (official SR, %).** Published values are transcribed from the cited papers and are not paired controls; † marks partial coverage or a distinct reproduction protocol. ‡ marks our matched-E1 reproduction: one task-specific training seed with mean ± sample standard deviation over evaluation manifests $\{0, 1, 42\}$ (100 episodes each), rather than training-seed uncertainty. INTACT values are mean ± sample standard deviation over three training seeds, each averaged over three 100-episode evaluation seeds. The INTACT block audits all eight exact objective–inference pairings on their matched checkpoint families.

Method / objective	Training	Deployment	Seq.	PushT	Cube	Reacher	TwoRoom	Macro
DINO-WM [3]	frozen E	CEM (published)	9k	74.0±4.5	86.0±4.7	79.0±5.1	100.0±.0	84.75
LeWM [4]	E2E 10 ep	CEM 300×(30/10)	9k/3k	96.0±4.0	74.0±3.0	86.0±5.0	87.0±2.5	85.75
Fast-LeWM [11]	E2E 10 ep	matched CEM	9k	96.0	80.0	88.0	98.0	90.50
Fast-LeWM + SC [11]	E2E 10 ep	CEM + score	9k	98.0	82.0	90.0	98.0	92.00
Qantara† [39]	E2E 10 ep	CEM 300×30, $K = 4$	9k	90.1±1.1	93.7±.7	80.9±1.8	100.0±.0	91.18
GC-IDM [34]	LeWM + 50-ep head	Direct	0	84.7±5.0	99.3±1.2	100.0±.0	100.0±.0	96.00
PRISM† [33]	LeWM + 50-ep head	MPPI 128×30	3,840	89±4	79±6	–	–	–
C-JEPA† [10]	frozen E + 30 ep	CEM 300×30	9k	88.67	–	–	–	–
Matched Math-SDPA E1 reproduction on a frozen task-specific E1 LeWM encoder								
GC-IDM (matched E1)‡	frozen E1 E + 1-ep head	Direct	0	9.33±2.31	69.33±7.02	84.33±2.08	99.33±.58	65.58
PRISM (matched E1)‡	frozen E1 E + 1-ep head	PoG-MPPI 128×30	3,840	18.67±5.51	86.67±1.15	66.33±1.15	88.33±1.53	65.00
Matched task-specific INTACT, E2E one epoch								
Waypoint INTACT	E2E 1 ep	Direct	0	77.67±.88	99.89±.19	88.11±.69	98.00±.33	90.92±.30
		Pure CEM 300×30	9k	89.67±1.45	67.67±1.33	82.67±1.20	77.33±2.03	79.33±.75
		Actor-on CEM 300×30	9k	92.89±.84	96.56±1.07	80.67±2.08	98.00±.67	92.03±1.06
		Guarded A 128×3	**384**	**89.89±.38**	**99.78±.19**	**88.56±1.26**	**97.89±.19**	**94.03±.25**
INTACT, goal displacement	**E2E 1 ep**	**Direct**	**0**	85.78±1.54	**100.00±.00**	**97.67±.00**	97.89±1.26	**95.33±.58**
		Pure CEM 300×30	9k	88.44±1.17	68.44±.77	83.67±.67	82.89±.84	80.86±.51
		Actor-on CEM 300×30	9k	**93.56±.96**	96.89±.19	86.67±.88	**98.00±1.15**	93.78±.77
		Guarded A 128×3	**384**	**92.22±.69**	**99.78±.19**	**97.44±.77**	**98.00±1.15**	**96.86±.38**

5×10^{-4}, weight decay 10^{-3}, SIGReg 0.03, and five epochs. Encoder and Forward attention are forced to deterministic Math-SDPA and every checkpoint stores backend and data fingerprints.

The controlled factorial contains forward-only LeWM, physical inverse-only, intent-only, and Full INTACT. Intent-only and Full are each trained with the waypoint and goal-displacement coordinates, producing six cells. All cells use training seeds $\{0, 42, 3072\}$ and are evaluated at E1–E5. At each epoch we run native Direct inference where available, actor-disabled pure CEM 300 × 30, Guarded A for actor-equipped cells, frozen probes, and latent diagnostics. The complete coverage contract and E1/epoch tables are reported in Section E.

For mechanism analysis, predicted–expert linear CKA, local kNN overlap, action R^2, and NLL are computed within each task's native action coordinates and averaged equally over tasks; heterogeneous action spaces are never zero-padded into one embedding. The primary cohort contains both Full INTACT interfaces, three seeds, and E1–E5. A separate task-conditioned gauge audit fits latent coordinate maps on paired, action-free calibration episodes and tests correct pairing, shuffling, calibration size, cross-seed ceilings, and reverse actor swaps under CLEAR Moderate. Full estimator definitions and controls appear in Sections E and G.

5 Results

5.1 Single-Task Results

One epoch makes search optional. Table 2 places the headline on a resource-aware axis. Published rows retain their own protocols and are not paired significance controls. Every row in this single-task comparison trains a separate visual model per domain; the shared-encoder study begins in Section 5.2.

INTACT uses the fewest reported full-data passes among methods with disclosed epoch schedules: one end-to-end epoch, no frozen-encoder phase, and no separately trained phase-2 controller. Direct reaches 95.33 macro SR with zero candidates and 2.9–5.5 ms planner-side latency. Guarded A evaluates 384 rather than 9,000 sequences and reaches 96.86 macro, placing INTACT at the state-of-the-art level among currently reported JEPA controllers.

The matched-E1 Math-SDPA rows ask a narrower question than the published headlines: how much control can a one-epoch head amortize from the same frozen, task-specific E1 LeWM representation? GC-IDM uses zero-search Direct control, whereas PRISM uses PoG-MPPI with 128×30 = 3,840 candidates. Their macros are 65.58 and 65.00, respectively, but the taskwise ordering reverses: PRISM is higher on PushT/Cube, while GC-IDM is higher on Reacher/TwoRoom. Because the objectives and inference budgets differ and each row uses one training seed, this is a controlled E1 stress test rather than a paired compute or significance ranking. Descriptively, the end-to-end one-epoch INTACT Direct row reaches 95.33 macro without candidate search.

The complete matched block in Table 2 separates training coordinates from deployment algorithms. Goal-displacement INTACT reaches 85.78/100.00/97.67/97.89% Direct SR on PushT/Cube/Reacher/TwoRoom, averaged over three independently trained models per task. Relative to published LeWM, it uses one tenth as many full-data passes and removes up to 9,000 candidate sequences. Goal displacement outperforms the waypoint coordinate under every matched interface: by 4.41/1.53/1.75/2.83 macro points for Direct, pure CEM, actor-on CEM, and Guarded A, respectively. The actor-disabled

表 2：已发表方法概览与匹配任务专用 INTACT 推理矩阵（官方成功率，%）。已发表值取自引用论文，不是配对对照；† 表示部分覆盖或不同复现方案。‡为匹配 E1 复现：每任务一个训练种子，报告评估清单 $\{0, 1, 42\}$（每个 100 回合）的平均 ± 样本标准差，不是训练种子不确定性。INTACT 报告三个训练种子的平均 ± 样本标准差，每种子再平均三个 100 回合评估种子。INTACT 组合审计全部八种精确目标–推理配对及匹配检查点族。

方法 / 目标	训练	部署	Seq.	PushT	Cube	Reacher	TwoRoom	宏平均
DINO-WM [3]	冻结 E	CEM（已发表）	9k	74.0±4.5	86.0±4.7	79.0±5.1	100.0±.0	84.75
LeWM [4]	端到端 10 轮	CEM 300×(30/10)	9k/3k	96.0±4.0	74.0±3.0	86.0±5.0	87.0±2.5	85.75
Fast-LeWM [11]	端到端 10 轮	匹配的 CEM	9k	96.0	80.0	88.0	98.0	90.50
Fast-LeWM +SC [11]	端到端 10 轮	CEM + 评分	9k	98.0	82.0	90.0	98.0	92.00
Qantara† [39]	端到端 10 轮	CEM 300×30, $K = 4$	9k	90.1±1.1	93.7±.7	80.9±1.8	100.0±.0	91.18
GC-IDM [34]	LeWM + 50 轮头部	Direct	0	84.7±5.0	99.3±1.2	100.0±.0	100.0±.0	96.00
PRISM† [33]	LeWM + 50 轮头部	MPPI 128×30	3,840	89±4	79±6	–	–	–
C-JEPA† [10]	冻结 E + 30 轮	CEM 300×30	9k	88.67	–	–	–	–
在冻结的任务专用 *E1 LeWM* 编码器上进行匹配的 *Math-SDPA E1* 复现								
GC-IDM（匹配 E1）‡	冻结 E1 E + 1 轮头部	Direct	0	9.33±2.31	69.33±7.02	84.33±2.08	99.33±.58	65.58
PRISM（匹配 E1）‡	冻结 E1 E + 1 轮头部	PoG-MPPI 128×30	3,840	18.67±5.51	86.67±1.15	66.33±1.15	88.33±1.53	65.00
匹配的任务专用 *INTACT*，端到端训练一轮								
航点 INTACT	端到端 1 轮	Direct	0	77.67±.88	99.89±.19	88.11±.69	98.00±.33	90.92±.30
		纯 CEM 300×30	9k	89.67±1.45	67.67±1.33	82.67±1.20	77.33±2.03	79.33±.75
		启用 actor 的 CEM 300×30	9k	92.89±.84	96.56±1.07	80.67±2.08	98.00±.67	92.03±1.06
		Guarded A 128×3	**384**	**89.89±.38**	**99.78±.19**	**88.56±1.26**	**97.89±.19**	**94.03±.25**
INTACT，目标位移，端到端训练一轮		**Direct**	**0**	85.78±1.54	**100.00±.00**	**97.67±.00**	97.89±1.26	**95.33±.58**
		纯 CEM 300×30	9k	88.44±1.17	68.44±.77	83.67±.67	82.89±.84	80.86±.51
		启用 actor 的 CEM 300×30	9k	**93.56±.96**	96.89±.19	86.67±.88	**98.00±1.15**	93.78±.77
		Guarded A 128×3	**384**	**92.22±.69**	**99.78±.19**	**97.44±.77**	**98.00±1.15**	**96.86±.38**

5×10^{-4}，权重衰减 10^{-3}，SIGReg 为 0.03，训练五轮。编码器与 Forward 注意力强制使用确定性 Math-SDPA，每个检查点保存后端和数据指纹。

受控析因含仅前向 LeWM、仅物理逆、仅意图、完整 INTACT。后两者均用航点和目标位移，形成六组合。全部训练种子 $\{0, 42, 3072\}$，评估 E1–E5。每轮运行可用的原生 Direct、关闭 actor 的纯 CEM 300 × 30、带 actor 组合的 Guarded A、冻结探测与潜变量诊断。完整覆盖约定与 E1／各轮表见附录 E。

机制分析中，预测–专家线性 CKA、局部 k 近邻重叠、动作 R^2、NLL，均在任务原生动作坐标内计算，再等权平均；异质动作空间从不补零拼成同一嵌入。主要组包含两种完整接口、三个种子、E1–E5。独立任务条件规范审计，在配对、无动作标注的标定回合上拟合潜在坐标映射，检验正确配对、打乱、标定大小、跨种子上限和 CLEAR Moderate 反向 actor 交换。完整定义及对照见附录 E、G。

5 结果

5.1 单任务结果

训练一轮便使搜索成为可选。表 2 从资源预算角度展示主要结果。已发表方法各行保留自身协议，不作为配对显著性对照。每一行该单任务比较为每个领域分别训练视觉模型；共享编码器实验从第 5.2 节开始。

披露轮次方案的方法中，INTACT 完整数据遍历最少：一轮端到端，无冻结编码器阶段，无独立第二阶段控制器。Direct 零候选，宏平均 95.33，规划器侧 2.9–5.5 毫秒；Guarded A 评估 384 而非 9000 条，宏平均 96.86，使 INTACT 达到目前报告 JEPA 控制器的最先进水平。

匹配 E1 的 Math-SDPA 各行考察比已发表主要结果更窄的问题：同一冻结、任务专用 E1 LeWM 表征，单轮训练头能摊销多少控制？GC-IDM 使用无搜索 Direct，PRISM 使用 PoG-MPPI、128×30 = 3,840 个候选。宏平均分别 65.58、65.00，但任务排序相反：PRISM 在 PushT／Cube 更高，GC-IDM 在 Reacher／TwoRoom 更高。因目标与推理预算不同，且每行只有一个训练种子，这属于受控 E1 压力测试，而非配对计算或显著性排名。描述性地，端到端单轮 INTACT Direct 无候选搜索达到宏平均 95.33。

表 2 完整匹配组合分离训练坐标与部署算法。目标位移 INTACT 在四任务 Direct 成功率为 85.78/100.00/97.67/97.89%，每任务三个独立训练模型取平均。相比已发表 LeWM，完整数据遍历仅其十分之一，消除最多 9000 条候选序列。所有匹配接口下目标位移均优于航点：Direct、纯 CEM、启用 actor 的 CEM、Guarded A 的宏平均分别高 4.41／1.53／1.75／2.83 个百分点。关闭 actor 时，

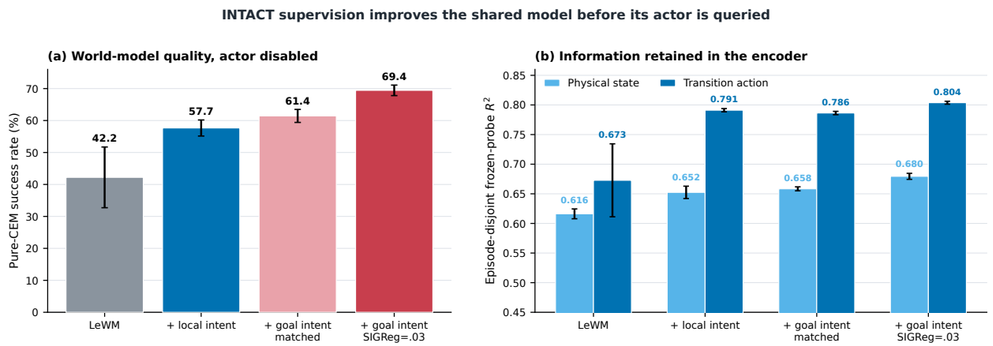


Figure 3: **Single-task action likelihoods shape the representation.** Actor-disabled CEM evaluates the encoder–Forward stack under a fixed optimizer; episode-disjoint probes measure readable state and transition action. Physical inverse supplies attached successor shaping, whereas the goal branch updates current-state coordinates and the shared actor with a stop-gradient goal anchor. Error bars are sample standard deviations over three PushT training seeds.

macros remain close (80.86 versus 79.33), showing that broad raw-action search largely discards the learned intent interface. More decisively, Goal actor-on CEM spends 9,000 sequences yet falls 1.55 points below zero-search Direct (93.78 versus 95.33), whereas Guarded A raises Direct by 1.53 points to 96.86 with only 384 sequences. The waypoint coordinate shows the same efficiency ordering: Guarded A reaches 94.03 versus 92.03 for actor-on CEM. Search is therefore useful as local verification around the learned plan, while unconstrained iterative search can recreate a train–deployment gap and waste the learned conditional.

The separate CLEAR-LeWM Moderate audit preserves the same ordering: Direct and Guarded A reach 83.33 ± 1.00 and 84.83 ± 1.32 macro, versus $65.47\pm.77$ for pure CEM. Reacher drops from 97.67 official to 49.56 Moderate after angle-topology correction, illustrating why CLEAR is reported separately rather than pooled. The complete taskwise table and protocol contract are in Table A3 and Section L.

The matched actor-sharing audit is deferred to Table A8 and Section F: separating the two calls loses 5.67 PushT points at matched capacity, and even doubling actor capacity remains 4.00 points below the fully shared conditional.

The encoder improves before the actor is used. On PushT, adding physical inverse supervision raises actor-disabled CEM 30×10 from 42.22 ± 9.50 to $57.67\pm2.52\%$ and frozen endpoint-action R^2 from $.673\pm.062$ to $.791\pm.003$. Adding matched waypoint supervision raises CEM to $61.44\pm2.04\%$; the larger Direct improvement primarily identifies the deployable conditional. The low-SIGReg waypoint model reaches $69.44\pm1.64\%$ actor-off CEM and $77.67\pm.88\%$ Direct. Its lower rank is not collapse: per-dimension standard deviation remains near one and no dimension collapses.

The taskwise gains isolate the learned interface. Replacing the waypoint coordinate by goal displacement changes Direct SR by $+8.11/+0.11/+9.56/-0.11$ points on PushT/Cube/Reacher/TwoRoom. The gain is therefore concentrated in contact-sensitive PushT and continuously actuated Reacher, while Cube and TwoRoom are already near saturation. Under actor-disabled pure CEM, the same replacement improves macro SR by only 1.53 points (79.33 to 80.86); after the learned actor is restored, the Direct gain is 4.41 points and Guarded A reaches 96.86. This separation matters: the objective improves the world representation, but its larger benefit comes from exposing a deployment condition that the shared actor can interpret directly.

Coordinate grammar, SIGReg, straightening, Huber, extra-epoch, interaction, and planner-radius controls are reported together in Sections D and E. They select the final model but are not repeated as headline experiments.

5.2 Multi-Task Results

We test the linked representation, control, and stability consequences on matched checkpoints below.

One shared encoder improves all four domains. At E5, goal-displacement INTACT reaches $89.39\pm.77$ macro Direct SR, versus 66.17 ± 2.67 for matched shared-encoder LeWM with CEM 300×30. It improves LeWM by 5.66/32.23/12.56/42.44 points on PushT/Cube/Reacher/TwoRoom and exceeds the 85.75 published task-specific LeWM macro. Against the matched intent-only cell, adding the attached physical likelihood raises macro SR by 8.78 ± 1.63 points; the gain is positive on three tasks and -0.44 on the saturated Cube task. This is the cleanest evidence that Full INTACT is not merely a goal-conditioned behavior-cloning head.

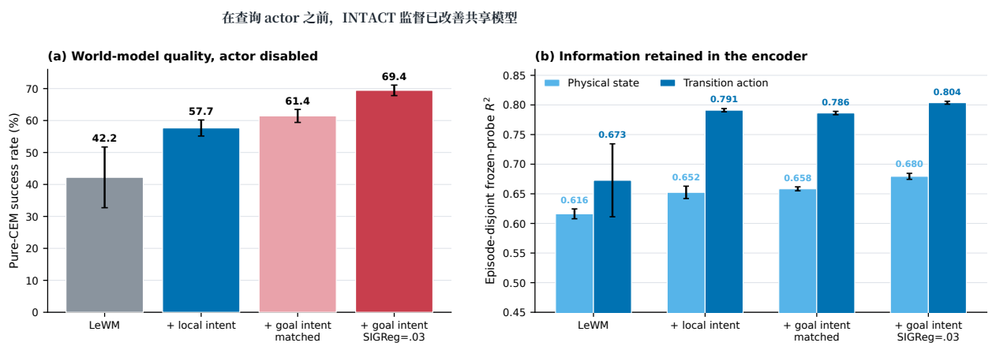


图 3：单任务动作似然塑造表征。禁用 actor 的 CEM 在固定优化器下评估编码器–Forward 组合；回合互不重叠的探针衡量可读出的状态与转移动作。物理逆向分支通过保留梯度的后继塑造表征；目标分支以停止梯度的目标锚点更新当前状态坐标及共享 actor。误差条为三个 PushT 训练种子的样本标准差。

宏平均接近（80.86 对 79.33），说明广泛原始动作搜索基本丢弃所学意图接口。更关键的是，目标模型启用 actor 的 CEM 用 9000 条序列，却低于无搜索 Direct 1.55 个百分点（93.78 对 95.33）；Guarded A 只用 384 条，将 Direct 提升 1.53 至 96.86。航点坐标呈同样效率排序：Guarded A 为 94.03，启用 actor 的 CEM 为 92.03。因此搜索适合在学习计划附近作局部验证，无约束迭代搜索却可能重新引入训练–部署差距，浪费所学条件关系。

独立 CLEAR Moderate 审计保持同样排序：Direct、Guarded A 宏平均 83.33 ± 1.00 、84.83 ± 1.32 ，纯 CEM 为 $65.47\pm.77$ 。修正角度拓扑后 Reacher 从官方 97.67 降至 Moderate 49.56，说明 CLEAR 必须单独而非合并报告。完整任务表与方案约定见表 A3、附录 L。

匹配的 actor 共享审计见表 A8 和附录 F：将两次调用分开，在匹配容量下损失 5.67 个 PushT 百分点；即使将 actor 容量翻倍，仍比完全共享条件模型低 4.00 个百分点。

使用 actor 前编码器已改善。PushT 加入物理逆监督，使关闭 actor 的 CEM30×10 从 42.22 ± 9.50 升至 $57.67\pm2.52\%$，冻结端点–动作 R^2 从 $.673\pm.062$ 升至 $.791\pm.003$。匹配航点监督使 CEM 达 $61.44\pm2.04\%$；更大的 Direct 改善主要说明可部署条件关系有效。低 SIGReg 航点模型，关闭 actor 的 CEM 为 $69.44\pm1.64\%$，Direct 为 $77.67\pm.88\%$。较低秩并非坍塌：各维标准差仍近一，没有维度坍塌。

各任务的增益隔离了学习所得接口的作用。将航点坐标替换为目标位移，使 Direct 成功率分别改变 $+8.11/+0.11/+9.56/-0.11$ 个百分点，任务为 PushT／Cube／Reacher／TwoRoom。因此收益集中于接触敏感的 PushT 和连续驱动 Reacher，Cube 与 TwoRoom 已近饱和。关闭 actor 的纯 CEM 中，同一替换只提高宏平均 1.53（79.33 至 80.86）；恢复 actor 后 Direct 高 4.41，Guarded A 达 96.86。该分离重要：目标改善世界表征，但更大收益来自暴露共享 actor 能直接解释的部署条件。

坐标特征语法、SIGReg、直线化、Huber、额外训练轮次、交互项与规划器半径对照集中报告于附录 D、E。这些实验用于选择最终模型，不作为主要实验重复展示。

5.2 多任务结果

下面在匹配检查点上检验相互关联的表征、控制与稳定性结果。

共享编码器改善全部四领域。E5 目标位移 INTACT Direct 宏平均为 $89.39\pm.77$ ，匹配共享编码器 LeWM 的 CEM300×30 为 66.17 ± 2.67 ；四任务分别提高 5.66／32.23／12.56／42.44 个百分点，并超过已发表任务专用 LeWM 的 85.75 宏平均。相较匹配仅意图组合，加入保留梯度的物理似然提高 8.78 ± 1.63 个宏平均百分点；三个任务为正，饱和 Cube 为 -0.44 。这是完整 INTACT 并非仅为目标条件行为克隆头的最清晰证据。

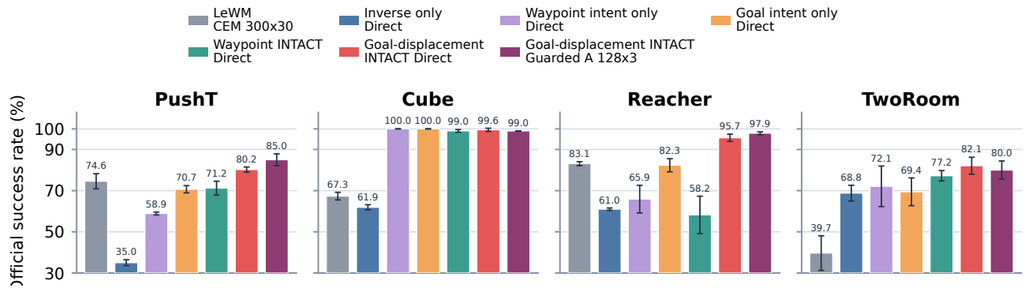


Figure 4: **Controlled E5 success across four tasks.** Bars visualize the native rows of Table 3: LeWM uses CEM 300×30, learned heads use Direct, and the final bar uses Guarded A 128×3. Values are means ± sample standard deviations over three training seeds. The ordinate starts at 30% to expose high-SR differences and is not a zero-based effect size.

Table 3: **Controlled Math-SDPA shared-encoder E5 result.** Native interfaces, actor-disabled pure CEM, and optional verification are evaluated on the same checkpoints. Values are mean ± sample standard deviation over three training seeds; each seed averages official evaluation seeds $\{0, 1, 42\}$ with 100 episodes each.

Training cell	Evaluation	PushT	Cube	Reacher	TwoRoom	Macro
Native interface (Direct where available)						
LeWM	CEM 300×30	74.56±3.67	67.33±1.86	83.11±.96	39.67±8.39	66.17±2.67
Inverse only	Direct	35.00±1.45	61.89±1.26	61.00±.58	68.78±3.83	56.67±.22
Waypoint intent only	Direct	58.89±.69	**100.00±.00**	65.89±6.71	72.11±9.83	74.22±3.23
Goal intent only	Direct	70.67±1.76	**100.00±.00**	82.33±3.18	69.44±6.75	80.61±2.11
Waypoint INTACT	Direct	71.22±3.36	99.00±.67	58.22±9.10	77.22±2.52	76.42±2.32
Goal-displacement INTACT	**Direct**	**80.22±1.26**	**99.56±.77**	**95.67±1.76**	**82.11±4.11**	**89.39±.77**
Actor-disabled pure CEM 300×30						
LeWM	Pure CEM	74.56±3.67	67.33±1.86	83.11±.96	39.67±8.39	66.17±2.67
Inverse only	Pure CEM	75.78±1.02	66.78±2.36	81.78±1.64	51.44±1.07	68.94±.43
Waypoint intent only	Pure CEM	78.44±4.72	68.67±1.76	83.22±2.69	39.11±8.17	67.36±4.06
Goal intent only	Pure CEM	**80.78±2.36**	68.00±3.18	**84.67±1.33**	40.11±8.30	68.39±2.22
Waypoint INTACT	Pure CEM	78.22±3.02	69.56±2.80	83.00±2.19	46.44±3.34	69.31±.68
Goal-displacement INTACT	**Pure CEM**	78.11±1.17	**70.33±3.00**	80.33±2.03	**51.56±4.53**	**70.08±1.13**
Optional local verification						
Goal-displacement INTACT	**Guarded A 128×3**	**85.00±2.85**	**99.00±.00**	**97.89±.69**	**80.00±4.41**	**90.47±.84**

Conclusion. Preserving and organizing the state-conditioned intent–action family relation jointly determines whether actions can be read out directly, whether the actor-disabled world model remains plannable, whether low-budget search is effective, and whether shared multi-task training is stable.

The preferred coordinate changes during shared training. At E1, Waypoint INTACT is the strongest first-pass cell at 42.69±7.19 macro, compared with 36.83±5.63 for Goal INTACT; its advantage is concentrated in PushT, Cube, and high-variance TwoRoom. By E5, the ordering reverses: goal displacement reaches 89.39±.77 versus 76.42±2.32 for waypoint, with per-task differences of $+9.00/+0.56/+37.45/+4.89$ points. Thus E1 measures optimization speed, whereas E5 reveals the stronger converged intent coordinate.

Actor-disabled controls separate representation from readout. Goal INTACT reaches 70.08±1.13 pure-CEM macro versus 66.17±2.67 for LeWM. Its native Guarded A result is 90.47±.84, only 1.08±.14 above Direct, again supporting search as a verifier. E1 matrices, all inference modes, CLEAR Moderate scores, and controlled E1/E5 probes are in Tables A5 to A7. The higher effective rank of E5 waypoint intent-only (93.87 versus 89.26) but lower SR (74.22 versus 89.39) confirms that rank is not a semantic certificate.

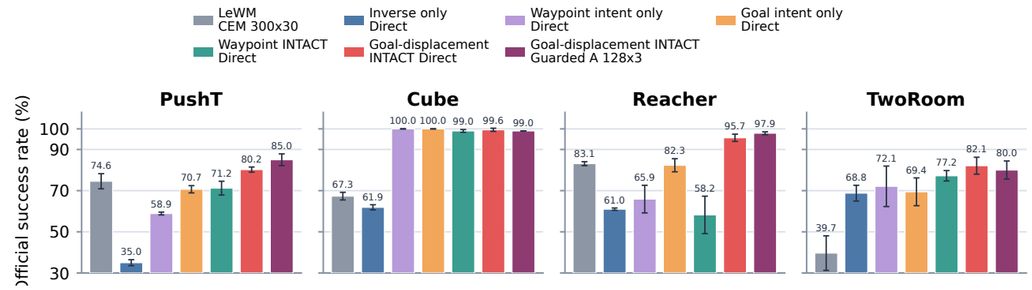


图 4：受控 E5 设置下四个任务的成功率。柱状图展示表 3 中各方法的原生接口：LeWM 使用 CEM 300×30，学习得到的头部使用 Direct，最后一根柱使用 Guarded A 128×3。数值为三个训练种子的均值 ± 样本标准差。纵轴从 30% 开始，以显示高成功率区域的差异，并非从零开始的效应量图。

表 3：受控 Math-SDPA 共享编码器 E5 结果。在相同检查点上评估原生接口、禁用 actor 的纯 CEM 及可选验证。数值为三个训练种子的均值 ± 样本标准差；每个训练种子对官方评估种子 $\{0, 1, 42\}$ 求平均，每个评估种子 100 回合。

训练单元	评估	PushT	Cube	Reacher	TwoRoom	宏平均
原生接口（可用时为 *Direct*）						
LeWM	CEM 300×30	74.56±3.67	67.33±1.86	83.11±.96	39.67±8.39	66.17±2.67
仅逆向	Direct	35.00±1.45	61.89±1.26	61.00±.58	68.78±3.83	56.67±.22
仅航点意图	Direct	58.89±.69	**100.00±.00**	65.89±6.71	72.11±9.83	74.22±3.23
仅目标意图	Direct	70.67±1.76	**100.00±.00**	82.33±3.18	69.44±6.75	80.61±2.11
航点 INTACT	Direct	71.22±3.36	99.00±.67	58.22±9.10	77.22±2.52	76.42±2.32
目标位移 INTACT	**Direct**	**80.22±1.26**	**99.56±.77**	**95.67±1.76**	**82.11±4.11**	**89.39±.77**
禁用 *actor* 的纯 *CEM* 300×30						
LeWM	纯 CEM	74.56±3.67	67.33±1.86	83.11±.96	39.67±8.39	66.17±2.67
仅逆向	纯 CEM	75.78±1.02	66.78±2.36	81.78±1.64	51.44±1.07	68.94±.43
仅航点意图	纯 CEM	78.44±4.72	68.67±1.76	83.22±2.69	39.11±8.17	67.36±4.06
仅目标意图	纯 CEM	**80.78±2.36**	68.00±3.18	**84.67±1.33**	40.11±8.30	68.39±2.22
航点 INTACT	纯 CEM	78.22±3.02	69.56±2.80	83.00±2.19	46.44±3.34	69.31±.68
目标位移 INTACT	**纯 CEM**	78.11±1.17	**70.33±3.00**	80.33±2.03	**51.56±4.53**	**70.08±1.13**
可选局部验证						
目标位移 INTACT	**Guarded A 128×3**	**85.00±2.85**	**99.00±.00**	**97.89±.69**	**80.00±4.41**	**90.47±.84**

结论。保留并组织以状态为条件的意图–动作族关系，共同决定动作能否被直接读出、禁用 actor 的世界模型是否仍可规划、低预算搜索是否有效，以及共享多任务训练是否稳定。

共享训练中最优坐标会改变。 E1 时，航点 INTACT 第一轮最强，宏平均 42.69±7.19，目标 INTACT 为 36.83±5.63；优势集中于 PushT、Cube、高方差 TwoRoom。E5 排序反转：目标位移 89.39±.77，航点 76.42±2.32，各任务差为

$+9.00/+0.56/+37.45/+4.89$ 个百分点。因此 E1 衡量优化速度，E5 则揭示收敛后更强的意图坐标。

禁用 actor 的对照将表征与 读出。目标 INTACT 纯 CEM 宏平均 70.08±1.13，LeWM 为 66.17±2.67；原生 Guarded A 为 90.47±.84，只比 Direct 高 1.08±.14，再次支持搜索作为验证。E1 矩阵、全部模式、CLEAR Moderate、受控 E1／E5 探测见表 A5–A7。E5 仅航点意图有效秩更高（93.87 对 89.26）、成功率却更低（74.22 对 89.39），证明秩不是语义证书。

Table 4: **Theory-linked evidence for shared four-task INTACT.** Correlations use official Direct SR; the actor-disabled and gauge rows are matched interventions that test representation and correspondence directly.

Evidence	Measured result	Interpretation
Pred.–expert kNN (45 eligible E1–E5 checkpoints)	Pooled $\boldsymbol{r = .954}$ [.928, .969]; adjusted $\boldsymbol{r = .902}$. Within Waypoint/Goal: $\boldsymbol{.968/.981}$.	Local action-family neighborhoods track deployable control.
Pred.–expert linear CKA (same 45 checkpoints)	Pooled $\boldsymbol{r = .897}$ [.837, .930]. Within Waypoint/Goal: $\boldsymbol{.979/.986}$. Leave-one-epoch-out: $\boldsymbol{[.888, .914]}$.	Global family geometry remains predictive across interfaces and epochs.
Pointwise action R^2 (same 45 checkpoints)	$r = .815$.	Recovering one expert action is weaker than preserving the family relation.
Effective-rank inversion (controlled E5 cells)	Rank $93.87 > 89.26$, while SR $74.22 < 89.39$.	Latent spread is a capacity check, not a semantic certificate.
Actor-disabled planning (matched E5 checkpoints)	Pure-CEM macro $66.17 \rightarrow \mathbf{70.08}$.	Action losses improve the Encoder–Forward stack before actor execution.
Paired gauge intervention (21,600 CLEAR Moderate episodes)	Shuffled $9.46\% \rightarrow \mathbf{68.04}\%$ paired ($+58.58$ pp).	Correct task-local correspondence, rather than coordinate identity, restores control.

Matched interfaces separate representation, readout, and search. With the actor disabled, Goal INTACT changes LeWM by $+3.55/+3.00/-2.78/+11.89$ points on PushT/Cube/Reacher/TwoRoom: representation shaping is positive overall but not uniformly sufficient. Restoring the native Direct interface raises macro SR from 70.08 to 89.39, and adding Guarded A contributes only another $1.08 \pm .14$ points. Conversely, Full Goal INTACT exceeds goal-intent-only Direct by $9.55/-0.44/13.34/12.67$ points. The three comparisons jointly rule out a single-factor explanation: physical likelihood improves the Encoder–Forward stack, shared intent readout supplies most closed-loop control, and low-budget search performs a final local check.

Summary. Physical inversion improves action-recoverable world features; deployment intent identifies a zero-search action interface; sharing forces both condition families to be interpreted by one conditional law.

Goal displacement is the strongest final coordinate without forcing latent trajectories to be linear. The single- and multi-task results agree: Direct is the native interface, actor-disabled planning remains useful, and small search is best understood as optional verification.

Why these measurements follow the theory. The quotient-to-law view in Propositions 1 and 3 predicts that control should follow agreement between the predicted and expert action-law families. kNN and CKA operationalize its local neighborhoods and global centered geometry; pointwise R^2 deliberately asks the weaker single-action question. Effective rank then serves as a capacity falsifier, actor-disabled planning tests representation before readout, and the paired gauge intervention tests correspondence while allowing coordinates to change. Their observed ordering matches these roles: family metrics are strongest, higher rank can accompany lower SR, physical supervision improves pure CEM, and correct pairing restores 58.58 points without requiring coordinate identity.

表 4：共享四任务 INTACT 的理论相关证据。相关系数使用官方 Direct 成功率；禁用 actor 与坐标规范行是匹配干预，直接检验表征和对应关系。

证据	测量结果	解释
预测–专家 kNN（45 个符合条件的 E1–E5 检查点）	汇总 $\boldsymbol{r = .954}$ [.928, .969]；调整后的 $\boldsymbol{r = .902}$。航点/目标内部：$\boldsymbol{.968/.981}$。	局部动作族邻域对应可部署的控制。
预测–专家线性 CKA（相同的 45 个检查点）	汇总 $\boldsymbol{r = .897}$ [.837, .930]。航点/目标内部：$\boldsymbol{.979/.986}$。留一训练轮次验证：$\boldsymbol{[.888, .914]}$。	全局族几何在不同设置下仍有预测力接口与训练轮次。
逐点动作 R^2（相同的 45 个检查点）	$r = .815$.	恢复一个专家动作弱于保留族关系。
有效秩排序反转（受控 E5 单元）	秩 $93.87 > 89.26$，而成功率 $74.22 < 89.39$。	潜变量展开是容量检查，而非语义证书。
禁用 actor 的规划（匹配的 E5 检查点）	纯 CEM 宏平均 $66.17 \rightarrow \mathbf{70.08}$。	动作损失在使用 actor 之前改善编码器–Forward 组合 actor 执行。
配对坐标规范干预（21,600 个 CLEAR Moderate 回合）	打乱配对 $9.46\% \rightarrow \mathbf{68.04}\%$ 正确配对（$+58.58$ 个百分点）。	正确的任务局部对应关系，而非坐标对应关系恢复控制。

匹配接口分离表征、读出与搜索。关闭 actor，目标 INTACT 相对 LeWM 在 PushT／Cube／Reacher／TwoRoom 改变 $+3.55/+3.00/-2.78/+11.89$ 个百分点：表征塑造总体积极，但未均匀地达到充分控制。恢复原生 Direct，宏平均成功率从 70.08 增至 89.39；加入 Guarded A 仅再贡献 $1.08 \pm .14$。反过来，完整目标 INTACT 比仅目标意图 Direct 高 $9.55/-0.44/13.34/12.67$ 个百分点。三个比较共同排除单因素解释：物理似然改善编码器–前向模型，共享意图读出承担大部分闭环控制，低预算搜索只作最后局部检查。

总结。物理求逆改善能够恢复动作的世界特征；部署意图辨识出无需搜索的动作接口；共享迫使同一个条件规律解释两类条件。

目标位移是最强的最终坐标，无需强制潜轨迹线性化。单任务与多任务结果一致：Direct 是原生接口，禁用 actor 的规划仍然有用，而小规模搜索应理解为可选验证。

为何这些测量遵循理论。命题 1 和 3 的商到规律观点预测：控制应随预测和专家动作规律族的一致性变化。k 近邻与 CKA 分别操作化局部邻域和全局中心化几何；逐点 R^2 刻意询问更弱的单动作问题。有效秩用于否证容量解释，关闭 actor 的规划在读出前测试表征，配对规范干预在允许坐标改变时测试对应。观测排序符合这些角色：族指标最强，更高秩可伴随更低成功率，物理监督改善纯 CEM，正确配对无需坐标相同即可恢复 58.58 个百分点。

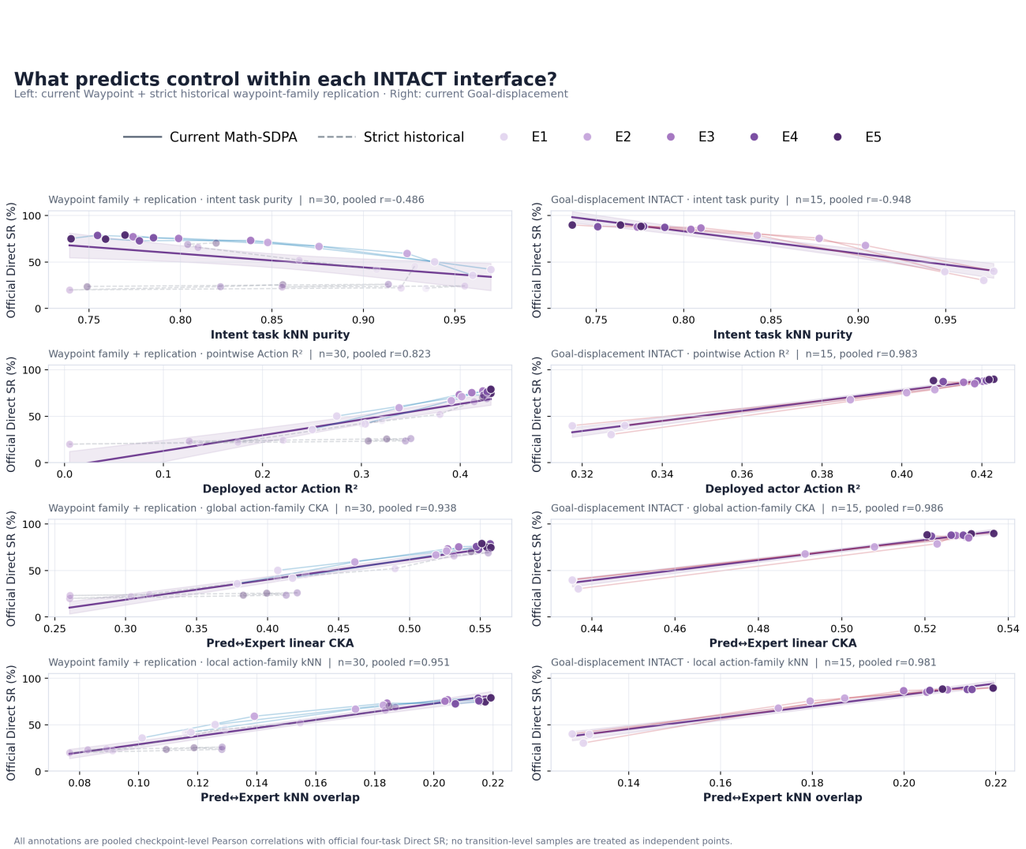


Figure 5: **Intent–action relation, rather than task clustering, tracks control.** Columns separate waypoint and goal-displacement interfaces; rows compare task purity, pointwise action R^2, predicted–expert CKA, and local kNN overlap. Each point is one checkpoint and each annotation is a pooled checkpoint-level Pearson correlation with official four-task Direct SR. Historical runs are lower opacity and never rank the controlled objectives.

Family conversion predicts closed-loop control. Across the 30 primary Math-SDPA Full-INTACT checkpoints, predicted–expert CKA and local kNN overlap correlate with official Direct SR at $r = .861$ and .926. Within complete waypoint/goal learning trajectories, CKA reaches $.979/.986$ and kNN $.968/.981$. Pooling the separately labeled strict replication cohort gives CKA $r = .897$ (95% run-cluster bootstrap CI $[.837, .930]$) and kNN $r = .954$ ($[.928, .969]$); adjusted kNN remains .902. By contrast, intent task purity is negatively associated with SR. Cleaner task clusters are therefore not the desired relation: what matters is whether the state-conditioned intent family is carried into the corresponding action family.

Gauge intervention separates correspondence from actor function. Correctly paired task-local calibration recovers 68.04% CLEAR Moderate SR, whereas identity and shuffled-pair maps obtain 7.96% and 9.46%. Same- objective swaps define a high alignment ceiling, and LeWM/inverse backbones can also align into a trained Full actor. The reverse direction is decisive: a Full backbone mapped into a LeWM actor remains at 5.08%, while an inverse actor recovers only inverse-level functionality. A coordinate map can restore a learned conditional but cannot create one absent from the target actor. The 72-cell, 21,600-episode audit and calibration curves are in Fig. A5 and Section E.

The eight panels reject a clustering-only explanation. Across both deployment coordinates, task purity has the wrong sign, pointwise action recovery is informative but weaker, and relation-aware CKA/kNN give the most consistent monotone trends. The agreement between current Math-SDPA runs and the separately marked historical cohort further shows that the result is not produced by one checkpoint family. Together with the gauge intervention, the figure supports a narrower claim than global latent isomorphism: successful control requires each state-conditioned intent neighborhood to retain the action-family correspondence that its task-specific head can read.

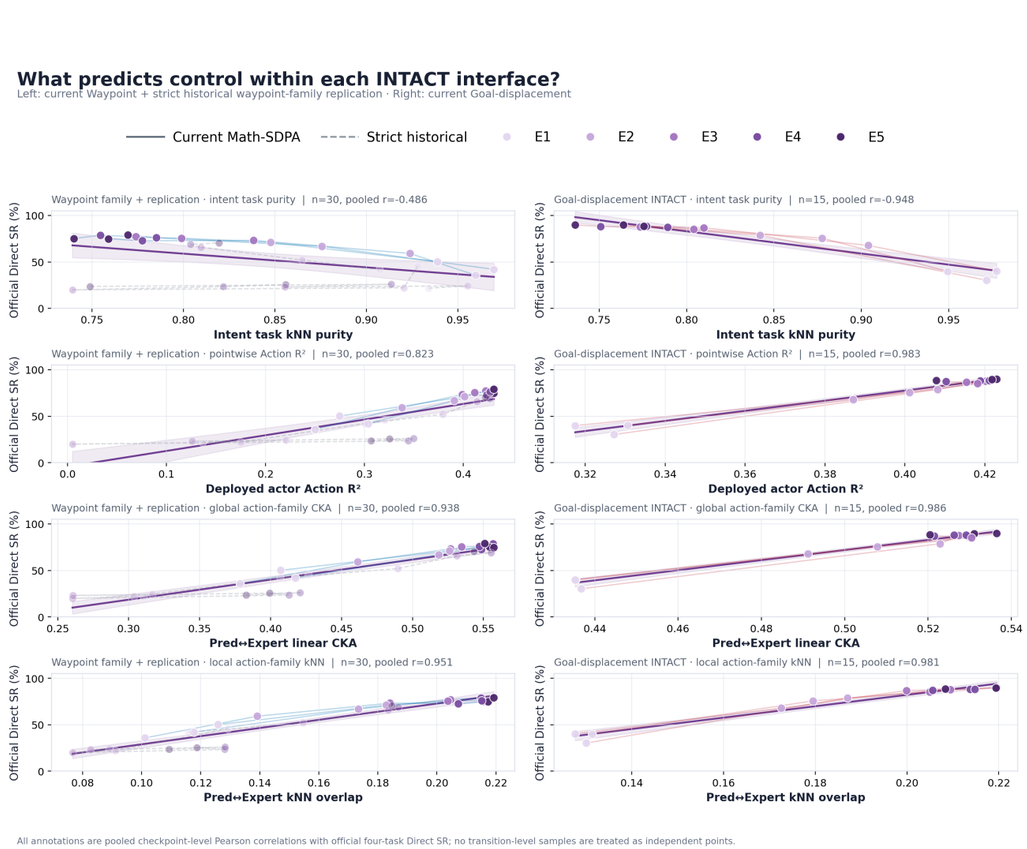


图 5：与控制表现相对应的是意图–动作关系，而非任务聚类。各列区分航点与目标位移接口；各行比较任务纯度、逐点动作 R^2、预测–专家 CKA 和局部 kNN 重叠。每个点代表一个检查点，标注为所有检查点与官方四任务 Direct 成功率的汇总 Pearson 相关系数。历史运行以更低透明度显示，不用于受控目标的排名。

族转换预测闭环控制。30 个主要 Math-SDPA 完整 INTACT 检查点中，预测–专家 CKA 和局部 k 近邻重叠，与官方 Direct 成功率相关为 $r = .861$ 和 .926。完整航点／目标学习轨迹内，CKA 达 $.979/.986$，k 近邻达 $.968/.981$。合并单独标记的严格复现组后，CKA 为 $r = .897$（95% 运行聚类 bootstrap 区间 $[.837, .930]$），k 近邻为 $r = .954$（$[.928, .969]$）；调整后 k 近邻仍 .902。相反，意图任务纯度与成功率负相关。更干净的任务簇并非目标关系；关键是状态条件意图族是否转换到对应动作族。

坐标规范干预分离对应关系与 actor 功能。正确配对的任务局部标定恢复 68.04% CLEAR Moderate 成功率，恒等和打乱配对为 7.96%、9.46%。同目标交换定义较高对齐上限，LeWM／逆模型骨干也能对齐到训练好的完整 actor。反向结果关键：完整骨干映射到 LeWM actor 仍仅 5.08%，逆 actor 仅恢复逆模型水平功能。坐标映射能恢复已学条件，却不能创造目标 actor 不具备的条件。72 组合、21600 回合审计与标定曲线见图 A5 和附录 E。

八个面板否定仅聚类解释。两种部署坐标中，任务纯度方向相反，逐点动作恢复有信息但较弱，关系敏感 CKA／k 近邻呈最一致单调趋势。当前 Math-SDPA 与单独标记历史组一致，说明结果不是某一检查点族所致。结合规范干预，图支持比全局潜在同构更窄的结论：成功控制要求各状态条件意图邻域保持专用头可读的动作族对应。

6 Limitations

Three seeds remain a coarse scaling estimate. The controlled multi-task matrix is complete for all six objective cells, three training seeds, three evaluation seeds per checkpoint, and E1–E5 under Math-SDPA. Official Direct, pure CEM, Guarded A, and CLEAR Moderate have full applicable coverage. The primary conversion audit is likewise complete for both Full INTACT interfaces, three seeds, and E1–E5. The separately marked historical replication cohort predates the backend audit and is used only for mechanism robustness, not objective ranking. The final single-task goal-displacement study now uses three training seeds on all four tasks, but three seeds still provide only a coarse estimate of training variability.

The action quotient is identified only on demonstrated support. Proposition 3 applies to endpoint conditions with positive sampling probability. It does not identify arbitrary counterfactual goals, nor does a single expert action reveal whether another action would have been equally valid. The full goal displacement preserves direction and distance but remains an approximate request around obstacles, contact switches, and multimodal demonstrations. The local-intent call physically anchors the representation, while the goal-intent call relies on demonstration support; neither construction proves correct behavior outside that support. Using the same goal-intent construction and INTACT Predictor at training and deployment removes the forward-model/action-proposal interface mismatch, not every train–test gap: after the first direct step, autoregressive control conditions on predicted latents whose distribution can drift from encoded demonstration states.

The current diagnostics are necessary checks, not certificates. Effective rank and SRS can reward isotropic noise. G_{sep} measures response to deterministic goal changes, but two different goals can require the same current action, so it is only a proxy for an action-distinct quotient. Expert NLL shares the INTACT Predictor used by Direct and is therefore not an independent causal variable. A stronger test would pair action-equivalent and action-distinct counterfactual conditions, report normalized Jacobian spectra, and predict held-out checkpoint failures without refitting thresholds. The t-SNE geometry is qualitative and is not used for model selection.

Direct Gaussian control can hide multimodality. The deployed action is the mean of a diagonal Gaussian. At junctions or contact transitions, this mean can lie between valid modes. Search can verify a proposal against forward dynamics, but then loses the zero-candidate identity. Mixture actors, sampling from the INTACT Predictor's learned conditional uncertainty instead of a unit-variance raw-action distribution, uncertainty-triggered verification, and quotient-aware mode selection are natural extensions; our current evidence supports a simple cross-task default rather than a universal action distribution.

Gauge equivalence is task-manifold conditioned. The base Full–goal-only swap uses two training seeds; the expanded 72-cell audit strengthens causal controls but does not establish exact conjugacy or a single global affine chart. Correct pairing is necessary, and same-objective cross-seed swaps define a high alignment ceiling. However, LeWM and inverse-only backbones can also be aligned into a trained Full actor, so alignability is not an INTACT-exclusive representation property. Conversely, reverse swaps show that a gauge cannot create a deployment map absent from the target actor. Cross-task and leave-one-task-out transfer still fail. This task-local scope is sufficient for task-specific control heads; a single universal actor would require explicit cross-task alignment, adapters, or stronger joint coverage.

Scope and evaluation. Experiments use four simulated tasks, fixed image goals, task-specific action heads, and offline expert trajectories. We have not established real-robot, out-of-distribution goal, distractor, or cross-embodiment generalization. The present headline uses the official LeWM benchmark to preserve direct comparability. Our stricter legality-aware audit is reported separately and should be validated independently. Finally, PRISM, GC-IDM, SMWM, QuoVLA, and related concurrent systems use different data, training schedules, and evaluators. We therefore compare their interfaces and assumptions, not unmatched success-rate numbers.

7 Conclusion

INTACT starts from a mismatch that is easy to overlook: a physical successor is a valid inverse-dynamics condition but is unavailable before acting, whereas a deployable goal intent is available but is not itself an observed transition. Their useful commonality is neither endpoint type nor Euclidean proximity. At a fixed world state, it is the expert action law they induce. This conditional action quotient leads to a deliberately small statistical unit: each supervised conditional consists of one demonstrated state, one supported endpoint condition, and one correct action. The forward JEPA and SIGReg remain responsible for preserving the richer world representation. In the full objective, the two action losses also remain distinct: local-intent likelihood shapes the encoder through the attached real successor, whereas goal-intent likelihood shapes the current-state encoding toward the deployable conditional and has no successor-gradient path. This is a branch-local stop-gradient, not a detached batch or encoder: the same future latent is plastic when used as a physical successor and fixed when reused as a deployment anchor. Taken separately, these are familiar objectives: local inverse dynamics and a GCSL-like hindsight action likelihood. INTACT begins at their irreducible coupling: one weight-shared, graph-isomorphic conditional operator must explain both families in action-law space, while asymmetric endpoint gradients preserve their distinct physical and deployment roles and the Forward Predictor retains richer world information. The method therefore does not reduce to placing two losses beside each other.

The controlled evidence supports the resulting predictions. In single-task training, the final goal-displacement model reaches 85.78%, 100.00%, 97.67%, and 97.89% official Direct SR after

6 局限性

三个种子仍只能粗略估计规模效应。受控多任务矩阵在 Math-SDPA 下覆盖六个目标组合、三个训练种子、每检查点三个评估种子及 E1–E5。官方 Direct、纯 CEM、Guarded A、CLEAR Moderate 均具有完整适用覆盖。主要转换审计也覆盖两种完整接口、三个种子与 E1–E5。单独标记的历史复现组早于后端审计，仅用于机制稳健性，不用于目标排名。最终单任务目标位移研究在四任务均用三个训练种子，但仍只粗略估计训练波动。

动作商关系仅在示范支持范围内被识别。命题 3 适用于采样概率为正的端点条件，不能识别任意反事实目标；单个专家动作也不能揭示其他动作是否同样有效。完整目标位移保留方向和距离，但在障碍物、接触切换及多模态示范附近，仍是近似请求。局部意图调用提供物理锚定，目标意图调用依赖示范支持；两者都不能证明支持范围外的正确行为。训练与部署使用同样目标意图和预测器，消除的是前向模型／动作提案接口不匹配，而非全部训练–测试差距：首个 Direct 步骤之后，自回归控制以预测潜变量为条件，其分布可能偏离示范状态编码。

当前诊断是必要检查，而非证书。有效秩和 SRS 可能奖励各向同性噪声。G_{sep} 测量对确定性目标变化的响应，但不同目标可要求同一当前动作，故仅是动作不同商类的代理。专家 NLL 与 Direct 共用预测器，不是独立因果变量。更强测试应配对动作等价和动作不同的反事实条件，报告归一化 Jacobian 谱，并不重拟阈值地预测未见检查点失败。t-SNE 仅定性，不用于模型选择。

Direct 高斯控制可能掩盖多模态。部署动作是对角高斯均值；交叉口或接触转移处，均值可能位于多个有效模式之间。搜索可通过前向动力学验证提案，但失去零候选性质。混合 actor、从预测器条件不确定性而非单位方差原始动作分布采样、不确定性触发验证、感知商类的模式选择，都是自然扩展。现有证据支持简单跨任务默认设置，而非通用动作分布。

坐标规范等价性以任务流形为条件。基础 Full–仅目标交换使用两个训练种子；扩展的 72 组合审计强化因果对照，但不证明精确共轭或单一全局仿射坐标图。正确配对必要，同目标跨种子交换定义高对齐上限。但 LeWM 和仅逆模型骨干也可对齐到完整 actor，故可对齐性非 INTACT 独有。反向交换说明规范变换不能创造目标 actor 缺失的部署映射。跨任务和留一任务迁移仍失败。任务局部范围足够支持专用头；通用 actor 则需显式跨任务对齐、适配器或更强联合覆盖。

范围与评估。实验使用四项仿真任务、固定图像目标、任务专用动作头、离线专家轨迹，尚未建立真实机器人、分布外目标、干扰物或跨形态泛化。主要结果沿用官方 LeWM 保持直接可比，更严格合法性审计单独报告，应独立验证。PRISM、GC-IDM、SMWM、QuoVLA 等同期系统数据、调度、评估器不同，因此比较接口与假设，不比较未匹配成功率数值。

7 结论

INTACT 从一个容易忽视的不匹配出发：物理后继是有效的逆动力学条件，却无法在动作前获得；可部署目标意图可以获得，却不是观测到的转移。两者有用的共同点，既不是端点类型，也不是欧氏距离接近，而是固定世界状态下所诱导的专家动作规律。该条件动作商关系形成一个刻意精简的统计单元：每个受监督条件由一个示范状态、一个受数据支持的端点条件及一个正确动作组成。前向 JEPA 与 SIGReg 仍负责保持更丰富的世界表征。完整目标中，两项动作损失仍有区别：局部意图似然通过保留梯度的真实后继塑造编码器；目标意图似然则将当前状态编码塑造成可部署条件，没有后继梯度路径。这是分支局部停止梯度，而非分离整个批次或编码器：同一个未来潜变量作为物理后继时可塑，复用为部署锚点时固定。单独看，两者都是熟悉的目标：局部逆动力学与类似 GCSL 的事后动作似然。INTACT 的关键是不可约的耦合：同一个权重共享、计算图同构的条件算子，必须在动作规律空间解释两类条件；非对称端点梯度保持各自物理和部署角色，前向预测器保留更丰富世界信息。因此，方法不能简化为并列放置两个损失。

受控证据支持由此得到的预测。在单任务训练中，最终目标位移模型在训练后达到官方 Direct 成功率 85.78%、100.00%、97.67% 和 97.89%，所需训练为

one epoch with no candidate search, with three training seeds on every task. The completed 2×4 task-specific matrix further shows that goal displacement exceeds the waypoint coordinate under Direct, pure CEM, actor-on CEM, and Guarded A. On the final coordinate, actor-on CEM spends 9,000 sequences yet falls from 95.33% Direct macro to 93.78%, whereas plan-centered Guarded A reaches 96.86% with 384 sequences. Controlled head-grammar interventions show a 7.00-point A→E gain along the matched-product path. The new G control improves A by only 0.89 points when centering the product alone, whereas matching the centered first-order intent with its product yields a 3.89-point residual–product interaction; under the final goal coordinate, removing that matched product costs 2.56 points. Earlier matched PushT attribution shows that the waypoint encoder reaches $69.44 \pm 1.64\%$ with the actor removed under pure CEM 30×10, separating representation shaping from proposal quality. In shared four-task training, the deterministic Math-SDPA factorial gives Full INTACT the strongest E1 Direct macro, $42.69 \pm 7.19\%$, and lower TwoRoom variation than waypoint-intent-only. At E5, Goal-displacement INTACT reaches $89.39 \pm 0.77\%$ Direct macro and exceeds matched three-seed goal-intent-only by 8.78 ± 1.63 points. Actor-disabled pure CEM improves from $66.17 \pm 2.67\%$ for LeWM to $70.08 \pm 1.13\%$, while Guarded A raises the Full model to $90.47 \pm 0.84\%$; search therefore remains a compatible verifier rather than the learned interface. Across 45 eligible simultaneous-dual-likelihood E1–E5 checkpoints, predicted–expert CKA/kNN correlate with official Direct SR at $r = 0.897/0.954$; kNN remains $r = 0.902$ after epoch/cohort adjustment. This mechanism audit complements, rather than replaces, the controlled performance comparison. On matched PushT waypoint training, parameter-matched independent actors lose 5.67 Direct points, and even double capacity remains 4.00 points below the fully shared operator. This directly supports sharing as a semantic coupling, not a parameter-count accident. The A–G head grammar supplies a complementary intra-call constraint: the first-order intent and its state–intent bilinear interaction must carry matched semantics inside each call. The deployment loss is still a goal-conditioned imitation likelihood; these controls show why the complete method is not reducible to goal-only or post-hoc imitation: it couples the GCSL-like deployment likelihood end to end with JEPA dynamics and the same shared conditional operator on attached one-chunk physical successors.

The 21,600-episode gauge audit supplies a direct causal check. Correct pairing recovers 68.04% CLEAR Moderate SR, versus 9.46% after pair shuffling, and the calibration curve saturates near 64/full episodes. Same-objective swaps set a high alignment ceiling, while LeWM/inverse backbones also align into a trained Full actor; alignability is therefore not unique to INTACT. In the reverse direction, Full-backbone-to-LeWM-actor remains at 5.08%, and an inverse actor recovers only inverse-level functionality. A coordinate gauge can restore a learned map but cannot invent a deployment conditional absent from the actor.

The complete 576-job planner audit adds a compatible efficiency result. Guarded A centers a 128×3 raw-action residual search on the deterministic INTACT plan and reaches 96.86% macro and 92.22% worst-task SR, exceeding matched pure CEM 300×30 by 16.00 macro points with $23.44\times$ fewer sampled candidate sequences. Ordinary $\sigma_0 = 0.25$ is only 0.08 macro point lower. A diagnostic L2-trust variant raises PushT to 92.78% but is less uniform across tasks, so it is not part of the recommended inference interface. The finite correction region improves contact-sensitive PushT without discarding the precision already present in Reacher, whereas iterative actor-particle sampling compounds mixture and model error.

The broader lesson is that amortized control need not be attached after world model learning or reduced to a planner-side sampling trick. A single JEPA can retain predictive world state, learn which conditions are equivalent for the current action, and expose that conditional directly at deployment. Search then becomes an optional verifier rather than the mandatory interface to the model. At a more general level, the construction is not tied to JEPA: whenever two condition families share supervised behavioral labels, a shared proper conditional operator can align their quotient semantics while asymmetric gradient routing preserves the role of each family. INTACT therefore offers a template for relation-preserving family mappings beyond world models, without requiring pointwise coordinate collapse.

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一轮训练、无候选搜索，每任务三个训练种子。完整 2×4 任务专用矩阵还表明，Direct、纯 CEM、启用 actor 的 CEM 和 Guarded A 下，目标位移均超过航点坐标。最终坐标中，启用 actor 的 CEM 使用 9000 条序列，却从 Direct 宏平均 95.33% 降至 93.78%；以计划为中心的 Guarded A 只用 384 条，达到 96.86%。受控动作头语法干预，在匹配乘积路径上产生 A→E 的 7.00 个百分点收益。新 G 对照仅中心化乘积，比 A 高 0.89；中心化一阶意图与乘积匹配，产生 3.89 个残差–乘积交互百分点；最终目标坐标下去除匹配乘积损失 2.56。较早 PushT 归因中，航点编码器关闭 actor、纯 CEM 30×10 达到 $69.44 \pm 1.64\%$，分离表征塑造与提案质量。共享四任务训练的确定性 Math-SDPA 析因实验，完整 INTACT E1 Direct 宏平均最高，为 $42.69 \pm 7.19\%$，TwoRoom 波动低于仅航点意图。E5 目标位移 INTACT 达 $89.39 \pm 0.77\%$ Direct 宏平均，比匹配三种子仅目标意图高 8.78 ± 1.63。关闭 actor 的纯 CEM 从 LeWM $66.17 \pm 2.67\%$ 增至 $70.08 \pm 1.13\%$，Guarded A 使完整模型达 $90.47 \pm 0.84\%$；搜索是兼容的验证器，而非学习接口。45 个合格的同时双似然 E1–E5 检查点中，预测–专家 CKA／k 近邻与官方 Direct 成功率相关为 $r = 0.897/0.954$；调整轮次和组别后 k 近邻仍为 $r = 0.902$。机制审计补充而非替代受控性能比较。匹配 PushT 航点训练中，参数数目匹配的独立 actor 比共享算子低 5.67 个 Direct 点，即使容量翻倍仍低 4.00，直接支持共享是语义耦合而非参数数目的偶然结果。A–G 头语法提供调用内部的互补约束：一阶意图与状态–意图双线性交互必须携带匹配语义。部署损失仍是目标条件模仿似然；这些对照解释为何完整方法不能归结为仅目标或事后模仿：它将类似 GCSL 的部署似然与 JEPA 动力学、保留梯度单块物理后继上的同一共享算子端到端耦合。

21600 回合规范审计提供直接因果检验。正确配对恢复 68.04% CLEAR Moderate 成功率，打乱仅 9.46%；标定曲线在 64／完整回合附近饱和。同目标交换设定高对齐上限，LeWM／逆骨干也可对齐到完整 actor，因此可对齐性并非 INTACT 独有。反向完整骨干到 LeWM actor 为 5.08%，逆 actor 仅恢复逆水平。坐标规范可以恢复已学映射，不能创造 actor 缺少的部署条件关系。

完整的 576 项规划器作业审计进一步证实效率优势。Guarded A 以确定性 INTACT 计划为中心进行 128×3 原始动作残差搜索，达到 96.86% 宏平均及 92.22% 最差任务成功率，比匹配纯 CEM 300×30 高 16.00 个宏平均点，采样序列少 $23.44\times$ 倍。普通 $\sigma_0 = 0.25$ 仅低 0.08。诊断 L2 信赖域变体使 PushT 达 92.78%，但跨任务更不均匀，不属于推荐接口。有限修正区域改善接触敏感 PushT，又不丢弃 Reacher 已有精度；迭代 actor 粒子采样则累积混合与模型误差。

更广泛的启示是，摊销控制无需在世界模型学习后附加，也不必简化为规划器采样技巧。同一个 JEPA 可保留预测世界状态，学习哪些条件对当前动作等价，并直接在部署暴露该条件关系。搜索成为可选验证器，而非模型必需接口。更一般地，此构造不依赖 JEPA：只要两类条件共享受监督行为标签，共享的适当条件算子就能对齐其商语义，非对称梯度路由则保留各类角色。因此，INTACT 为世界模型之外保持关系的族映射提供模板，无须逐点坐标坍塌。

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Appendix

A Proofs and Statistical Interpretation

Proof of Proposition 2. Let $S = \mathrm{supp}(z, z^+)$. The successor-only risk is an integral of $-\log p_\eta(a \mid z, y)$ over triples with $(z, y) \in S$. If two actors agree on S, their integrands and therefore their risks are equal almost everywhere. Changing either actor on a goal intent m outside S cannot change that integral. Thus the successor-only objective does not identify deployment behavior at m.

Proof of Proposition 3. Condition on an endpoint pair (z, y) with positive probability. Its expected negative log likelihood decomposes as

$$\begin{aligned}&\mathbb{E}_{a\sim p_E^\star(\cdot|z,y)}[-\log p_\eta(a \mid z, y)] \\ &\quad = H[p_E^\star(\cdot \mid z, y)] + D_{\mathrm{KL}}(p_E^\star \| p_\eta).\end{aligned} \tag{19}$$

The entropy is constant in η and the KL term is nonnegative, with equality only when the modeled and expert conditional laws agree almost everywhere. Applying this argument to every supported endpoint proves the result. If $y \sim_z y'$, both recovered laws equal the same expert law; if the expert laws differ, a proper likelihood preserves that distinction.

For a Bernoulli endpoint sample, linearity of expectation gives Eq. (30). A paired estimator of $(1-p)\ell(a; z^+) + p\ell(a; z + m^{\mathrm{goal}})$ can have the same population expectation. Our claim is consequently not that sampling changes the population optimum by itself. It keeps one conditional observation per transition, fixes the action-loss scale, and changes finite-batch gradient covariance under adaptive optimization. The paired controls test this finite-sample prediction.

B Implementation and Evaluation Identities

The encoder is ViT-Tiny/14 followed by a two-layer projection to 192 dimensions. The forward model is a six-layer causal transformer with 16 attention heads and history length three. One model action contains five environment actions. The shared INTACT Predictor is a three-layer width-1024 MLP with LayerNorm and GELU; it emits a diagonal-Gaussian mean and log standard deviation clipped to $[-5, 2]$. Images use ImageNet normalization and actions use training-split statistics.

The backbone has 18.03M trainable parameters and the shared INTACT Predictor adds 3.11M. Task heads are separate; the encoder and projector are shared across all four tasks. Training calls `seed_everything(seed, workers=True)` before dataset and model construction. Checkpoints are loaded in strict mode and every evaluation records the checkpoint hash, manifest hash, evaluator fingerprint, task, protocol, solver identity, and seed.

Table A1: Inference identities enforced by the evaluator. A zero in Direct is an assertion checked from the trace, not merely a requested setting.

Mode	Actor	Candidates/iter.	Cost calls
Direct	✓	0	0
Pure CEM 5×2	✗	5 per iteration	> 0
Pure CEM 30×10	✗	30 per iteration	> 0
Pure CEM 300×30	✗	300 per iteration	> 0

C Extended Method Derivations and Control Pseudocode

C.1 Waypoint route and equal-progress assumption

The waypoint attribution route solves, at teacher step k with $r = H - k$,

$$\min_{\delta_k,\ldots,\delta_{H-1}} \sum_{j=k}^{H-1} \|\delta_j\|_2^2 \quad \text{s.t.} \quad \sum_{j=k}^{H-1} \delta_j = z_g - z_{t+k}. \tag{20}$$

Strict convexity gives the equal-progress request

$$m^{\mathrm{waypoint}}_{t+k,H-k} = \frac{\mathrm{sg}(z_g) - z_{t+k}}{H-k}, \tag{21}$$

implemented in the endpoint grammar as

$$q_{t+k,H-k} = z_{t+k} + m^{\mathrm{waypoint}}_{t+k,H-k}. \tag{22}$$

It is paired with the demonstrated one-chunk action,

$$\mathcal{L}^{\mathrm{waypoint}}_{\mathrm{goal}} = \mathbb{E}[-\log p_\eta(a_{t+k} \mid {}^{z_{t+k}, q_{t+k,H-k}}_{a_{t+k-1}})]\,. \tag{23}$$

This route narrows train–deployment support but divides away remaining-distance magnitude. The final raw-goal route instead uses $m^{\mathrm{goal}}_t = \mathrm{sg}(z_g) - z_t$ and lets the nonlinear actor choose state-dependent progress.

The two routes share the same branch-local gradient rule. The future endpoint is detached only in the deployment occurrence; it remains attached when it is the real successor of the last physical transition. Neither a complete batch nor the encoder is detached.

C.2 Why intent coordinates do not straighten trajectories

A realized transition under the goal-conditioned actor may be written descriptively as

$$\Delta z_t^{\mathrm{real}} = \alpha_t(z_t, m_t^{\mathrm{goal}}, a_{t-1}) m_t^{\mathrm{goal}} + r_t, \tag{24}$$

where r_t can contain detours, contact changes, and multimodal residuals. The Forward Predictor updates reachability and the goal residual,

$$a_t \sim \pi_\eta(\cdot \mid z_t, m_t^{\mathrm{goal}}, a_{t-1}), \tag{25}$$
$$\widehat{z}_{t+1} = F_\phi(\widehat{z}_t, a_t), \tag{26}$$
$$m_{t+1}^{\mathrm{goal}} = z_g - \widehat{z}_{t+1}. \tag{27}$$

附录

A 证明与统计解释

命题 2 证明。令 $S = \mathrm{supp}(z, z^+)$。仅后继风险是在满足 $(z, y) \in S$ 的三元组上，对 $-\log p_\eta(a \mid z, y)$ 积分。若两 actor 在 S 一致，被积函数及风险几乎处处相等。改变 S 外目标意图 m 上的任一 actor，不影响积分。因此仅后继目标不识别 m 的部署行为。

命题 3 的证明。给定具有正概率的端点对 (z, y)，其期望负对数似然分解为

$$\begin{aligned}&\mathbb{E}_{a\sim p_E^\star(\cdot|z,y)}[-\log p_\eta(a \mid z, y)] \\ &\quad = H[p_E^\star(\cdot \mid z, y)] + D_{\mathrm{KL}}(p_E^\star \| p_\eta).\end{aligned} \tag{19}$$

熵相对于 η 为常数，KL 项非负，且仅当模型与专家的条件规律几乎处处一致时取等号。将该论证用于每个支持的端点即可证明结果。若 $y \sim_z y'$，两个恢复的规律都等于同一专家规律；若专家规律不同，适当似然就会保留这种差异。

对 Bernoulli 端点采样，期望线性性给出式（30）。$(1-p)\ell(a;\ z^+) + p\ell(a;\ z + m^{\mathrm{goal}})$ 的配对估计器可有相同总体期望。因此并非声称采样本身改变总体最优；它使每转移保留一个条件观测，固定动作损失尺度，在自适应优化下改变有限批次梯度协方差。配对对照检验这一有限样本预测。

B 实现与评估属性

编码器为 ViT-Tiny/14，后接两层投影至 192 维。前向模型为六层因果 Transformer，16 头、历史长度三。一个模型动作包含五个环境动作。共享预测器为三层、宽 1024 的 MLP，带 LayerNorm 和 GELU，输出对角高斯均值及对数标准差，后者截断至 $[-5, 2]$。图像用 ImageNet 归一化，动作用训练集统计。

骨干网络包含 18.03M 个可训练参数，共享的 INTACT Predictor 再增加 3.11M。各任务的头部独立；编码器和投影器由四个任务共享。训练在构建数据集和模型之前调用 seed_everything(seed, workers=True)。检查点以严格模式加载；每次评估都会记录检查点哈希、清单哈希、评估器指纹、任务、协议、求解器标识和随机种子。

表 A1：评估器强制保证的推理属性。Direct 中的零是从执行轨迹检查的断言，不只是请求的设置。

Mode	Actor	每轮候选数	代价调用
Direct	✓	0	0
纯 CEM 5×2	✗	每轮 5 个	> 0
纯 CEM 30×10	✗	每轮 30 个	> 0
纯 CEM 300×30	✗	每轮 300 个	> 0

C 扩展方法推导与控制伪代码

C.1 航点路径与等进度假设

航点归因路径在教师步骤 k、$r = H - k$ 时求解，

$$\min_{\delta_k,\ldots,\delta_{H-1}} \sum_{j=k}^{H-1} \|\delta_j\|_2^2 \quad \text{s.t.} \quad \sum_{j=k}^{H-1} \delta_j = z_g - z_{t+k}. \tag{20}$$

严格凸性给出等进度请求

$$m^{\mathrm{waypoint}}_{t+k,H-k} = \frac{\mathrm{sg}(z_g) - z_{t+k}}{H-k}, \tag{21}$$

在端点特征语法中实现为

$$q_{t+k,H-k} = z_{t+k} + m^{\mathrm{waypoint}}_{t+k,H-k}. \tag{22}$$

它与示范的单动作块配对，

$$\mathcal{L}^{\mathrm{waypoint}}_{\mathrm{goal}} = \mathbb{E}[-\log p_\eta(a_{t+k} \mid {}^{z_{t+k}, q_{t+k,H-k}}_{a_{t+k-1}})]\,. \tag{23}$$

该路径缩小训练与部署的支持集差异，却除去了剩余距离的大小。最终原始目标路径改用 $m_t^{\mathrm{goal}} = \mathrm{sg}(z_g) - z_t$，让非线性 actor 选择依赖状态的推进程度。

两条路径共享相同的分支局部梯度规则。未来端点仅在部署调用中停止梯度；当它是最后一次物理转移的真实后继时，仍保留梯度。没有对整批数据或整个编码器停止梯度。

C.2 为什么意图坐标不使轨迹直线化

目标条件 actor 下已实现的转移可描述为

$$\Delta z_t^{\mathrm{real}} = \alpha_t(z_t, m_t^{\mathrm{goal}}, a_{t-1}) m_t^{\mathrm{goal}} + r_t, \tag{24}$$

其中 r_t 可包含绕行、接触变化和多模态残差。Forward Predictor 更新可达性与目标残差，

$$a_t \sim \pi_\eta(\cdot \mid z_t, m_t^{\mathrm{goal}}, a_{t-1}), \tag{25}$$
$$\widehat{z}_{t+1} = F_\phi(\widehat{z}_t, a_t), \tag{26}$$
$$m_{t+1}^{\mathrm{goal}} = z_g - \widehat{z}_{t+1}. \tag{27}$$

This does not impose the Euclidean trajectory penalty

$$\mathcal{L}_{\text{lin}} = \sum_{k=1}^{H-1} \|z_{t+k} - [(1-\alpha_k)z_t + \alpha_k z_g]\|_2^2, \qquad \alpha_k = k/H, \tag{28}$$

whose matched negative control appears in Section H.

C.3 Alternative endpoint sampling and end-to-end necessity

One may draw $b \sim \text{Bernoulli}(p)$ and use one endpoint per transition,

$$y = (1-b)z^+ + b(z + m^{\text{goal}}), \tag{29}$$
$$\mathcal{L}_{\text{end}} = \mathbb{E}[-\log p_\eta(a \mid z, y, a_{t-1})]. \tag{30}$$

Its expectation is $(1-p)\mathcal{L}_{z^+} + p\mathcal{L}_{\text{goal}}$, although paired evaluation of both calls changes finite-batch gradient covariance. More importantly, if $E(o) = E(o')$ for observations requiring different action laws under the same intent, no frozen downstream actor can undo that aliasing. Backpropagating the proper likelihood through E exposes the missing controllable distinction during representation learning.

C.4 Endpoint grammar and interaction prediction

The waypoint experiments use

$$h_t^{\text{end}}(y_t) = [z_t; y_t; z_t \odot y_t; A_\omega(a_{t-1})]. \tag{31}$$

For an endpoint $q_t = z_t + \delta_t$, $z_t \odot q_t = z_t \odot z_t + z_t \odot \delta_t$. The controlled A–G grammar study distinguishes a matched displacement/product pair from changing only the product slot. If the product acts as a state–intent interaction, paired Direct scores obey the finite-capacity prediction

$$\Delta_{\text{match}} = (S_C - S_D) - (S_A - S_B) > 0, \qquad S_E - S_F > 0, \tag{32}$$

while $S_C > S_G$ rejects the explanation that centering the product alone is sufficient. The complete experiment is in Section D.

C.5 General endpoint conditioning

For heterogeneous tasks, a more general endpoint design may expose an optional relation r,

$$\mathcal{L}^\tau_{\text{act}}(\nu_\tau) = \mathbb{E}_{(z,a),(y,r)\sim\nu_\tau}[-\log p_{\eta_\tau}(a \mid z, y, r, a_{t-1})]. \tag{33}$$

The final raw-goal policy is not explicitly horizon-conditioned: training spans remaining offsets $r = 1, \ldots, 5$, but its input contains only the current state, raw goal displacement, interaction feature, and previous action.

Table A2: PushT intent-coordinate grammar under the matched three-seed protocol. Values are official Direct SR (%).

ID	Main condition	Interaction	Direct SR
A	q_t	$z_t \odot q_t$	78.78±0.96
B	q_t	omitted	80.56±0.96
G	q_t	$z_t \odot (q_t - z_t)$	79.67±1.76
C	$q_t - z_t$	$z_t \odot (q_t - z_t)$	83.11±1.39
D	$q_t - z_t$	omitted	81.00±1.45
F	$z_g - z_t$	omitted	83.22±0.51
E	$z_g - z_t$	$z_t \odot (z_g - z_t)$	**85.78±1.84**

D Design Selection and Ablation Logic

This appendix separates model definition from design selection. Each study is organized as a question, a matched intervention, and the conclusion used to fix the final configuration; none changes the headline evaluation protocol.

D.1 Which intent coordinate should the shared operator receive?

The waypoint endpoint q_t, centered waypoint displacement $q_t - z_t$, and full goal displacement $z_g - z_t$ successively remove an unnecessary absolute coordinate and the hand-set equal-progress scale. Product-off controls test whether the gain comes from the coordinate or its state-dependent interaction; G centers only the interaction while leaving the endpoint main effect intact. All cells use one full-data pass, three training seeds, the same evaluation episodes, and deterministic Math-SDPA.

The matched endpoint-to-centered change A→C is +4.33 points and the centered-to-full-goal change C→E is +2.67 points; A→E is +7.00 points (exact $p = 4.17 \times 10^{-9}$). The interaction prediction also closes quantitatively: the matched difference-in-differences is +3.89 points, $S_E - S_F = +2.56$, and $S_C - S_G = +3.44$. G shows that merely deleting the $z_t \odot z_t$ component is insufficient. The supported inductive bias is the matched intent-dependent sub-grammar $[z_t, m_t, z_t \odot m_t]$ within the full four-slot input; the shared action-history slot $A_\omega(a_{t-1})$ remains unchanged. Here m_t explicitly carries the intended displacement.

D.2 Does the result require trajectory straightening or a particular optimizer?

For the selected full-goal variants, re-encoding intervals 1/2/5 give 85.67/86.00/86.67% for E and 81.50/81.67/83.00% for F; the endpoint differences are not significant ($p = 0.392/0.253$). The waypoint curve is not a clean drift test because every reset also restarts its $1/h$ phase.

Joint goal-intent gradients reach 78.78%; detaching that branch reaches 79.67%. Adding a weight-0.01 strict-dynamic SmoothL1 chord penalty lowers validation chord MSE from 0.2173 to 0.2064 but gives 79.22% SR ($p = 0.764$). A stronger flexible linearity weight of 0.3 collapses Direct SR to 14.11%. Geometry can therefore become visibly straighter without becoming a better controller.

The waypoint SIGReg sweep gives 78.56/77.67/76.56% at 0.02/0.03/0.04 and 68.22% at 0.09. We use the stable low-

这没有施加欧氏轨迹惩罚

$$\mathcal{L}_{\text{lin}} = \sum_{k=1}^{H-1} \|z_{t+k} - [(1-\alpha_k)z_t + \alpha_k z_g]\|_2^2, \qquad \alpha_k = k/H, \tag{28}$$

其匹配负对照见附录 H。

C.3 替代端点采样与端到端的必要性

可以采样 $b \sim \text{Bernoulli}(p)$，每次转移使用一个端点，

$$y = (1-b)z^+ + b(z + m^{\text{goal}}), \tag{29}$$
$$\mathcal{L}_{\text{end}} = \mathbb{E}[-\log p_\eta(a \mid z, y, a_{t-1})]. \tag{30}$$

其期望为 $(1-p)\mathcal{L}_{z^+} + p\mathcal{L}_{\text{goal}}$，但配对评估两次调用会改变有限批次的梯度协方差。更重要的是，如果在相同意图下需要不同动作规律的观测满足 $E(o) = E(o')$，冻结的下游 actor 就无法消除这种混叠。让适当似然经由 E 反向传播，可在表征学习期间暴露缺失的可控差异。

C.4 端点特征语法与交互项预测

航点实验使用

$$h_t^{\text{end}}(y_t) = [z_t; y_t; z_t \odot y_t; A_\omega(a_{t-1})]. \tag{31}$$

对于端点 $q_t = z_t + \delta_t$，$z_t \odot q_t = z_t \odot z_t + z_t \odot \delta_t$。受控 A–G 特征语法实验区分匹配的位移/乘积组合与仅修改乘积槽位。如果乘积充当状态–意图交互项，那么配对的 Direct 得分应满足以下有限容量预测：

$$\Delta_{\text{match}} = (S_C - S_D) - (S_A - S_B) > 0, \qquad S_E - S_F > 0, \tag{32}$$

而 $S_C > S_G$ 排除了仅中心化乘积就足够的解释。完整实验见附录 D。

C.5 一般端点条件化

对于异构任务，更一般的端点设计可提供可选关系 r，

$$\mathcal{L}^\tau_{\text{act}}(\nu_\tau) = \mathbb{E}_{(z,a),(y,r)\sim\nu_\tau}[-\log p_{\eta_\tau}(a \mid z, y, r, a_{t-1})]. \tag{33}$$

最终的原始目标策略不显式以时域为条件：训练覆盖剩余偏移 $r = 1, \cdots, 5$，但输入只包含当前状态、原始目标位移、交互特征与上一动作。

表 A2：匹配的三训练种子协议下的 PushT 意图坐标特征语法。数值为官方 Direct 成功率（%）。

ID	主条件	交互	Direct 成功率
A	q_t	$z_t \odot q_t$	78.78±0.96
B	q_t	省略	80.56±0.96
G	q_t	$z_t \odot (q_t - z_t)$	79.67±1.76
C	$q_t - z_t$	$z_t \odot (q_t - z_t)$	83.11±1.39
D	$q_t - z_t$	省略	81.00±1.45
F	$z_g - z_t$	省略	83.22±0.51
E	$z_g - z_t$	$z_t \odot (z_g - z_t)$	**85.78±1.84**

D 设计选择与消融逻辑

本附录将模型定义与设计选择分开。每项实验均按问题、匹配干预和用于确定最终配置的结论组织；它们不改变主要评估协议。

D.1 共享算子应该接收哪种意图坐标?

航点端点 q_t、中心化航点位移 $q_t - z_t$ 和完整目标位移 $z_g - z_t$ 依次去掉了多余的绝对坐标及手工设定的等进度尺度。关闭乘积的对照检验增益来自坐标还是依赖状态的交互；G 只将交互项中心化，保留端点主效应。所有单元均完整遍历数据一次，使用三个训练种子、相同评估回合及确定性 Math-SDPA。

匹配端点到中心化变更 A→C 为 +4.33 个百分点，中心化到完整目标 C→E 为 +2.67，A→E 为 +7.00（精确 $p = 4.17 \times 10^{-9}$）。交互预测也定量闭合：匹配双重差分为 +3.89 个百分点，$S_E - S_F = +2.56$，以及 $S_C - S_G = +3.44$。G 表明仅删除 $z_t \odot z_t$ 分量不够。受证据支持的归纳偏置，是完整四槽输入内匹配的意图依赖子语法 $[z_t, m_t, z_t \odot m_t]$；共享动作历史槽 $A_\omega(a_{t-1})$ 不变。这里 m_t 显式携带期望位移。

D.2 结果是否需要轨迹直线化或特定优化器?

对于所选的完整目标变体，重编码间隔为 1/2/5 时，E 的成功率为 85.67/86.00/86.67%，F 为 81.50/81.67/83.00%；端点之间的差异不显著（$p = 0.392/0.253$）。航点曲线并不能单独检验漂移，因为每次重置也会重新启动其 $1/h$ 阶段。

联合目标意图梯度得 78.78%，分离该分支得 79.67%。加入权重 0.01 严格动力学 SmoothL1 弦惩罚，使验证弦 MSE 从 0.2173 降至 0.2064，成功率为 79.22%（$p = 0.764$）；较强灵活线性权重 0.3，使 Direct 跌至 14.11%。几何可以明显更直，却不成为更好控制器。

航点 SIGReg 扫描在 0.02/0.03/0.04 时分别得到 78.56/77.67/76.56%，在 0.09 时为 68.22%。我们使用稳定的低

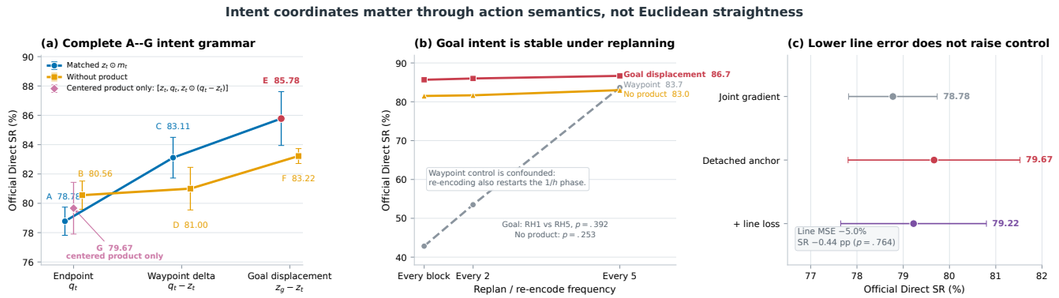


Figure A1: **Controls used to select the intent grammar.** (a) Coordinate and interaction interventions. (b) Re-encoding every 1, 2, or 5 action blocks. (c) Detaching the goal anchor and explicitly reducing chord error separate action-field learning from Euclidean latent straightening. Error bars are training-seed standard deviations.

regularization region rather than claim monotonic benefit. A single-seed Huber screening run obtains 55.0%, whereas the formal probabilistic NLL settings obtain 68.22±1.26% and 77.67± 0.88%; protocols are not paired, so this only rejects the claim that pointwise regression is already known to be preferable. Forward weights 1.0/0.3/0.1/0 give 77.67/73.11/59.56/25.22%, confirming that the action interface complements rather than replaces world prediction.

D.3 Why is Guarded A the optional verifier?

Figure A2 visualizes the parameter-selection audit; the full 576-job table is in Table A12. PushT benefits from a finite correction radius, rising from 85.78% Direct to 92.33% at $\sigma_0 = 0.25$, whereas Reacher degrades at larger radii. Guarded A retains the Direct reference and global best, reaches 96.86% macro with 384 sampled sequences, and is statistically near the simpler unguarded $\sigma_0 = 0.25$ rule at 96.78%. Doubling to $K = 256$ reaches 96.83% and provides no resolved gain. Hence Direct is the primary interface and Guarded A is a bounded verifier, not a replacement controller.

D.4 Qualitative multi-task geometry

The fixed-sample projections below illustrate collapse, partial expansion, and healthy SIGReg-supported spread. A spherical or shell-like t-SNE projection is not a success criterion: it is a nonlinear view of a high-dimensional cloud and is reported only to diagnose gross failure morphology.

Across these controls, the selection rule is consistent: preserve the world objective, expose a distance-preserving intent coordinate and its matched state interaction, learn both condition families through one probabilistic operator, and add search only as bounded local verification.

E Extended Controlled Results

This section preserves the complete protocol tables and selection evidence summarized in the main text. It contains no additional model selection based on test outcomes: the single-task grammar and verifier were locked before the final matrices, and every multi-task row uses the audited deterministic Math-SDPA contract.

E.1 Single-task evaluator and attribution details

The Forward Predictor remains necessary: weights 1.0/0.3/0.1/0 give PushT Direct SR 77.67/73.11/59.56/25.22%. Increasing the one-epoch peak learning rate to 7.5×10^{-4} gives 77.00± 2.03%; a second epoch gives 77.22±1.07%, versus 77.67±.88 at the selected setting. The second epoch raises effective rank from about 43.7 to 58.8 without raising SR, directly separating geometric spread from action-aligned representation quality.

E.2 Complete shared-encoder low-budget result

Waypoint INTACT is the strongest first-pass cell, whereas goal-displacement INTACT becomes the strongest E5 cell. This reversal is why the E1 table is a low-budget optimization diagnostic rather than the final objective ranking.

E.3 Matched E5 inference decomposition

E.4 Representation and conversion diagnostics

Across 45 formula-eligible checkpoints, pooled predicted–expert CKA/kNN correlations with official Direct SR are .897/.954; pointwise action R^2 is .815 and NLL is −.786. Leave-one-epoch-out values remain $[.888, .914]$ for CKA and $[.946, .963]$ for kNN. These analyses use one point per checkpoint, not one point per transition. Intent-only, inverse-only, and LeWM cells are excluded because a transition must evaluate both local and goal likelihoods through the same actor to instantiate the claimed family map.

Fixed-sample t-SNE is retained only as a qualitative diagnostic. Under SIGReg, healthy high-dimensional clouds can project to an interleaved shell; this appearance neither computes nor ranks SR. The quantitative analyses use original-space rank,

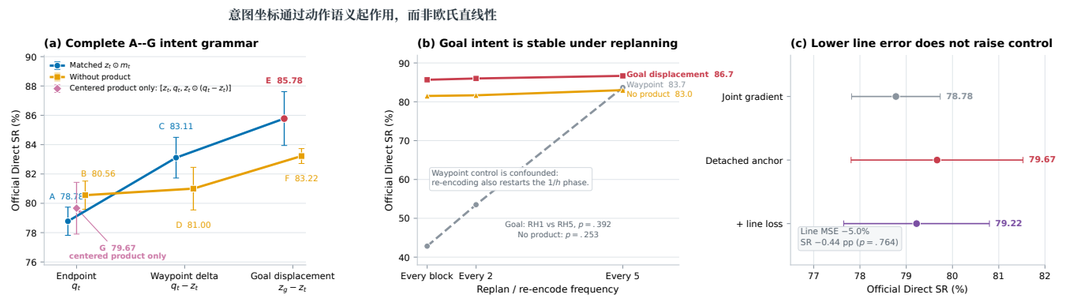


图 A1：用于选择意图特征语法的对照实验。(a) 坐标与交互项干预。(b) 每 1、2 或 5 个动作块重编码一次。(c) 对目标锚点停止梯度，并显式减小弦误差，用以区分动作场学习与潜空间的欧氏直线化。误差条为不同训练种子的标准差。

正则化区域，而非声称单调获益。单种子 Huber 筛选得 55.0%，正式概率 NLL 设置得 68.22±1.26%、 77.67± 0.88%；方案不配对，故只能否定逐点回归已知更优的说法。前向权重 1.0／0.3／0.1／0 的成功率 77.67／73.11／59.56／25.22%，确认动作接口补充而非替代世界预测。

D.3 为什么选择 Guarded A 作为可选验证器?

图 A2 展示参数选择审计，576 项完整表见 A12。PushT 受益于有限修正半径，从 Direct 85.78% 升至 $\sigma_0 = 0.25$ 时的 92.33%；Reacher 在较大半径退化。Guarded A 保留 Direct 参考与全局最优，用 384 条序列达到宏平均 96.86%，与更简单无保护 $\sigma_0 = 0.25$ 规则的 96.78% 统计上接近。翻倍至 $K = 256$ 得 96.83%，没有可分辨提升。因此 Direct 是主要接口，Guarded A 是有界验证器而非替代控制器。

D.4 多任务几何的定性分析

下面的固定样本投影展示坍塌、部分展开及 SIGReg 支持的健康分布。球形或壳状的 t-SNE 投影不是成功标准：它只是高维点云的非线性视图，仅用于诊断明显的失效形态。

这些对照给出一致的选择规则：保留世界目标，提供保留距离的意图坐标及匹配的状态交互，通过同一个概率算子学习两类条件，并且只将搜索作为有界的局部验证。

E 扩展受控结果

本节保留正文概述的完整协议表与选择证据，不包含额外的基于测试结果选择模型：单任务特征语法和验证器在最终矩阵之前已经锁定，各多任务行均遵循经过审计的确定性 Math-SDPA 约定。

E.1 单任务评估器与归因细节

前向预测器仍必要：权重 1.0／0.3／0.1／0 对应 PushT Direct 成功率 77.67／73.11／59.56／25.22%。提高单轮峰值学习率至 7.5×10^{-4} 得 77.00±2.03%；第二轮得 77.22±1.07%，所选设置为 77.67±.88。第二轮将有效秩约 43.7 提至 58.8，却未提高成功率，直接分离几何扩展与动作对齐表征质量。

E.2 完整共享编码器低预算结果

航点 INTACT 是首次遍历数据时最强的单元，而目标位移 INTACT 在 E5 时最强。这种排序反转说明 E1 表反映低预算下的优化情况，而非最终目标排名。

E.3 匹配的 E5 推理分解

E.4 表征与转换诊断

45 个公式适用检查点中，汇总预测–专家 CKA／k 近邻与官方 Direct 成功率相关为 .897/.954；逐点动作 R^2 为 .815，NLL 为 −.786。留一轮分析仍为 CKA $[.888, .914]$ 、k 近邻 $[.946, .963]$ 。每检查点一个点，而非每转移一个点。仅意图、仅逆及 LeWM 被排除，因为所声称族映射要求每转移经过同一 actor 评估局部和目标似然。

固定样本 t-SNE 仅作为定性诊断。使用 SIGReg 时，健康的高维点云可能投影为交错的壳；这种外观既不计算成功率，也不用于成功率排名。定量分析使用原空间中的秩、

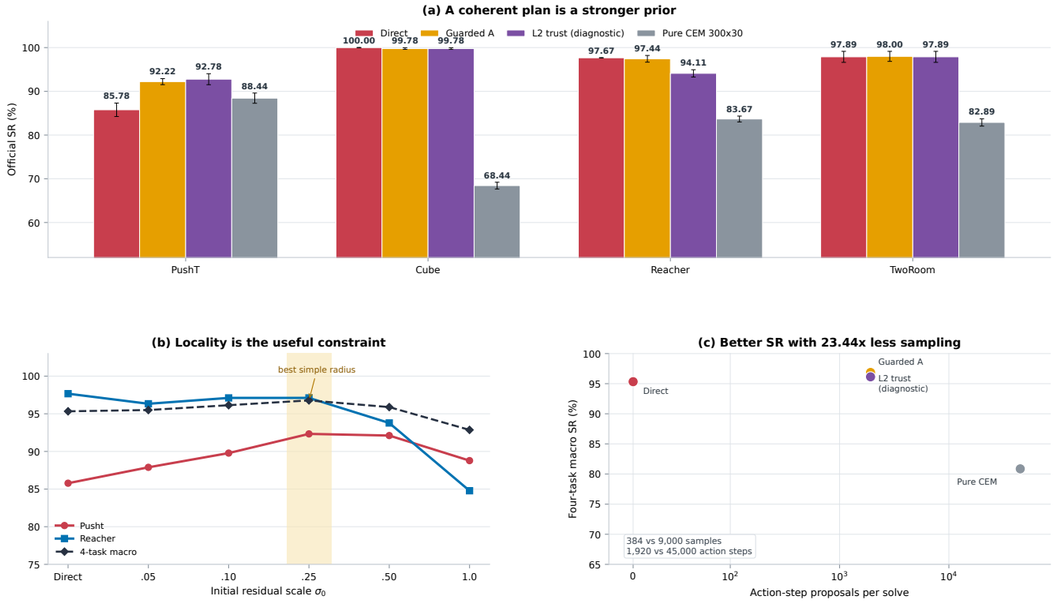


Figure A2: **Verifier-selection controls.** The complete three-seed audit compares Direct, guarded and unguarded plan-centered search, trust penalties, actor particles, and actor-disabled CEM.

Table A3: **Official versus CLEAR-LeWM v0.5.1 Moderate single-task SR (%).** Every row evaluates the same three one-epoch goal-displacement checkpoints per task; each checkpoint averages evaluation seeds 0/1/42 with 100 episodes per seed. Values are mean ± sample standard deviation across training seeds. Protocol scores are never pooled.

Protocol	Deployment	PushT	Cube	Reacher	TwoRoom	Macro
Official	Pure CEM 300×30	88.44±1.17	68.44±.77	83.67±.67	82.89±.84	80.86±.51
Official	Direct	85.78±1.54	100.00±.00	97.67±.00	97.89±1.26	95.33±.58
Official	**Guarded A 128×3**	**92.22±.69**	**99.78±.19**	**97.44±.77**	**98.00±1.15**	**96.86±.38**
CLEAR Moderate	Pure CEM 300×30	90.22±2.27	49.44±3.60	50.11±1.17	72.11±2.79	65.47±.77
CLEAR Moderate	Direct	88.22±.39	99.89±.19	49.56±2.52	95.67±2.33	83.33±1.00
CLEAR Moderate	**Guarded A 128×3**	**92.78±1.71**	**99.44±.20**	**51.67±3.18**	**95.44±2.22**	**84.83±1.32**

Table A4: PushT waypoint attribution chain. Pure CEM disables the intent-conditioned actor; probes are frozen and episode-disjoint. The final row also changes SIGReg and is not an intent-only attribution.

Training	Direct	CEM 30×10	State R^2	Action R^2	Rank
LeWM	–	42.22±9.50	.616±.008	.673±.062	63.7±.9
+ physical inverse	35.78±.69	57.67±2.52	.652±.010	.791±.003	67.0±.8
+ matched waypoint intent	68.22±1.26	61.44±2.04	.658±.003	.786±.003	67.0±.4
Waypoint INTACT, SIGReg 0.03	**77.67±.88**	**69.44±1.64**	**.680±.005**	**.804±.002**	43.7±.6

standard deviation, forward error, probes, and predicted–expert action-family relations.

E.5 Task-conditioned gauge controls

Correct pairing raises CLEAR Moderate SR from 9.46% shuffled to 68.04%, a +58.58 point gain (8-cluster bootstrap 95% CI $[48.83, 67.58]$), positive for all 24 protocol units. Calibration with 4/16/64/full episodes gives 26.67/56.00/64.42/64.83% SR. Same-objective cross-seed swaps define the alignment ceiling; LeWM and inverse backbones aligned into a trained Full actor are positive alignability controls. In reverse, Full-backbone-to-LeWM-actor stays at 5.08%, and the inverse actor recovers only inverse-level function.

Cross-task and leave-one-task-out actor-output transfer remain negative, while pooled calibration reaches only .484/.539 R^2. The shared visual backbone can therefore support several task-local charts; the task-specific heads do not require one global canonical action space.

F Actor-Sharing Mechanism Audit

The closed-loop result is paired with mechanism measurements on fixed PushT waypoint batches. "Shared fraction" is the fraction of action-network parameters touched by both local and

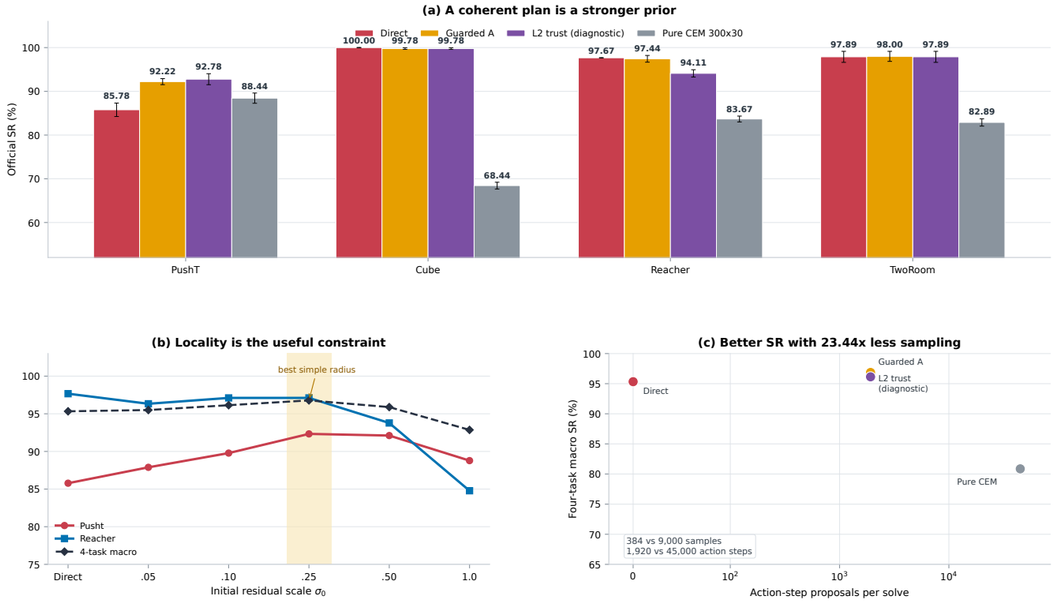


图 A2：验证器选择对照。完整的三训练种子审计比较 Direct、带守护及不带守护的计划中心搜索、信赖域惩罚、actor 粒子和禁用 actor 的 CEM。

表 A3：官方协议与 CLEAR-LeWM v0.5.1 Moderate 的单任务成功率（%）。每行评估各任务相同的三个单轮目标位移检查点；每个检查点对评估种子 0/1/42 求平均，每个种子 100 回合。数值为训练种子的均值 ± 样本标准差，不混合不同协议的分数。

协议	部署	PushT	Cube	Reacher	TwoRoom	宏平均
官方	纯 CEM 300×30	88.44±1.17	68.44±.77	83.67±.67	82.89±.84	80.86±.51
官方	Direct	85.78±1.54	100.00±.00	97.67±.00	97.89±1.26	95.33±.58
官方	**Guarded A 128×3**	**92.22±.69**	**99.78±.19**	**97.44±.77**	**98.00±1.15**	**96.86±.38**
CLEAR Moderate	纯 CEM 300×30	90.22±2.27	49.44±3.60	50.11±1.17	72.11±2.79	65.47±.77
CLEAR Moderate	Direct	88.22±.39	99.89±.19	49.56±2.52	95.67±2.33	83.33±1.00
CLEAR Moderate	**Guarded A 128×3**	**92.78±1.71**	**99.44±.20**	**51.67±3.18**	**95.44±2.22**	**84.83±1.32**

表 A4：PushT 航点归因链。纯 CEM 禁用意图条件 actor；探针冻结且回合互不重叠。最后一行还改变 SIGReg，因此不能把变化仅归因于意图。

训练	Direct	CEM 30×10	状态 R^2	动作 R^2	Rank
LeWM	–	42.22±9.50	.616±.008	.673±.062	63.7±.9
+ 物理逆向	35.78±.69	57.67±2.52	.652±.010	.791±.003	67.0±.8
+匹配的航点意图	68.22±1.26	61.44±2.04	.658±.003	.786±.003	67.0±.4
航点 INTACT，SIGReg 0.03	**77.67±.88**	**69.44±1.64**	**.680±.005**	**.804±.002**	43.7±.6

标准差、前向误差、探针及预测–专家动作族关系。

E.5 以任务为条件的坐标规范对照

正确配对使 CLEAR Moderate 成功率从打乱配对时的 9.46% 提高到 68.04%，增益为 +58.58 个百分点（8 簇 bootstrap 的 95% 置信区间为 $[48.83, 67.58]$），且在全部 24 个协议单元中均为正。校准使用 4／16／64／完整回合，成功率为 26.67／56.00／64.42／64.83%。同目标跨种子交换定义对齐上限，LeWM 和逆骨干对齐至完整 actor 为正向可对齐对照。反向完整骨干到 LeWM actor 保持 5.08%，逆 actor 仅恢复逆水平功能。

跨任务及留一任务 actor 输出迁移仍呈负结果，合并校准也仅达到 .484/.539R^2。因此，共享视觉骨干可支持多个任务局部坐标图；任务专用头部不要求一个全局规范动作空间。

F actor 共享机制审计

闭环结果配有固定 PushT 航点批次上的机制测量。“Shared fraction”是同时受局部与

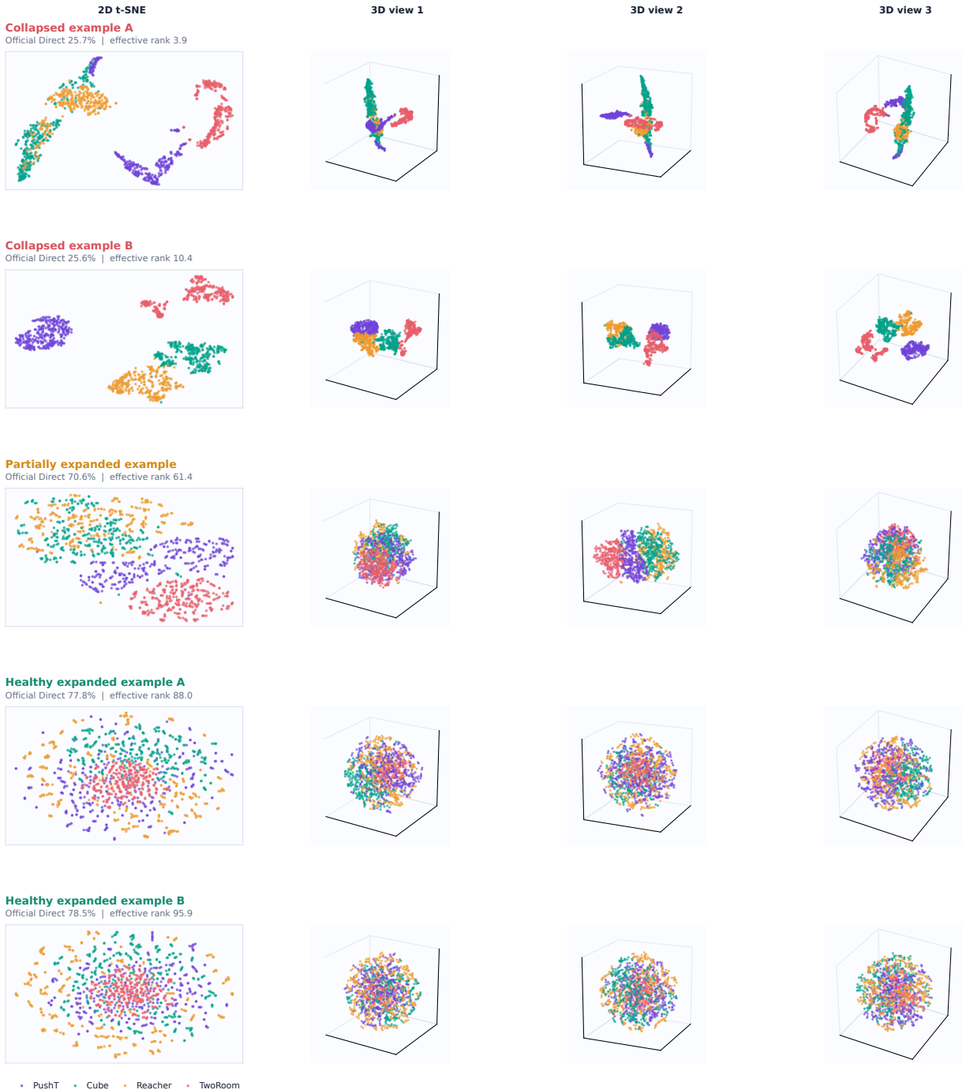


Figure A3: Exploratory E5 collapse morphologies from fixed samples. These legacy non-Math-SDPA runs are qualitative diagnostics and do not enter performance ranking or headline correlations.

deployment losses. Gradient cosine is computed in the complete action-network parameter coordinates; it is zero by construction when the two actors are disjoint. Shuffle response is the normalized-action MAE between the original goal-conditioned output and the output after permuting goals across the batch. Consistency is $\|F_\phi(z_t, \mu_\eta(z_t, q_t)) - q_t\|_2^2$.

The fully shared model receives positively aligned physical/deployment gradients (cosine 0.575). As sharing is weak-

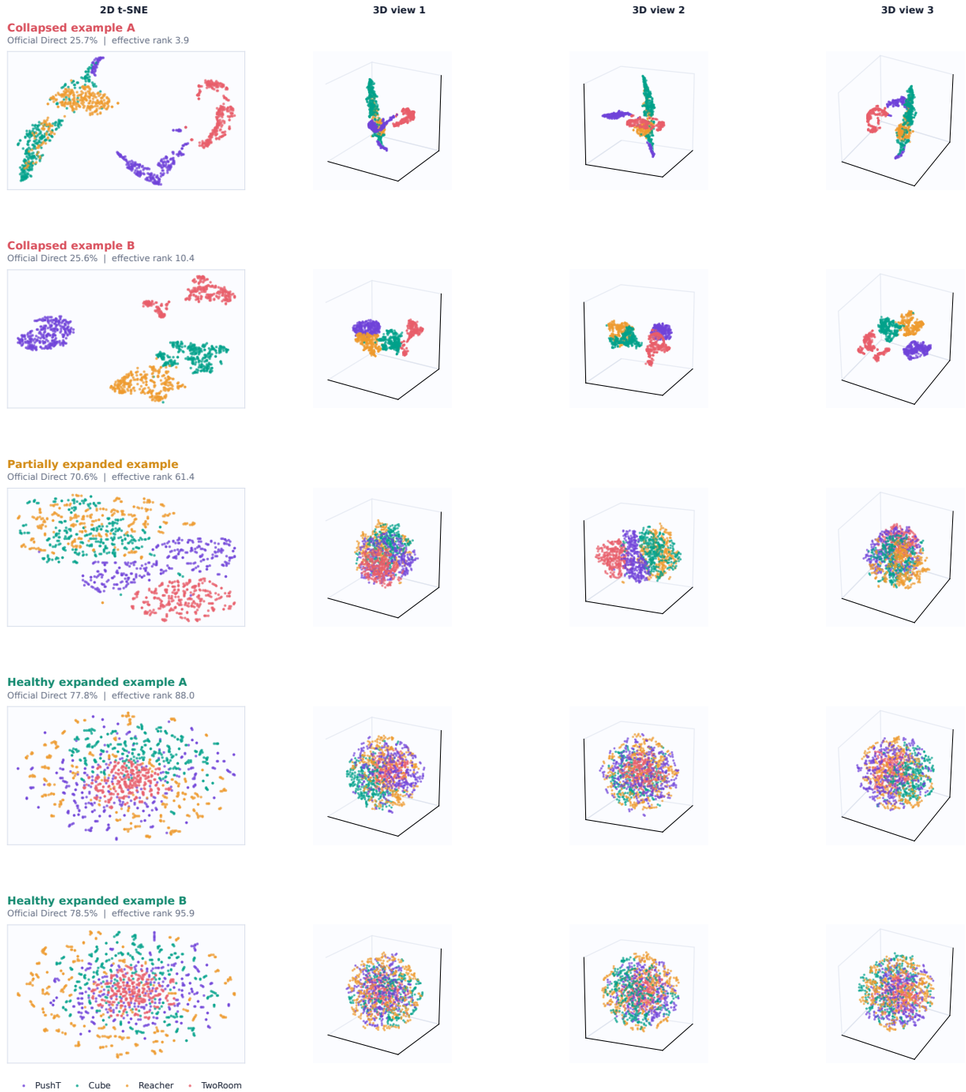


图 A3：固定样本中的探索性 E5 坍塌形态。这些历史非 Math-SDPA 运行仅用于定性诊断，不参与性能排名或主要相关分析。

部署损失。梯度余弦在完整动作网络的参数坐标中计算；若两个 actor 不相交，则按构造必然为零。打乱响应是原始目标条件

输出与在批次内置换目标后输出之间的归一化动作 MAE。一致性为 $\|F_\phi(z_t, \mu_\eta(z_t, q_t)) - q_t\|_2^2$。

完全共享模型的物理/部署梯度正向对齐（余弦为 0.575）。随着共享程度减弱，

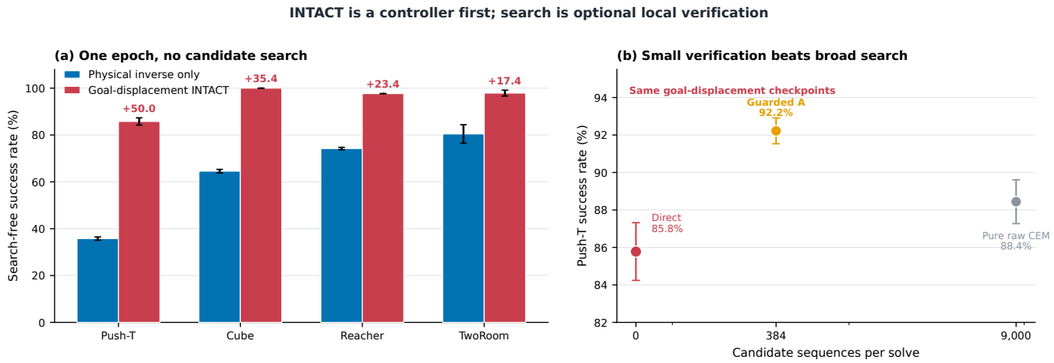


Figure A4: **Search-free control and optional verification.** Left: physical-successor inversion versus goal-displacement INTACT after one epoch. Right: Direct uses no candidate sequence, Guarded A locally verifies 384, and broad actor-disabled CEM evaluates 9,000. More search is not automatically better once the action conditional is aligned.

Table A5: **Controlled Math-SDPA shared-encoder E1 result.** Values are mean ± sample standard deviation over three training seeds; each seed averages official evaluation seeds $\{0, 1, 42\}$ with 100 episodes each. LeWM has no INTACT Predictor and uses CEM 300×30; other rows use Direct.

Training cell	Evaluation	n	PushT	Cube	Reacher	TwoRoom	Macro
LeWM	CEM 300×30	3	17.00±7.17	52.89±3.95	**25.00±4.26**	34.33±1.76	32.31±2.34
Inverse only	Direct	3	24.89±5.97	48.22±3.83	9.44±3.01	51.33±13.32	33.47±3.98
Waypoint intent only	Direct	3	17.67±2.40	68.78±5.50	11.89±2.78	44.67±31.13	35.75±7.58
Goal intent only	Direct	3	24.00±4.36	**73.22±3.67**	11.56±2.04	48.22±17.20	39.25±5.40
Waypoint INTACT	**Direct**	3	**30.56±9.34**	69.89±10.99	10.33±5.03	**60.00±10.58**	**42.69±7.19**
Goal INTACT	Direct	3	26.89±6.26	61.44±10.46	10.33±3.18	48.67±21.31	36.83±5.63

ened, gradient coordination and SR decline together. Across the six designs, shuffle-goal response has exploratory Spearman $\rho = 0.943$ with SR ($n = 6$), indicating that stronger controllers use the goal condition more actively; the small design count precludes a universal claim. Two negative controls are especially informative. Goal-intent-only has the lowest consistency MSE but the worst SR, so self-consistency with the Forward Predictor does not establish physical correctness. Conversely, the double-capacity independent model obtains real/deployment NLLs of $-1.347/-1.160$ yet reaches only 72.89% SR. Separately fitting both likelihoods is therefore not equivalent to learning one deployable inverse map.

G Exploratory Diagnostic Definitions and Correlations

For centered latent samples $Z \in \mathbb{R}^{n\times D}$ with normalized covariance eigenvalues $\bar{\lambda}_j$, entropy effective rank is

$$d_{\text{eff}} = \exp\left(-\sum_j \bar{\lambda}_j \log(\bar{\lambda}_j + \epsilon)\right). \tag{34}$$

Let $r_i = \|Z_i - \bar{Z}\|_2$, let CV_r be the coefficient of variation of these radii, and let $\text{CV}_{\chi(d)}$ be that of a chi distribution with d degrees of freedom. We define

$$S_{\text{rank}} = d_{\text{eff}}/D, \tag{35}$$

$$S_{\text{shell}} = \exp\left[-\left|\log \frac{\text{CV}_r}{\text{CV}_{\chi(d_{\text{eff}})}}\right|\right], \tag{36}$$

$$\text{SRS} = 100\sqrt{S_{\text{rank}} S_{\text{shell}}}. \tag{37}$$

SRS is high when the spectrum uses many dimensions and sample radii resemble the Gaussian shell expected at that effective dimension; it does not measure what information those dimensions encode.

For the actor diagnostic, write $(\mu_t^{\text{dep}}, \sigma_t^{\text{dep}})$ for the Gaussian under the deployed goal condition and $(\tilde{\mu}_t, \tilde{\sigma}_t)$ for a deterministic different-goal condition on the same current state. The reported condition response is the per-sample RMS standardized mean shift

$$G_{\text{sep}} = \mathbb{E}_t\left[\left(\frac{1}{d_a}\sum_{j=1}^{d_a} \frac{(\mu_{t,j}^{\text{dep}} - \tilde{\mu}_{t,j})^2}{\frac{1}{2}[(\sigma_{t,j}^{\text{dep}})^2 + \tilde{\sigma}_{t,j}^2] + \epsilon}\right)^{1/2}\right]. \tag{38}$$

Thus G_{sep} asks whether the actor uses its condition relative to its own predictive uncertainty. It does not ask whether the changed goal requires a different expert action, nor whether the response has the correct direction.

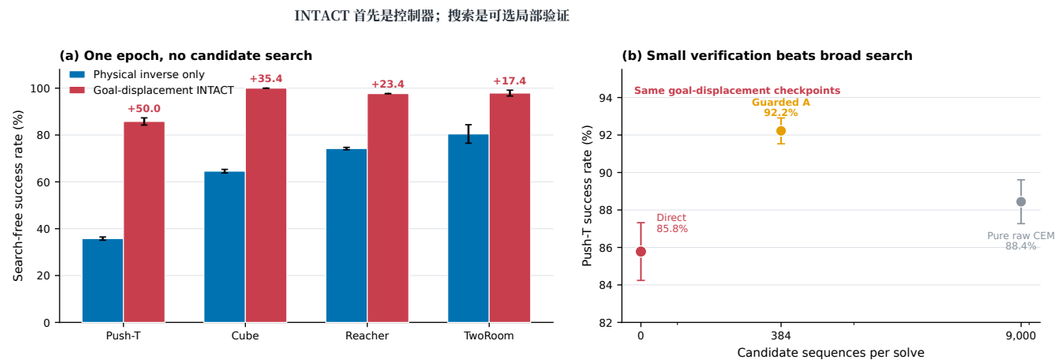


图 A4：无需搜索的控制与可选验证。左：训练一轮后，物理后继求逆与目标位移 INTACT 的比较。右：Direct 不使用候选序列，Guarded A 在局部验证 384 条，广域的禁用 actor 的 CEM 评估 9,000 条。当动作条件模型已经对齐时，更多搜索并不必然更好。

表 A5：受控 Math-SDPA 共享编码器 E1 结果。数值为三个训练种子的均值 ± 样本标准差；每个训练种子对官方评估种子 $\{0, 1, 42\}$ 求平均，每个评估种子 100 回合。LeWM 没有 INTACT Predictor，使用 CEM 300×30；其他行使用 Direct。

训练单元	评估	n	PushT	Cube	Reacher	TwoRoom	宏平均
LeWM	CEM 300×30	3	17.00±7.17	52.89±3.95	**25.00±4.26**	34.33±1.76	32.31±2.34
仅逆向	Direct	3	24.89±5.97	48.22±3.83	9.44±3.01	51.33±13.32	33.47±3.98
仅航点意图	Direct	3	17.67±2.40	68.78±5.50	11.89±2.78	44.67±31.13	35.75±7.58
仅目标意图	Direct	3	24.00±4.36	**73.22±3.67**	11.56±2.04	48.22±17.20	39.25±5.40
航点 INTACT	**Direct**	3	**30.56±9.34**	69.89±10.99	10.33±5.03	**60.00±10.58**	**42.69±7.19**
目标 INTACT	Direct	3	26.89±6.26	61.44±10.46	10.33±3.18	48.67±21.31	36.83±5.63

减弱时，梯度协调与成功率一起下降。六种设计中，打乱目标响应与成功率的探索性 Spearman 为 $\rho = 0.943$（$n = 6$），说明较强控制器更主动使用目标条件；设计数少，不能作普遍断言。两项负对照尤其有用：仅目标意图的一致性 MSE 最低、成功率最差，因此与前向预测器自一致不能证明物理正确。反之，双容量独立模型实际／部署 NLL 为 $-1.347/-1.160$，成功率却仅 72.89%。分别拟合两个似然，不等于学习一个可部署逆映射。

G 探索性诊断定义与相关性

对于中心化潜变量样本 $Z \in \mathbb{R}^{n\times D}$，归一化协方差特征值为 $\bar{\lambda}_j$，熵有效秩为

$$d_{\text{eff}} = \exp\left(-\sum_j \bar{\lambda}_j \log(\bar{\lambda}_j + \epsilon)\right). \tag{34}$$

设 $r_i = \|Z_i - \bar{Z}\|_2$，$\text{CV}_r$ 为这些半径的变异系数，$\text{CV}_{\chi(d)}$ 为具有 d 的卡方分布所对应的卡分布变异系数自由度。我们定义

$$S_{\text{rank}} = d_{\text{eff}}/D, \tag{35}$$

$$S_{\text{shell}} = \exp\left[-\left|\log \frac{\text{CV}_r}{\text{CV}_{\chi(d_{\text{eff}})}}\right|\right], \tag{36}$$

$$\text{SRS} = 100\sqrt{S_{\text{rank}} S_{\text{shell}}}. \tag{37}$$

当谱利用较多维度、样本半径类似该有效维度下预期的高斯壳时，SRS 较高；它不衡量这些维度编码了什么信息。

在 actor 诊断中，用 $(\mu_t^{\text{dep}}, \sigma_t^{\text{dep}})$ 表示部署目标条件下的高斯分布，用 $(\tilde{\mu}_t, \tilde{\sigma}_t)$ 表示相同当前状态下确定性选择的另一目标条件。报告的条件响应为每个样本的均方根标准化均值位移：

$$G_{\text{sep}} = \mathbb{E}_t\left[\left(\frac{1}{d_a}\sum_{j=1}^{d_a} \frac{(\mu_{t,j}^{\text{dep}} - \tilde{\mu}_{t,j})^2}{\frac{1}{2}[(\sigma_{t,j}^{\text{dep}})^2 + \tilde{\sigma}_{t,j}^2] + \epsilon}\right)^{1/2}\right]. \tag{38}$$

因此 G_{sep} 检验 actor 相对于自身预测不确定性是否使用了条件。它不检验改变目标是否需要不同专家动作，也不检验响应方向是否正确。

Table A6: **Matched E5 inference decomposition.** Values are mean ± sample standard deviation over three training seeds. Panel (a) removes all learned action heads; panel (b) locally verifies the learned plan; panel (c) uses CLEAR Moderate with each cell's native interface.

(a) Official actor-disabled pure CEM 300×30

Training cell	PushT	Cube	Reacher	TwoRoom	Macro
LeWM	74.56±3.67	67.33±1.86	83.11±.96	39.67±8.39	66.17±2.67
Inverse only	75.78±1.02	66.78±2.36	81.78±1.64	51.44±1.07	68.94±.43
Waypoint intent only	78.44±4.72	68.67±1.76	83.22±2.69	39.11±8.17	67.36±4.06
Waypoint INTACT	78.22±3.02	69.56±2.80	83.00±2.19	46.44±3.34	69.31±.68
Goal intent only	**80.78±2.36**	68.00±3.18	**84.67±1.33**	40.11±8.30	68.39±2.22
Goal INTACT	78.11±1.17	**70.33±3.00**	80.33±2.03	**51.56±4.53**	**70.08±1.13**

(b) Official Guarded A 128×3

Training cell	PushT	Cube	Reacher	TwoRoom	Macro
Inverse only	55.67±4.10	65.89±1.02	78.67±1.20	66.33±6.08	66.64±.88
Waypoint intent only	74.67±3.79	**99.78±.19**	87.78±1.07	74.78±10.63	84.25±3.58
Waypoint INTACT	83.22±2.36	99.11±.51	85.33±.58	74.89±7.15	85.64±2.04
Goal intent only	80.44±.96	99.33±.33	94.67±.33	69.33±5.93	85.94±1.57
Goal INTACT	**85.00±2.85**	99.00±.00	**97.89±.69**	**80.00±4.41**	**90.47±.84**

(c) CLEAR-LeWM v0.5.1 Moderate

Training cell	PushT	Cube	Reacher	TwoRoom	Macro
LeWM (pure CEM)	78.78±2.52	51.33±1.20	46.00±4.06	33.33±13.86	52.36±2.97
Inverse only	32.67±.88	42.67±1.67	31.67±2.33	70.11±2.17	44.28±1.33
Waypoint intent only	60.89±2.91	**99.78±.19**	35.78±7.58	67.11±12.01	65.89±2.79
Waypoint INTACT	74.78±4.34	99.11±.84	32.22±5.43	73.33±3.28	69.86±1.06
Goal intent only	73.00±3.18	**100.00±.00**	39.44±2.71	60.67±5.51	68.28±1.17
Goal INTACT	**84.33±2.00**	99.11±1.54	**47.44±2.91**	**79.33±5.03**	**77.56±.39**

Table A7: **Controlled shared-encoder representation snapshots.** $\bar{\sigma}$ is mean per-dimension latent standard deviation, cosine is mean pairwise cosine, and action R^2 is an episode-disjoint frozen probe.

E1

Training cell	n	Rank	$\bar{\sigma}$	Cos.	Act. R^2
LeWM	3	25.53	.830	.554	.309
Inverse only	3	24.03	.902	.590	.372
Waypoint intent only	3	24.83	.846	.582	.328
Goal intent only	3	24.45	.839	.604	.349
Waypoint INTACT	3	**25.73**	.877	.587	**.383**
Goal INTACT	3	25.34	.888	.616	.382

E5

Training cell	n	Rank	$\bar{\sigma}$	Cos.	Act. R^2
LeWM	3	**94.06**	.964	.039	.366
Inverse only	3	85.24	.956	.087	.401
Waypoint intent only	3	93.87	.959	.049	.393
Goal intent only	3	93.47	.965	.047	.390
Waypoint INTACT	3	87.17	.960	.080	.406
Goal INTACT	3	89.26	.958	.081	**.409**

We pair rank, SRS, and G_{sep} with forward MSE, deployment NLL/RMSE, symmetric KL between action conditionals, and shuffled-goal ΔNLL. Every method uses the same 64 sequences per task. Correlations first subtract the mean within each task and then pool the 56 method–task cells. These endpoint runs predate the controlled Math-SDPA factorial. Their matched sampling makes the geometric and conversion metrics informative as exploratory mechanism diagnostics, but their SR values are not used to rank training objectives in the main performance comparison.

SRS correlates with CLEAR v0.3 E5 Moderate Direct SR at Pearson $r = 0.978$ and Spearman $\rho = 0.652$, and with Strict Direct SR at $r = 0.977$ and $\rho = 0.809$, over the ten semantic endpoint configurations. This smaller analysis is useful for geometry auditing but is not treated as a universal threshold.

H Single-Task Latent-Linearity Stress Test

Table A11 is a separately trained four-frame PushT isolation, not a row from the controlled multi-task factorial. Every model uses full data for one epoch, learning rate 10^{-4}, batch size 256, physical-inverse weight 0.1, and SIGReg 0.09; only the latent-linearity term changes. Each cell pools three training seeds (3072–3074), three evaluation seeds (0, 1, 42), and 100 episodes per pair, for 900 episodes. The flexible penalty permits a learned local adjustment, whereas the strict penalty directly targets Euclidean waypoints. Machine-readable values come from experiment `linearity4_exact_isolation_pusht_e1`, specifically the Direct and pure-CEM summary artifacts.

The strict losses reduce latent waypoint error but do not improve either Direct or actor-disabled planning. At flexible weight 0.30, effective rank falls to 41.96 and the frozen action probe falls to $R^2 = 0.111$. This negative control is why INTACT uses the Euclidean construction only to form an actor condition and never as a target for the encoder trajectory. Its 34.00% control should not be compared numerically to the 35.78% single-task physical-inverse row in Table 2: the stress test uses an independently trained seed set and a four-frame isolation protocol.

I Historical Latent-Perturbation Diagnostic

Before INTACT, we tested whether replacing the deterministic latent predictor with five-step flow matching improved LeWM.

表 A6：匹配的 E5 推理分解。数值为三个训练种子的均值 ± 样本标准差。(a) 移除全部学习得到的动作头部；(b) 在局部验证学习所得计划；(c) 在 CLEAR Moderate 下使用各单元的原生接口。

(a) 官方禁用 actor 的纯 CEM 300×30

训练单元	PushT	Cube	Reacher	TwoRoom	宏平均
LeWM	74.56±3.67	67.33±1.86	83.11±.96	39.67±8.39	66.17±2.67
仅逆向	75.78±1.02	66.78±2.36	81.78±1.64	51.44±1.07	68.94±.43
仅航点意图	78.44±4.72	68.67±1.76	83.22±2.69	39.11±8.17	67.36±4.06
航点 INTACT	78.22±3.02	69.56±2.80	83.00±2.19	46.44±3.34	69.31±.68
仅目标意图	**80.78±2.36**	68.00±3.18	**84.67±1.33**	40.11±8.30	68.39±2.22
目标 INTACT	78.11±1.17	**70.33±3.00**	80.33±2.03	**51.56±4.53**	**70.08±1.13**

(b) 官方 Guarded A 128×3

训练单元	PushT	Cube	Reacher	TwoRoom	宏平均
仅逆向	55.67±4.10	65.89±1.02	78.67±1.20	66.33±6.08	66.64±.88
仅航点意图	74.67±3.79	**99.78±.19**	87.78±1.07	74.78±10.63	84.25±3.58
航点 INTACT	83.22±2.36	99.11±.51	85.33±.58	74.89±7.15	85.64±2.04
仅目标意图	80.44±.96	99.33±.33	94.67±.33	69.33±5.93	85.94±1.57
目标 INTACT	**85.00±2.85**	99.00±.00	**97.89±.69**	**80.00±4.41**	**90.47±.84**

(c) CLEAR-LeWM v0.5.1 Moderate

训练单元	PushT	Cube	Reacher	TwoRoom	宏平均
LeWM（纯 CEM）	78.78±2.52	51.33±1.20	46.00±4.06	33.33±13.86	52.36±2.97
仅逆向	32.67±.88	42.67±1.67	31.67±2.33	70.11±2.17	44.28±1.33
仅航点意图	60.89±2.91	**99.78±.19**	35.78±7.58	67.11±12.01	65.89±2.79
航点 INTACT	74.78±4.34	99.11±.84	32.22±5.43	73.33±3.28	69.86±1.06
仅目标意图	73.00±3.18	**100.00±.00**	39.44±2.71	60.67±5.51	68.28±1.17
目标 INTACT	**84.33±2.00**	99.11±1.54	**47.44±2.91**	**79.33±5.03**	**77.56±.39**

表 A7：受控共享编码器表征快照。$\bar{\sigma}$ 为各维潜变量标准差的均值，cosine 为成对余弦均值，action R^2 为回合互不重叠的冻结探针。

E1

训练单元	n	Rank	$\bar{\sigma}$	Cos.	动作 R^2
LeWM	3	25.53	.830	.554	.309
仅逆向	3	24.03	.902	.590	.372
仅航点意图	3	24.83	.846	.582	.328
仅目标意图	3	24.45	.839	.604	.349
航点 INTACT	3	**25.73**	.877	.587	**.383**
目标 INTACT	3	25.34	.888	.616	.382

E5

训练单元	n	Rank	$\bar{\sigma}$	Cos.	动作 R^2
LeWM	3	**94.06**	.964	.039	.366
仅逆向	3	85.24	.956	.087	.401
仅航点意图	3	93.87	.959	.049	.393
仅目标意图	3	93.47	.965	.047	.390
航点 INTACT	3	87.17	.960	.080	.406
目标 INTACT	3	89.26	.958	.081	**.409**

将秩、SRS、G_{sep} 与前向 MSE、部署 NLL／RMSE、动作条件间对称 KL、打乱目标 ΔNLL 配对。各方法每任务均用相同 64 条序列。相关分析先减去任务内均值，再汇总 56 个方法–任务组合。这些端点实验早于受控 Math-SDPA 析因；匹配采样使几何和转换指标可作探索机制诊断，但成功率不用于主要比较的训练目标排名。

在十种语义端点配置上，SRS 与 CLEARv0.3 E5 Moderate Direct 成功率的 Pearson、Spearman 相关系数分别为 $r = 0.978$、$\rho = 0.652$；与 Strict Direct 成功率的相关系数分别为 $r = 0.977$、$\rho = 0.809$。这项较小规模的分析有助于审查几何结构，但不应视为通用阈值。

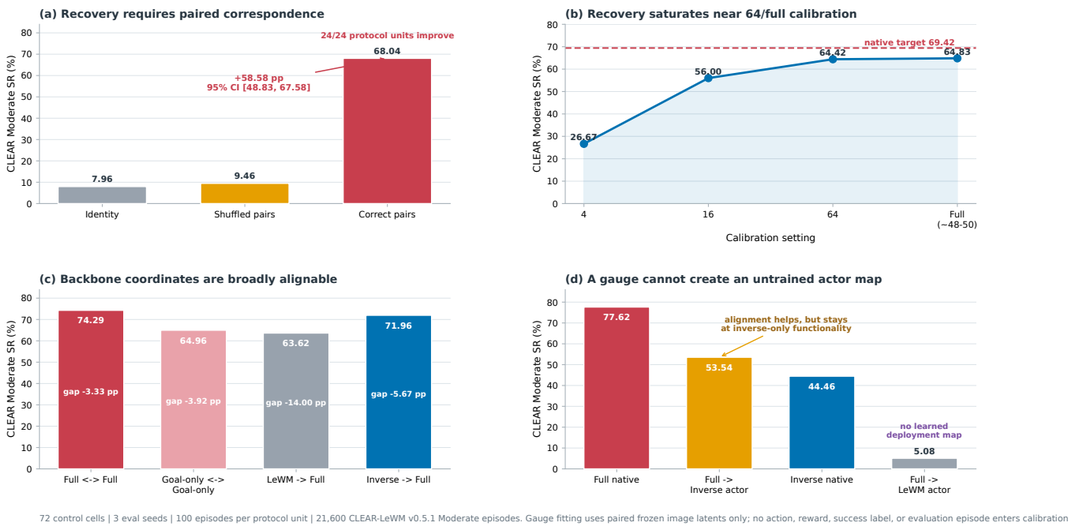
H 单任务潜空间线性化压力测试

表 A11 是单独训练的四帧 PushT 隔离实验，并非受控多任务析因实验中的一行。所有模型使用全部数据训练一轮，学习率 10^{-4}，批量大小 256，物理逆向权重 0.1，SIGReg 权重 0.09；仅潜空间线性项变化。每格汇总三个训练种子（3072–3074）、三个评估种子（0、1、42）、每对 100 回合，共 900 回合。灵活惩罚允许学习局部调整，严格惩罚直接以欧氏航点为目标。机器可读值来自实验 linearity4_exact_isolation_pusht_e1 的 Direct 和纯 CEM 汇总文件。

严格损失减少潜在航点误差，却不改善 Direct 或关闭 actor 的规划。灵活权重 0.30 时有效秩降至 41.96，冻结动作探测器降至 $R^2 = 0.111$。这一负对照解释为何只用欧氏构造形成 actor 条件，从不将其作为编码器轨迹目标。其 34.00% 控制不能与表 2 单任务物理逆 35.78% 数值比较：压力测试使用独立训练种子组及四帧隔离方案。

I 历史潜变量扰动诊断

在提出 INTACT 之前，我们测试过将确定性潜变量预测器替换为五步流匹配能否改善 LeWM。



72 control cells | 3 eval seeds | 100 episodes per protocol unit | 21,600 CLEAR-LeWM v0.5.1 Moderate episodes. Gauge fitting uses paired frozen image latents only; no action, reward, success label, or evaluation episode enters calibration

Figure A5: **Gauge controls isolate correspondence from actor functionality.** Correctly paired calibration restores control; recovery saturates near 64/full episodes; same-objective swaps define a ceiling; reverse swaps show that a coordinate map cannot create an untrained deployment conditional. The 72-cell audit contains 21,600 CLEAR Moderate episodes and excludes all evaluation episodes from calibration.

Table A8: **Sharing is the operative inductive bias.** Matched PushT waypoint Direct SR (%). Except for the explicit double-capacity control, parameter counts match within 0.12%. p is an episode-paired exact McNemar comparison with the fully shared model.

Action-conditional design	Params (M)	Direct SR	Δ vs shared	Paired p
Waypoint intent only	3.110	65.56±1.35	−11.33	4.23×10^{-15}
Independent, parameter matched	3.114	71.22±2.27	−5.67	1.15×10^{-5}
Independent, double capacity	6.220	72.89±2.17	−4.00	0.00362
Shared trunk, two outputs	3.110	74.67±1.53	−2.22	0.0927
Condition-token shared	3.111	77.11±2.12	+0.22	0.923
Fully shared (default)	3.110	76.89±.69	0	–

On PushT, the deterministic model reached 26%, 86%, 94%, and 96% official SR at E1, E5, E10, and E15. The best x-prediction/loss-x flow model reached 4%, 76%, 88%, and 92%; a second flow run with matched SIGReg 0.09 reached 4%, 44%, 72%, and 76%. At matched 139,319 optimizer steps, the strongest flow run used 0.568 seconds/step versus 0.479 for the deterministic predictor, an 18.4% increase.

This historical control is deliberately not a headline ablation: the strongest flow run changes SIGReg from 0.09 to 0.03, and there is no matched RGB-noise control. It supports only the narrower observation that perturbing an already compressed latent did not improve low-budget learning in these runs. This is consistent with GC-IDM's independent latent-noise ablation, whose taskwise optimum occurs at zero noise [34]. It does not establish that latent diffusion is universally harmful or that it is directly comparable to pixel-space diffusion.

J Complete Goal-Displacement Planner Matrix

Table A12 reports every completed planner cell used in Fig. A2. Each method contains 36 jobs: three training checkpoints per task, each evaluated with seeds $\{0, 1, 42\}$ and 100 episodes. Macro is the unweighted task mean. The 12 base rules contribute 432 jobs and the four guarded rules contribute 144, each under a separate PASS audit.

K Episode-Disjoint Anti-Retrieval Audit

The fixed PushT manifest contains 14,948/1,868/1,869 train/validation/test episodes with zero overlap. Normalization, optimization, and checkpoint selection exclude the test

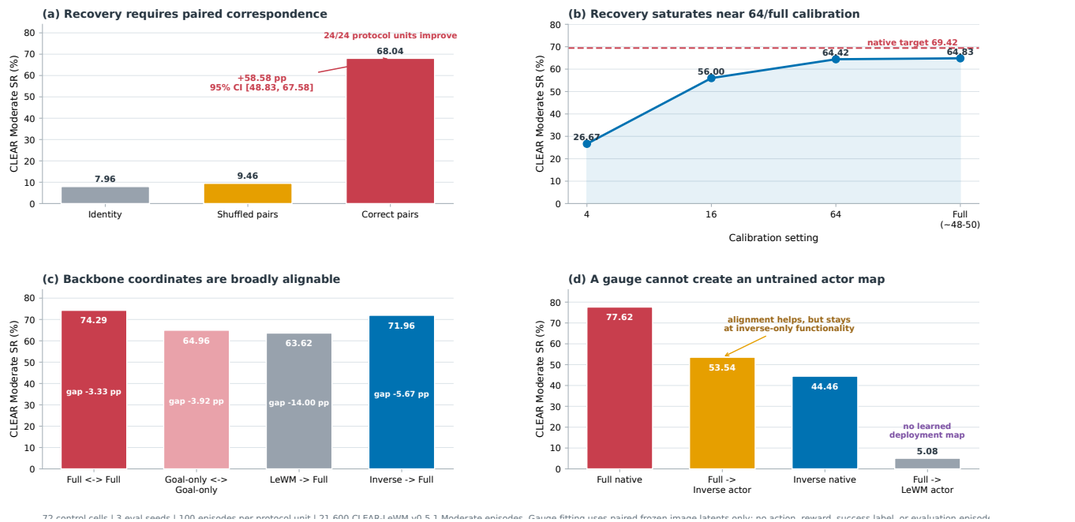


图 A5：坐标规范对照将对应关系与 actor 功能分离。正确配对的校准恢复控制；恢复程度在约 64 个/全部回合附近饱和；同目标交换定义上限；反向交换表明坐标映射不能创造未经训练的部署条件模型。72 个单元的审计包含 21,600 个 CLEAR Moderate 回合；校准不使用任何评估回合。

表 A8：共享是起作用的归纳偏置。匹配的 PushT 航点 Direct 成功率（%）。除显式双容量对照外，参数量差异不超过 0.12%。 p 是与完全共享模型进行回合配对的精确 McNemar 比较。

动作条件模型设计	参数量（M）	Direct 成功率	Δ 与共享比较	配对 p
仅航点意图	3.110	65.56±1.35	−11.33	4.23×10^{-15}
独立，参数匹配	3.114	71.22±2.27	−5.67	1.15×10^{-5}
独立，容量翻倍	6.220	72.89±2.17	−4.00	0.00362
共享主干，两个输出	3.110	74.67±1.53	−2.22	0.0927
条件 token 共享	3.111	77.11±2.12	+0.22	0.923
完全共享（默认）	3.110	76.89±.69	0	–

PushT 上，确定性模型 E1、E5、E10、E15 官方成功率为 26%、86%、94%、96%。最佳 x 预测／loss-x 流模型为 4%、76%、88%、92%；另一匹配 SIGReg 0.09 的流运行为 4%、44%、72%、76%。匹配 139319 个优化步骤时，最强流模型每步 0.568 秒，确定性预测器 0.479 秒，增加 18.4%。

历史对照刻意不作主要消融：最强流运行把 SIGReg 0.09 改为 0.03，没有匹配 RGB 噪声对照。只支持较窄观察：这些运行中，扰动已压缩潜变量未改善低预算学习。与 GC-IDM 独立潜在噪声消融一致，其任务最优在零噪声 [34]。不证明潜在扩散普遍有害，也不表明可直接与像素扩散比较。

J 完整目标位移规划器矩阵

表 A12 列出图 A2 使用的全部已完成规划器单元。每种方法包含 36 项作业：每个任务的三个训练检查点分别以种子 $\{0, 1, 42\}$ 评估，每次运行 100 个回合。宏平均为任务均值的等权平均。12 种基础规则共 432 项作业，四种守护规则共 144 项；各组均通过独立 PASS 审计。

K 回合互不重叠的反检索审计

固定 PushT 清单包含训练/验证/测试回合 14,948/1,868/1,869 个，完全无重叠。归一化、优化及检查点选择均排除测试

Table A9: Mechanism diagnostics for the matched PushT actor-sharing ablation. These measurements explain the closed-loop table but are not substitutes for SR.

Action-conditional design	Shared fraction	Grad. cosine	Shuffle-goal MAE	Consistency MSE
Condition-token shared	1.000	0.506	1.256	0.1320
Fully shared	1.000	**0.575**	**1.263**	0.1275
Shared trunk, two outputs	0.987	0.138	1.200	0.1309
Independent, double capacity	0	0	1.030	0.1382
Independent, parameter matched	0	0	1.029	0.1370
Goal-intent only	1.000	0.321	0.889	**0.1133**

Table A10: Task-centered Pearson correlation with E5 Direct SR. These are diagnostic associations, not independent causal effects.

Metric	Moderate	Strict
Effective rank	0.888	0.846
G_{sep}	0.819	0.731
Deployment NLL	-0.640	-0.766

Table A11: Separate matched PushT E1 latent-linearity stress test, official SR (%). The control is physical-inverse LeWM with no linearity loss, not forward-only LeWM.

Linearity term	Weight	Direct	Pure CEM 30×10
Inverse control (no $\mathcal{L}_{\text{lin}}$)	0	34.00	**43.78**
Flexible	0.01	**35.89**	41.11
Strict	0.01	35.11	41.22
Strict	0.03	35.11	41.78
Flexible	0.30	14.11	–

episodes. We retrain transition-only and full goal-intent models, then evaluate the same 100 test-only start–goal pairs in each of three training-seed by three evaluation-seed cells.

The full model reaches 77.56%, only 0.12 point below the 77.67% standard-protocol estimate. Among 900 paired episodes, 358 are solved only by the goal-intent model and 22 only by transition inverse (descriptive exact McNemar $p = 2.36 \times 10^{-79}$; checkpoint clustering precludes treating all episodes as independent). Every training-seed mean improves: 39.33→77.00, 40.33→80.33, and 41.00→75.33. Thus the Direct gain survives both unseen expert episodes and independent optimization runs rather than retrieving a training episode.

L Evaluation Protocol Scope

The official LeWM benchmark remains the headline protocol because it preserves comparability with prior work. CLEAR-LeWM is a separate, versioned, MIT-licensed software benchmark rather than a contemporaneously reviewed paper; we cite the public v0.5.1 release and immutable commit as a software artifact [41]. Official and CLEAR scores are never pooled. The INTACT audit pins v0.5.1 commit `32f4416c`, canonical manifests, batch one, Math-only SDPA, deterministic algorithms, full checkpoint hashes, and complete episode outcomes.

L.1 Why a corrected protocol is informative

The historical sampler draws rows uniformly and retains initially successful start–goal pairs. This creates both episode-length bias and automatic-success mass. Across the released datasets, 38.38% of valid Cube pairs and 8.82% of TwoRoom +25 pairs are already within the historical endpoint threshold. Random control consequently reaches 49% and 26%, respectively, in the three-run historical audit. Moderate removes initially solved pairs, samples episodes uniformly, restores the final valid row, and fixes only demonstrated evaluator defects. This lowers the Cube and TwoRoom random floors to 15.67% and 6.67% while retaining strong official-checkpoint performance (Table A15). A lower floor increases usable dynamic range; it does not guarantee that every model score must decrease because the sampled pair distribution also changes.

L.2 Moderate and Strict are different claims

Moderate is the closest corrected continuation of the released task. Strict builds on the same data hygiene and physics but asks for precise task-semantic completion. They are reported separately rather than averaged (Table A16).

The task-specific repairs matter. Reacher's released all-periodic comparison can wrap its physically bounded wrist coordinate; Moderate wraps only the unbounded shoulder. The rewritten TwoRoom runtime previously checked requested segment endpoints and could admit motion through a wall. Both modes now use continuous swept-disk collision for the full agent radius; Strict additionally requires a legal cross-room route and goal-side arrival. Cube Moderate retains the OGBench 4 cm position task so that compatibility is explicit, whereas Strict adds object orientation modulo 24 proper cube rotations. PushT Moderate retains the released full pusher–object goal state; Strict removes the task-irrelevant terminal pusher pose and scores precise T-object completion.

L.3 Pinned references and public checkpoint bundles

Table A17 summarizes the publicly auditable reference set; INTACT has completed Moderate only, so no Strict score is imputed.

表 A9：匹配的 PushT actor 共享消融的机制诊断。这些测量解释闭环结果表，但不能代替成功率。

动作条件模型设计	共享比例	梯度余弦	打乱目标 MAE	一致性 MSE
条件 token 共享	1.000	0.506	1.256	0.1320
完全共享	1.000	**0.575**	**1.263**	0.1275
共享主干，两个输出	0.987	0.138	1.200	0.1309
独立，容量翻倍	0	0	1.030	0.1382
独立，参数匹配	0	0	1.029	0.1370
仅目标意图	1.000	0.321	0.889	**0.1133**

表 A10：与 E5 Direct 成功率的任务中心化 Pearson 相关系数。这些是诊断关联，而非独立因果效应。

指标	Moderate	Strict
有效秩	0.888	0.846
G_{sep}	0.819	0.731
部署负对数似然	-0.640	-0.766

表 A11：独立且匹配的 PushT E1 潜空间线性化压力测试，官方成功率（%）。对照为不含线性化损失的物理逆向 LeWM，而非仅前向 LeWM。

线性化项	权重	Direct	纯 CEM 30×10
逆向对照（不含 $\mathcal{L}_{\text{lin}}$）	0	34.00	**43.78**
灵活	0.01	**35.89**	41.11
Strict	0.01	35.11	41.22
Strict	0.03	35.11	41.78
灵活	0.30	14.11	–

回合。我们重新训练仅转移和完整目标意图模型，然后在三个训练种子与三个评估种子的各组合中，评估相同的 100 对仅来自测试集的起始–目标状态。

完整模型达 77.56%，只比标准方案 77.67% 低 0.12 个百分点。900 个配对回合中，358 个仅目标意图模型成功，22 个仅转移逆模型成功（描述性精确 McNemar$p = 2.36 \times 10^{-79}$；检查点聚类不能把全部回合当独立样本）。每个训练种子平均均改善：39.33→77.00、40.33→80.33、41.00→75.33。因此 Direct 收益在未见专家回合和独立优化运行中仍成立，并非检索训练回合。

L 评估协议范围

官方 LeWM 仍为主要方案，以保持可比。
CLEAR-LeWM 是独立、版本化、MIT 许可的软件基准，不是同期评审论文；引用公开 v0.5.1 及不可变提交作为软件产物 [41]。官方与 CLEAR 分数绝不合并。
INTACT 审计固定 v0.5.1 提交 32f4416c、规范清单、清单、批量大小为 1、仅 Math SDPA、确定性算法、完整检查点哈希及完整回合结果。

L.1 修正后的协议为何有参考价值

历史采样器均匀抽取数据行，并保留初始已成功的起点–目标对，产生轨迹长度偏差与自动成功比例。发布数据中，38.38% 的有效 Cube 对和 8.82% 的 TwoRoom +25 对，已处于历史端点阈值内。因此三次历史审计中随机控制成功率为 49% 和 26%。Moderate 移除初始已解对、均匀采样轨迹、恢复最后有效行，并仅修复已证实评估缺陷。随机基线降至 15.67% 和 6.67%，官方检查点仍保持高性能（表 A15）。较低基线扩大可用动态范围，却不保证每个模型分数都下降，因为采样对分布也改变了。

L.2 Moderate 与 Strict 是不同的结果主张

Moderate 是对发布任务进行修正后最接近原任务的延续。Strict 使用相同的数据清洁规则与物理设置，但要求精确完成任务语义。两者分别报告，不取平均（表 A16）。

任务专用修复很重要。Reacher 发布版把全部坐标按周期比较，可能环绕物理有界的腕坐标；Moderate 只环绕无界肩坐标。重写 TwoRoom 运行时此前只检查请求线段端点，可能容许穿墙；如今两种模式均对完整智能体半径进行连续扫掠圆盘碰撞检测，Strict 还要求合法跨房间路线及抵达目标侧。Cube Moderate 保留 OGBench 的 4 cm 位置任务以明确兼容，Strict 加入模 24 种正当立方体旋转的朝向条件。PushT Moderate 保留发布版完整推手–物体目标状态，Strict 删去任务无关的终端推手姿态，评分精确 T 物体完成情况。

L.3 固定版本参考结果与公开检查点包

表 A17 汇总可公开审计的参考结果；INTACT 只完成了 Moderate，因此不推算 Strict 分数。

Table A12: Complete 576-job MATH-only planner matrix, official SR (%). "Plan" means that the deterministic goal-displacement action sequence initializes the factorized raw-action distribution; trust variants use coefficient one. Guarded rules preserve the Direct reference and globally best observed sequence across refits.

Planner family	Setting	PushT	Cube	Reacher	TwoRoom	Macro	Worst
Direct	Goal-displacement mean	85.78	**100.00**	**97.67**	97.89	95.33	85.78
Actor particles	K128-I1	91.44	99.67	82.78	98.00	92.97	82.78
Actor particles	K128-I3	91.89	99.67	78.44	97.44	91.86	78.44
Plan + raw CEM	$\sigma_0 = 0.05$	87.89	**100.00**	96.33	97.78	95.50	87.89
Plan + raw CEM	$\sigma_0 = 0.10$	89.78	**100.00**	97.11	97.67	96.14	89.78
Plan + raw CEM	$\sigma_0 = 0.25$	92.33	99.78	97.11	97.89	96.78	92.33
Plan + raw CEM	$\sigma_0 = 0.50$	92.11	99.78	93.78	97.89	95.89	92.11
Plan + raw CEM	$\sigma_0 = 1.00$	88.78	99.56	84.78	**98.33**	92.86	84.78
Plan + L2 trust	$\sigma_0 = 0.50$	**92.78**	99.78	94.11	97.89	96.14	**92.78**
Plan + actor-NLL trust	$\sigma_0 = 0.50$	92.11	99.89	94.22	97.89	96.03	92.11
Guarded B	K256-I3, $\sigma_0 = 0.25$	92.00	99.89	**97.67**	97.78	96.83	92.00
Guarded C	K128-I5, smooth	91.56	99.89	97.56	97.89	96.72	91.56
Guarded D	K128-I3, actor-cov + L2	91.67	99.89	97.22	98.00	96.69	91.67
Guarded A	**K128-I3, $\sigma_0 = 0.25$**	**92.22**	**99.78**	**97.44**	**98.00**	**96.86**	**92.22**
Pure raw CEM	K128-I3	58.00	65.44	73.44	80.56	69.36	58.00
Pure raw CEM	K300-I30	88.44	68.44	83.67	82.89	80.86	68.44

Table A13: PushT Direct SR (%) on the held-out 10% episode split. Each cell contains 100 paired test-only episodes.

Training objective	eval 0	eval 1	eval 42	pooled
Transition inverse	35.67	44.33	40.67	40.22
+ deployment goal intent	75.33	81.33	76.00	**77.56**

Table A14: PushT successes on held-out expert episodes (100 per cell).

	Transition inverse			Goal intent		
Training seed	e0	e1	e42	e0	e1	e42
3072	35	42	41	74	83	74
3073	36	44	41	81	81	79
3074	36	47	40	71	80	75

The comparison illustrates the division of labor between protocols. The official benchmark supplies historical continuity. Moderate removes automatic success and known implementation defects while changing as little task semantics as possible. Strict supports a stronger physical-completion claim. For single-task INTACT Direct, Moderate preserves high PushT/Cube/TwoRoom performance but changes Reacher from 97.67% official SR to 49.56%; this is exactly the kind of near-ceiling ambiguity that the corrected topology is intended to reveal.

Table A15: Historical automatic-success audit and CLEAR Moderate calibration. Historical random is the mean of three runs under the released stack. CLEAR values use canonical seeds 0/1/42 and 100 episodes per seed.

Task	Initially solved	Historical random	Moderate random
PushT	0.22	4.00	4.00
Cube	38.38	49.00	15.67
Reacher	0.57	13.00	4.33
TwoRoom	8.82	26.00	6.67

Table A16: CLEAR-LeWM v0.5.1 task contracts. "Hold" is the number of consecutive successful environment steps. The 24-fold Cube comparison minimizes orientation error over the proper rotational symmetry group of a cube.

Task	Moderate: minimal compatibility correction	Strict: task-semantic precision
PushT	Released pusher + T position < 20 px and T angle $< 20^\circ$; first hit.	T object only, < 10 px and $< 10^\circ$; hold 3.
Cube	Initially unsolved, cube-center distance ≤ 4 cm; first hit.	Cube center ≤ 3 cm and symmetry-aware orientation $\leq 15^\circ$; hold 3.
Reacher	Periodic unbounded shoulder, raw bounded wrist, joint error < 0.05 rad; first hit.	Physical fingertip endpoint ≤ 1 cm; hold 2.
TwoRoom	Clean cross-room pair, continuous swept-disk collision, endpoint < 16 px.	Legal doorway crossing, valid route, goal-side arrival, endpoint < 8 px.

Table A17: Public CLEAR-LeWM v0.5.1 results. Each task cell is Moderate/Strict SR (%). INTACT uses audited Direct inference; official LeWM uses pinned CEM 300×30. A dash denotes an unsubmitted mode. INTACT Moderate averages three training seeds, each over canonical evaluation seeds 0/1/42; official LeWM and random average those three evaluation seeds; community rows are self-reported CEM seed-42 bundles and did not submit Reacher.

Model	PushT	Cube	Reacher	TwoRoom
INTACT goal, Direct	88.22/–	99.89/–	49.56/–	95.67/–
Official LeWM	86.33/70.33	50.33/26.33	46.00/43.00	84.00/58.33
Paired random	4.00/5.00	15.67/6.00	4.33/5.00	6.67/1.67
DINOv2 No-Proprio	8/7	43/17	–/–	55/26
GCBC Joint	9/9	16/3	–/–	15/9

表 A12：完整的 576 项仅 MATH 规划器作业矩阵，官方成功率（%）。"Plan"表示用确定性的目标位移动作序列初始化因子化的原始动作分布；信赖域变体使用系数 1。守护规则在重新拟合的各轮间保留 Direct 参考及历史观测到的全局最佳序列。

规划器族	设置	PushT	Cube	Reacher	TwoRoom	宏平均	最差
Direct	目标位移均值	85.78	**100.00**	**97.67**	97.89	95.33	85.78
actor 粒子	K128-I1	91.44	99.67	82.78	98.00	92.97	82.78
actor 粒子	K128-I3	91.89	99.67	78.44	97.44	91.86	78.44
计划 + 原始 CEM	$\sigma_0 = 0.05$	87.89	**100.00**	96.33	97.78	95.50	87.89
计划 + 原始 CEM	$\sigma_0 = 0.10$	89.78	**100.00**	97.11	97.67	96.14	89.78
计划 + 原始 CEM	$\sigma_0 = 0.25$	92.33	99.78	97.11	97.89	96.78	92.33
计划 + 原始 CEM	$\sigma_0 = 0.50$	92.11	99.78	93.78	97.89	95.89	92.11
计划 + 原始 CEM	$\sigma_0 = 1.00$	88.78	99.56	84.78	**98.33**	92.86	84.78
计划 +L2 信赖域	$\sigma_0 = 0.50$	**92.78**	99.78	94.11	97.89	96.14	**92.78**
计划 + actor 负对数似然信赖域	$\sigma_0 = 0.50$	92.11	99.89	94.22	97.89	96.03	92.11
Guarded B	K256-I3, $\sigma_0 = 0.25$	92.00	99.89	**97.67**	97.78	96.83	92.00
Guarded C	K128-I5，平滑	91.56	99.89	97.56	97.89	96.72	91.56
Guarded D	K128-I3，actor 协方差 + L2	91.67	99.89	97.22	98.00	96.69	91.67
Guarded A	**K128-I3，$\sigma_0 = 0.25$**	**92.22**	**99.78**	**97.44**	**98.00**	**96.86**	**92.22**
纯原始 CEM	K128-I3	58.00	65.44	73.44	80.56	69.36	58.00
纯原始 CEM	K300-I30	88.44	68.44	83.67	82.89	80.86	68.44

表 A13：保留 10% 回合划分上的 PushT Direct 成功率（%）。每个单元包含 100 个配对、仅来自测试集的回合。

训练目标	评估 0	评估 1	评估 42	汇总
转移逆向	35.67	44.33	40.67	40.22
+部署目标意图	75.33	81.33	76.00	**77.56**

表 A14：保留专家回合上的 PushT 成功数（每个单元 100 回合）。

	转移逆向			目标意图		
训练种子	e0	e1	e42	e0	e1	e42
3072	35	42	41	74	83	74
3073	36	44	41	81	81	79
3074	36	47	40	71	80	75

比较体现方案分工。官方基准提供历史连续性；Moderate 移除自动成功和已知实现缺陷，尽量少改任务语义；Strict 支持更强物理完成结论。单任务 Direct 在 Moderate 保持高 PushT／Cube／TwoRoom 性能，但 Reacher 从官方 97.67% 变为 49.56%，正是修正拓扑要揭示的近上限歧义。

表 A15：历史自动成功审计与 CLEAR Moderate 校准。历史随机值是在发布的系统上运行三次的均值。CLEAR 使用规范种子 0/1/42，每个种子 100 回合。

Task	初始已解决	历史随机	Moderate 随机
PushT	0.22	4.00	4.00
Cube	38.38	49.00	15.67
Reacher	0.57	13.00	4.33
TwoRoom	8.82	26.00	6.67

表 A16：CLEAR-LeWM v0.5.1 任务约定。"Hold"表示连续成功的环境步数。Cube 的 24 重比较在立方体的正旋转对称群上最小化姿态误差。

Task	Moderate：最小兼容性修正	Strict：任务语义精度
PushT	发布的推杆 +T 位置 < 20 像素与 T 角度 $< 20^\circ$；首次命中。	仅 T 形物体，< 10 像素且 $< 10^\circ$；保持 3 步。
Cube	初始未解决，立方体中心距离 ≤ 4 厘米；首次命中。	Cube 中心 ≤ 3 厘米及考虑对称性的姿态 $\leq 15^\circ$；保持 3 步。
Reacher	周期性无界肩关节，原始有界腕部，关节误差 < 0.05 弧度；首次命中。	物理指尖端点 ≤ 1 厘米；保持 2 步。
TwoRoom	清洁跨房间配对，连续扫掠圆盘碰撞，端点 < 16 像素。	合法穿门、有效路径、到达目标侧抵达，端点 < 8 像素。

表 A17：公开 CLEAR-LeWM v0.5.1 结果。每格为 Moderate／Strict 成功率（%）。INTACT 使用审计 Direct，官方 LeWM 使用固定 CEM 300×30；横线为未提交模式。INTACT Moderate 平均三个训练种子，各再平均规范评估种子 0／1／42；官方 LeWM 和随机平均这三个评估种子。社区行是自行报告的 CEM 种子 42 结果包，未提交 Reacher。

模型	PushT	Cube	Reacher	TwoRoom
INTACT 目标，Direct	88.22/–	99.89/–	49.56/–	95.67/–
官方 LeWM	86.33/70.33	50.33/26.33	46.00/43.00	84.00/58.33
配对随机	4.00/5.00	15.67/6.00	4.33/5.00	6.67/1.67
DINOv2，无本体感觉	8/7	43/17	–/–	55/26
GCBC 联合	9/9	16/3	–/–	15/9